

The Sturm-Liouville problem on the half-line



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Abstract

This thesis aims to study the Sturm-Liouville problem on the real half-line :

$$L(u) = \frac{d}{ds} \left(p(s) \frac{du}{ds}(s) \right) - q(s)u(s) = g(s), \quad s \geq 0$$

with boundary condition

$$\begin{cases} R_1 u := \alpha_1 \cdot u(0) + \alpha_2 \cdot p(a)u'(0) = 0 \\ u \in L^2([0, +\infty)) \end{cases}$$

where the second condition can be thought as a boundary condition at $+\infty$. In Chapter 1, we briefly recall the theory of the regular Sturm-Liouville problem on a compact interval $[a, b]$ and, in particular, the expansion theorem for $L^2([a, b])$ -functions in terms of eigenfunctions which is obtained as a corollary of Hilbert-Schmidt theorem. Our main goal in this thesis is to generalize this theorem for the Sturm-Liouville problem on the half-line.

To that end, our first goal will be to study the feasibility of the above problem. This is described by Weyl's limit-circle and limit-point dichotomy and will be proven in Chapter 2 following the original proof of H. Weyl in [Wey10]. The treatment of the limit-circle case is analogous to that of the compact case and is done in Chapter 3, still following the 1910 paper of H. Weyl. The general case however is more complex : the expansion theorem takes an integral form instead of a discrete one. This is studied in Chapter 4, where we define the integral transform of a function $f \in L^2([0, +\infty))$ and obtain the general expansion theorem as a consequence of an inversion theorem. We later define the spectrum of the operator L and observe that, contrary to the compact or limit-circle case, the spectrum can be much more complex. Section 5.1 will for example show that we can find a Sturm-Liouville operator L on $[0, +\infty)$ such that the spectrum of L is exactly $[0, +\infty)$ but no point of the spectrum is an eigenvalue.

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Notation

1. $\mathbb{F}$ denotes $\mathbb{R}$ or $\mathbb{C}$.
2. Let $c \in \mathbb{C}$. We write c_1 or $\Re c$ the real part of c and write c_2 or $\Im c$ the imaginary part of c .
3. Let $U \subset \mathbb{R}^n$ be an open set, $I \subset \mathbb{R}$ be any interval and μ be a σ -finite measure on I . For $p \in [1, +\infty)$ and $k \in \mathbb{N}$, we denote by

(a)

$$C^k(U, \mathbb{F}) = \{f : U \rightarrow \mathbb{F} \mid f \text{ has continuous partial derivatives up to order } k\}$$

(b)

$$L^p(I, \mathbb{F}, \mu) = \{f : I \rightarrow \mathbb{F} \mid f \text{ is } \mu\text{-measurable, } \int_I |f(s)|^p d\mu < +\infty\} / \sim$$

where $f \sim g$ if $f = g$ μ -almost everywhere on I .

(c)

$$C_{L^p}^k(I, \mathbb{F}, \mu) = \{f \in C^k(I, \mathbb{F}) \mid \int_I |f(s)|^p d\mu < +\infty\}$$

Moreover, if $\mathbb{F}$ is not specified, then it is assumed to be $\mathbb{R}$ and if k is not specified, then it is assumed to be 0. For example, $C(I)$ will mean $C^0(I, \mathbb{R})$.

4. Let $[a, b] \subset \mathbb{R}$ be a compact interval and let $f : [a, b] \rightarrow \mathbb{C}$. Let $\mathcal{P}$ be the set of partitions $\sigma = \{a = x_0 < x_1 < \dots < x_{n-1} < x_n = b\}$ of $[a, b]$. The *total variation* of f on $[a, b]$ is denoted by

$$V_a^b(f) = \sup_{\sigma \in \mathcal{P}} \sum_{i=1}^{n-1} |f(x_{i+1}) - f(x_i)|$$

If $V_a^b(f)$ is finite, we say that f is of *bounded variation* on $[a, b]$.

5. Let $U \subset \mathbb{R}^n$ be open and $f \in C^1(U)$. Writing $x = (x_1, \dots, x_n)$ an element of $\mathbb{R}^n$, we denote by

$$f_{x_i} := \frac{\partial f}{\partial x_i}$$

the partial derivative of f with respect to the variable x_i . If $n = 1$, we write

$$f' = \frac{df}{dx}$$

the derivative of f .

1 The regular Sturm-Liouville problem

We briefly recall the theory of the Sturm-Liouville problem on a compact interval $[a, b]$. See [Wal98] Section 27 and 28 for more details.

1.1 The general Sturm-Liouville problem on $[a, b]$

Let $p(s) \in C^1([a, b])$, $p(s) > 0 \forall s \in [a, b]$, and $q(s) \in C^0([a, b])$. Let (α_1, α_2) and $(\beta_1, \beta_2) \in \mathbb{R}^2 \setminus \{0\}$. We consider the following nonhomogeneous *linear* differential equation

$$L(u) = \frac{d}{ds} \left(p(s) \frac{du}{ds}(s) \right) - q(s)u(s) = g(s) \in C^0([a, b]), \quad s \in [a, b]$$

with real boundary conditions

$$\begin{cases} R_1 u := \alpha_1 \cdot u(a) + \alpha_2 \cdot p(a)u'(a) = \alpha \in \mathbb{R} \\ R_2 u := \beta_1 \cdot u(b) + \beta_2 \cdot p(b)u'(b) = \beta \in \mathbb{R} \end{cases}$$

This is called the *general Sturm-Liouville (SL) problem* on $[a, b]$ and L is called a *Sturm-Liouville (SL) operator* on $[a, b]$.

The boundary conditions can be normalized by dividing the first condition by $\alpha_1^2 + \alpha_2^2 > 0$ and the second one by $\beta_1^2 + \beta_2^2 > 0$.

Proposition 1.1

Let L be a SL operator on $[a, b]$. Two solutions $u^{(1)}, u^{(2)}$ of $Lu = 0$ on $[a, b]$ are linearly independent if and only if $(u^{(1)}, \frac{d}{ds}u^{(1)})$ and $(u^{(2)}, \frac{d}{ds}u^{(2)})$ are linearly independent everywhere on $[a, b]$.

In that case, $\{u^{(1)}, u^{(2)}\}$ is called a fundamental system for $Lu = 0$.

Proof. The Cauchy-Lipschitz-Picard-Lindelöf theorem about existence and unicity of solu-

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tions to differential equations implies that for all fixed $\tau \in [a, b]$, the map

$$u(s) \mapsto \left(u(\tau), \frac{du}{ds}(\tau) \right)$$

is an isomorphism between the two-dimensional real vector space of solutions to $Lu = 0$ and $\mathbb{R}^2$. $\square$

Proposition 1.2

Let L be a SL operator on $[a, b]$ with boundary conditions $R_1 u, R_2 u$. Two solutions $u^{(1)}, u^{(2)}$ of $Lu = 0$ on $[a, b]$ both satisfying either $R_1 u^{(1)} = R_1 u^{(2)} = 0$ or $R_2 u^{(1)} = R_2 u^{(2)} = 0$ are colinear.

Proof. For example, if $R_1 u^{(1)} = R_1 u^{(2)} = 0$ and $\alpha_2 \neq 0$, then $(u^{(i)}(a), \frac{d}{ds} u^{(i)}(a))$, $i = 1, 2$, are both colinear to $(1, -\alpha_1/\alpha_2)$. The proof is similar for the other cases. $\square$

Proposition 1.3

Let L be a SL operator on $[a, b]$ and let $\{u^{(1)}, u^{(2)}\}$ be a fundamental system for $Lu = 0$. Then the general SL problem on $[a, b]$ with boundary conditions $R_1 u, R_2 u$ has a unique solution of class $C^2([a, b])$ $\forall g \in C^0([a, b])$ and $\forall \alpha, \beta \in \mathbb{R}$ if and only if the matrix

$$\begin{pmatrix} R_1 u^{(1)} & R_1 u^{(2)} \\ R_2 u^{(1)} & R_2 u^{(2)} \end{pmatrix}$$

is invertible. Moreover, this does not depend on the chosen fundamental system.

In that case, the problem is called non-degenerate.

Proof. Choose an arbitrary solution $\tilde{u}$ to $Lu = g$ on $[a, b]$ and then modify it by considering $u^* = \tilde{u} + r_1 u^{(1)} + r_2 u^{(2)}$ for some $r_1, r_2 \in \mathbb{R}$ chosen so that $R_1 u^* = R_2 u^* = 0$. The coefficients r_1, r_2 exist and are unique if and only if the above matrix is invertible.

For another fundamental system $v^{(1)} = \gamma_1 u^{(1)} + \gamma_2 u^{(2)}$, $v^{(2)} = \gamma_3 u^{(1)} + \gamma_4 u^{(2)}$ for some $\gamma_i \in \mathbb{R}$, we have

$$\begin{pmatrix} R_1 v^{(1)} & R_1 v^{(2)} \\ R_2 v^{(1)} & R_2 v^{(2)} \end{pmatrix} = \begin{pmatrix} R_1 u^{(1)} & R_1 u^{(2)} \\ R_2 u^{(1)} & R_2 u^{(2)} \end{pmatrix} \cdot \begin{pmatrix} \gamma_1 & \gamma_3 \\ \gamma_2 & \gamma_4 \end{pmatrix}$$

where the rightmost matrix is a change of basis matrix. $\square$

Proposition 1.4

Let L be a SL operator on $[a, b]$ with non-degenerate boundary conditions. Let $\{u^{(1)}, u^{(2)}\}$ be two non trivial solutions to $Lu = 0$ such that $R_1 u^{(1)} = R_2 u^{(2)} = 0$. Then $\{u^{(1)}, u^{(2)}\}$ forms a fundamental system.

Proof. If $u^{(2)}$ is colinear to $u^{(1)}$, then $R_1 u^{(2)} = R_2 u^{(2)} = 0$ and the non-degeneracy hypothesis implies $u^{(2)} \equiv 0$. $\square$

Proposition 1.5 (Green's identity)

Let $U \subset \mathbb{R}$ be open and let $p \in C^1(U)$, $u, v \in C^2(U, \mathbb{C})$. Then

$$\frac{d}{ds} \left[p \cdot \left(u \frac{dv}{ds} - v \frac{du}{ds} \right) \right] = uL(v) - vL(u) \text{ on } U$$

In particular, if $Lu = Lv = 0$, then $p \cdot \left(u \frac{dv}{ds} - v \frac{du}{ds} \right)$ is constant on each connected component of U .

Proof. This is easily checked by expanding both sides. □

Proposition 1.6

Assume that $u, v \in C^2([a, b])$ satisfy $R_1 u = R_1 v = 0$ (resp. $R_2 u = R_2 v = 0$). Then

$$p(s) \left(u(s) \frac{dv}{ds}(s) - v(s) \frac{du}{ds}(s) \right)$$

evaluated at $s = a$ (resp. $s = b$) is equal to zero.

In particular, if $R_1 u = R_1 v = 0$ and $R_2 u = R_2 v = 0$, then

$$\int_a^b uL(v) - vL(u) dt = 0$$

Proof. If $\alpha_2 = 0$, then $u(a) = v(a) = 0$ and the equality is true. Else, $\alpha_2 \neq 0$ and

$$p(a) \left(u(a) \frac{dv}{ds}(a) - v(a) \frac{du}{ds}(a) \right) = \alpha_2^{-1} ((R_1 v)u(a) - (R_1 u)v(a)) = 0$$

Proceed similarly for $s = b$. □

Theorem 1.7

Let L be a SL operator on $[a, b]$ with non-degenerate boundary conditions. Let $\{u^{(1)}, u^{(2)}\}$ be two non trivial solutions of $Lu = 0$ such that $R_1 u^{(1)} = R_2 u^{(2)} = 0$. We set

$$c \equiv p \cdot \left(u^{(1)} \frac{du^{(2)}}{ds} - u^{(2)} \frac{du^{(1)}}{ds} \right)$$

and define the Green function on $[a, b]^2$:

$$G(s, t) = \begin{cases} c^{-1} u^{(1)}(s) u^{(2)}(t) & s \leq t \\ c^{-1} u^{(2)}(s) u^{(1)}(t) & s > t \end{cases} \in C^0([a, b] \times [a, b])$$

For $g \in C^0([a, b])$,

$$u(s) = \int_a^b G(s, t) g(t) dt$$

is of class $C^2([a, b])$ and uniquely solves $Lu = g$ on $[a, b]$ with $R_1 u = R_2 u = 0$.

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Proof. We have

$$\begin{aligned}
 u(s) &= c^{-1} u^{(2)}(s) \int_a^s u^{(1)}(t) g(t) dt + c^{-1} u^{(1)}(s) \int_s^b u^{(2)}(t) g(t) dt \in C^1([a, b]) \\
 \frac{du}{ds}(s) &= c^{-1} \frac{du^{(2)}}{ds}(s) \int_a^s u^{(1)}(t) g(t) dt + c^{-1} \frac{du^{(1)}}{ds}(s) \int_s^b u^{(2)}(t) g(t) dt \in C^1([a, b]) \\
 \frac{d}{ds} \left(p(s) \frac{du}{ds}(s) \right) &= c^{-1} \frac{d}{ds} \left(p(s) \frac{du^{(1)}}{ds}(s) \right) \int_a^s u^{(1)}(t) g(t) dt + c^{-1} \frac{d}{ds} \left(p(s) \frac{du^{(2)}}{ds}(s) \right) \int_s^b u^{(2)}(t) g(t) dt \\
 &\quad + c^{-1} p(s) \cdot \left(\frac{du^{(2)}}{ds}(s) u^{(1)}(s) - \frac{du^{(1)}}{ds}(s) u^{(2)}(s) \right) g(s) \in C^0([a, b])
 \end{aligned}$$

so that

$$\begin{aligned}
 L(u)(s) &= c^{-1} L(u^{(2)})(s) \int_a^s u^{(1)}(t) g(t) dt + c^{-1} L(u^{(1)})(s) \int_s^b u^{(2)}(t) g(t) dt \\
 &\quad + c^{-1} p(s) \cdot \left(\frac{du^{(2)}}{ds}(s) u^{(1)}(s) - \frac{du^{(1)}}{ds}(s) u^{(2)}(s) \right) g(s) \\
 &= g(s)
 \end{aligned}$$

Moreover,

$$\begin{aligned}
 u(a) &= c^{-1} u^{(1)}(a) \int_a^b u^{(2)}(t) g(t) dt, & u(b) &= c^{-1} u^{(2)}(b) \int_a^b u^{(1)}(t) g(t) dt \\
 \frac{du}{ds}(a) &= c^{-1} \frac{du^{(1)}}{ds}(a) \int_a^b u^{(2)}(t) g(t) dt, & \frac{du}{ds}(b) &= c^{-1} \frac{du^{(2)}}{ds}(b) \int_a^b u^{(1)}(t) g(t) dt
 \end{aligned}$$

so that $R_1 u = R_2 u = 0$. The uniqueness follows from the non-degeneracy of L . $\square$

Definition 1.1 (Eigenfunction)

Let $u \in C^2([a, b])$ be non-trivial. We call u an *eigenfunction* for the SL operator L on $[a, b]$ with boundary conditions $R_1 u, R_2 u$ if $Lu + \lambda u = 0$ and $R_1 u = R_2 u = 0$ for some $\lambda \in \mathbb{R}$. In that case, we call λ an *eigenvalue* of L and the set of eigenvalues is called the *spectrum* of L .

Remark 1.8 (About multiplicity of eigenvalues)

Because of Proposition 1.2, each eigenvalue of L has multiplicity 1.

Remark 1.9 (About orthogonality of eigenfunctions)

Because of Proposition 1.6, any two eigenfunctions u, v associated to distinct eigenvalues are $L^2([a, b])$ -orthogonal. In particular, since $L^2([a, b])$ is separable, the number of eigenvalues is at most countable.

Remark 1.10 (About the zero eigenvalue)

The SL operator L on $[a, b]$ with boundary conditions $R_1 u, R_2 u$ is degenerate if and only if 0 is an eigenvalue of L . In the following, we will assume that L is non-degenerate, i.e. 0 is not an eigenvalue of L , but note that the expansion Theorems 1.18 and 1.19 still hold if 0 is an eigenvalue.

Indeed, if (λ, u) is a pair of eigenvalue and eigenfunction for L on $[a, b]$ with boundary conditions $R_1 u, R_2 u$, then $(\lambda + \eta, u)$ is a pair of eigenvalue and eigenfunction for the SL operator $L + \eta \cdot \text{id}$ on $[a, b]$ with boundary conditions $R_1 u, R_2 u$. Since the number of eigenvalues of L is at most countable, it is clear that we can choose η so that 0 is not an eigenvalue of $L + \eta \cdot \text{id}$.

The Green function allows us to characterize the spectrum of the linear operator L .

Lemma 1.11

Let L be a SL operator on $[a, b]$ with non-degenerate boundary conditions.

If $\lambda \in \mathbb{R}$ is an eigenvalue of L with associated eigenfunction u , then $\lambda \neq 0$ and $\mu := -\lambda^{-1}$ is an eigenvalue of the linear operator

$$T : C^0([a, b]) \rightarrow C^0([a, b])$$

$$f(t) \mapsto (Tf)(s) := \int_a^b G(s, t) f(t) dt$$

with associated eigenvector u .

Conversely, if $\mu \in \mathbb{R}$ is an eigenvalue of T with eigenvector $u \in C^0([a, b])$, then $\mu \neq 0$, $u \in C^2([a, b])$, $R_1 u = R_2 u = 0$ and $Lu + \lambda u = 0$ for $\lambda := -\mu^{-1}$.

Proof. Assume that $Lu + \lambda u = 0$, $R_1 u = R_2 u = 0$, for some non-trivial $u \in C^2([a, b])$. Then $\lambda \neq 0$ by non-degeneracy. Moreover, if we set $g := -\lambda u$, then Theorem 1.7 implies that $v := -\lambda Tu$ is the unique solution to $Lv = g$ on $[a, b]$ with boundary conditions $R_1 v = R_2 v = 0$. Hence, $u = -\lambda Tu$ by uniqueness.

Conversely, assume that $Tu = \mu u$ for some non-trivial $u \in C^0([a, b])$. By Theorem 1.7, $v := Tu \in C^2([a, b])$ is the unique solution to $Lv = u$ with boundary conditions $R_1 v = R_2 v = 0$. In particular, Lv is non-trivial so $\mu \neq 0$. From $v = \mu u$, we finally get $u \in C^2([a, b])$, $R_1 u = R_2 u = 0$, $\mu Lu = u$. $\square$

The linear operator T has properties that allows us to describe topologically the spectrum of L and obtain an expansion of arbitrary functions in terms of eigenfunctions.

Definition 1.2

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner-product space on $\mathbb{F}$ and let $T : V \rightarrow V$ be a linear operator. We say that T is

1. *bounded* if $\sup\{\|Tv\| \mid \|v\| \leq 1\} < +\infty$. This is equivalent to continuity of T .

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2. *compact* if $\overline{T(B_1(0))}$ is compact. This implies boundedness.
3. *self-adjoint* if it is bounded and $\langle Tv, w \rangle = \overline{\langle v, Tw \rangle} \forall v, w \in V$. In that case, the spectrum of T is real and any two eigenvectors not belonging to the same eigenspace are orthogonal.

Proposition 1.12

The linear operator

$$T : (C^0([a, b]), \langle \cdot, \cdot \rangle_{L^2}) \rightarrow (C^0([a, b]), \langle \cdot, \cdot \rangle_{L^2})$$

$$f(t) \mapsto (Tf)(s) := \int_a^b G(s, t) f(t) dt$$

is bounded, compact and self-adjoint. Furthermore,

$$T(C^0([a, b])) = \{u \in C^2([a, b]) \mid R_1 u = R_2 u = 0\}$$

Proof. Boundedness follows from Cauchy-Schwarz inequality, self-adjointness follows from the symmetry $G(s, t) = G(t, s)$ of the kernel and compactness follows from Arzela-Ascoli theorem (see [Wal98] Section 28, Lemma XI or the proof of Proposition 2.23 which can easily be adapted and does not require proving L^2 -compactness of T first). The inclusion $T(C^0([a, b])) \subset \{u \in C^2([a, b]) \mid R_1 u = R_2 u = 0\}$ follows directly from Theorem 1.7 and the reverse inclusion follows from the same theorem applied to $g := Lu$ for $u \in \{v \in C^2([a, b]) \mid R_1 v = R_2 v = 0\}$. $\square$

Proposition 1.13 (Bessel's inequality and Parseval's identity)

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner-product space on $\mathbb{F}$, $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, and let $(u_n)_{n=1}^N \subset V$, where $N \in \mathbb{N} \cup \{+\infty\}$, be an orthonormal sequence. Then for any $w \in V$:

$$\sum_{k=1}^N |\langle w, u_k \rangle|^2 \leq \|w\|^2 \quad (\text{Bessel's inequality})$$

Furthermore, this is an equality (i.e. Parseval's identity holds) if and only if w can be represented as the following limit in $(V, \|\cdot\|_{\langle \cdot, \cdot \rangle})$:

$$w = \lim_{n \rightarrow N} \sum_{k=1}^n \langle w, u_k \rangle u_k$$

i.e.

$$\lim_{n \rightarrow N} \|w - \sum_{k=1}^n \langle w, u_k \rangle u_k\| = 0$$

Proof. Observe that by orthonormality of $(u_n)_{n=1}^N$:

$$\|w - \sum_{k=1}^n \langle w, u_k \rangle u_k\|^2 = \|w\|^2 - \sum_{k=1}^n |\langle w, u_k \rangle|^2 \geq 0 \quad \forall n \in \mathbb{N}, n \leq N$$

□

Remark 1.14

The orthonormality of $(u_n)_{n=1}^N$ and Bessel's inequality imply that for any $w \in V$, the sequence

$$\left(\sum_{k=1}^n \langle w, u_k \rangle u_k \right)_{n=1}^N \subset V$$

is a Cauchy sequence. If $(V, \|\cdot\|_{\langle \cdot, \cdot \rangle})$ is a complete space, then this sequence converges in V .

Theorem 1.15 (Hilbert-Schmidt)

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner-product space on $\mathbb{F}$ and let $T : V \rightarrow V$ be a linear, compact and self-adjoint operator. Then there exists a countable sequence of eigenvalues $(\mu_n)_{n=1}^N$, where $N \in \mathbb{N} \cup \{+\infty\}$, such that

$$|\mu_{n+1}| \leq |\mu_n| \quad \forall n \in \mathbb{N} \quad \text{and} \quad \lim_{n \rightarrow +\infty} \mu_n = 0 \quad \text{if } N = +\infty$$

as well as a sequence of associated eigenvectors $(u_n)_{n=1}^N \subset V \setminus \{0\}$ such that $U = \{u_n\}_{n=1}^N$ forms an orthonormal set and such that any $w \in \text{Im } T$ can be represented as the following limit in $(V, \|\cdot\|_{\langle \cdot, \cdot \rangle})$:

$$w = \lim_{n \rightarrow N} \sum_{k=1}^n \langle w, u_k \rangle u_k$$

Moreover, if $(\overline{V}, \langle \cdot, \cdot \rangle_{\overline{V}})$ denotes the completion of $(V, \langle \cdot, \cdot \rangle_V)$ and $T(V) \subset \overline{V}$ is dense, then this representation holds for every $w \in \overline{V}$.

Remark 1.16

$T(V) \subset V$ is dense if and only if $T(V) \subset \overline{V}$ is dense.

Proof. See [Wal98] Section 28, Theorem VII for the first part and [Sar98] Theorem 1.5.2 for the second part.

The difficult part of the proof is to construct an eigenvector u_1 of T , $\|u_1\| = 1$, associated to an eigenvalue $\mu_1 = \pm \|T\|$. Then, we restrict T to the subspace $V_1 = \text{span}(u_1)^\perp$ and proceed the same way to find an eigenvector u_2 of $T|_{V_1}$, $\|u_2\| = 1$, associated to an eigenvalue μ_2 such that $|\mu_2| = \|T|_{V_1}\| \leq \|T\|$. Next, we define $V_2 = \text{span}(u_1, u_2)^\perp$ and proceed similarly by induction to get our sequences $(\mu_n)_{n=1}^N$, $(u_n)_{n=1}^N$. The compactness of T applied to the uniformly bounded sequence $(u_n)_{n=1}^N$ shows that $\lim_{n \rightarrow +\infty} \mu_n = 0$ if $N = +\infty$. Then, for $w = Tv \in \text{Im } T$, we have

$$\begin{aligned} w - \sum_{n=1}^k \langle w, u_n \rangle u_n &= Tv - \sum_{n=1}^k \langle Tv, u_n \rangle u_n \\ &= Tv - \sum_{n=1}^k \langle v, Tu_n \rangle u_n \end{aligned}$$

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$$= Tv - \sum_{n=1}^k \mu_n \langle v, u_n \rangle u_n \in V_k = \text{span}(u_1, \dots, u_k)^\perp$$

so that

$$\|w - \sum_{n=1}^k \langle w, u_n \rangle u_n\|_V \leq \|T|_{V_k}\| \cdot \|v - \sum_{n=1}^k \langle v, u_n \rangle u_n\|_V \leq 2|\mu_k| \cdot \|v\|_V \rightarrow 0 \text{ as } k \rightarrow +\infty$$

where we used the boundedness of T , $|\mu_k| = \|T|_{V_k}\|$ and Bessel's inequality. This concludes the proof of the first part of the theorem.

As for the second part, this is a simple density argument. By Proposition 1.13, it suffices to show that Parseval's identity holds for elements of $\overline{V}$. Let $w \in \overline{V}$ be fixed and let $(w_m)_{m=1}^\infty \subset T(V)$ be a sequence for which :

$$\lim_{m \rightarrow +\infty} \|w_m - w\|_{\overline{V}} = 0$$

By the first part of the theorem, Parseval's identity holds for the w_m 's :

$$\sum_{n=1}^{+\infty} \langle w_m, u_n \rangle_{\overline{V}}^2 = \sum_{n=1}^{+\infty} \langle w_m, u_n \rangle_V^2 = \|w_m\|_V^2 = \|w_m\|_{\overline{V}}^2 \quad \forall m \in \mathbb{N}$$

Since w_m converges to w in $(\overline{V}, \langle \cdot, \cdot \rangle_{\overline{V}})$, the right-hand side converges to $\|w\|_{\overline{V}}^2$.

As for the left-hand side, observe that by Bessel's inequality, $\sum_{n=1}^\infty \langle w, u_n \rangle_{\overline{V}}^2$ and $\sum_{n=1}^\infty \langle w_m, u_n \rangle_{\overline{V}}^2$ converge absolutely for any $m \in \mathbb{N}$ and that

$$\begin{aligned} \sum_{n=1}^\infty \langle w_m, u_n \rangle_{\overline{V}}^2 - \sum_{n=1}^\infty \langle w, u_n \rangle_{\overline{V}}^2 &= \sum_{n=1}^\infty (\langle w_m, u_n \rangle_{\overline{V}} - \langle w, u_n \rangle_{\overline{V}}) \cdot (\langle w_m, u_n \rangle_{\overline{V}} + \langle w, u_n \rangle_{\overline{V}}) \\ &= \sum_{n=1}^\infty \langle w_m - w, u_n \rangle_{\overline{V}} \cdot \langle w_m + w, u_n \rangle_{\overline{V}} \\ &\leq \underbrace{\left(\sum_{n=1}^\infty \langle w_m - w, u_n \rangle_{\overline{V}}^2 \right)^{1/2}}_{(A)} \cdot \underbrace{\left(\sum_{n=1}^\infty \langle w_m + w, u_n \rangle_{\overline{V}}^2 \right)^{1/2}}_{(B)} \end{aligned}$$

where we used Cauchy-Schwarz inequality and wrote $\langle \cdot, \cdot \rangle = \langle \cdot, \cdot \rangle_{\overline{V}}$ for simplicity. By Bessel's inequality, (B) is uniformly bounded and

$$(A)^2 = \sum_{n=1}^\infty \langle w_m - w, u_n \rangle_{\overline{V}}^2 \leq \|w_m - w\|_{\overline{V}}^2 \rightarrow 0 \text{ as } m \rightarrow +\infty$$

□

Remark 1.17 (Consequences of Hilbert-Schmidt Theorem)

In particular, the theorem implies that

1. $N < +\infty$ if and only if $\text{Im } T$ is finite-dimensional.

2. Each non-zero eigenvalue μ of T appears in the sequence $(\mu_n)_{n=1}^N$. Indeed, if that were not the case, for $w \in V \setminus \{0\}$ such that $Tw = \mu w$, we would have $w \in \text{Im } T$ so that

$$w = \lim_{n \rightarrow N} \sum_{k=1}^n \langle w, u_k \rangle u_k = \lim_{n \rightarrow N} \sum_{k=1}^n 0 \cdot u_k = 0$$

by orthogonality of eigenvectors belonging to different eigenspaces.

3. $\|T\| = |\mu_1| = \max\{|\mu| \mid \mu \text{ is an eigenvalue of } T\}$.
4. Each eigenspace $\ker(T - \mu I)$ of T with $\mu \neq 0$ is finite-dimensional. Observe however that this result holds for any compact linear operator from a Banach space into itself. This is a consequence of Riesz's Lemma.
5. The dimension of an eigenspace $\ker(T - \mu I)$ with $\mu \neq 0$ is equal to the number of occurrences of μ in $(\mu_n)_{n=1}^N$. Indeed, for an eigenvector w of T with eigenvalue $\mu \neq 0$, we have $Tw = \mu w$ so that $w \in \text{Im } T$ and

$$w = \lim_{n \rightarrow N} \sum_{k=1}^n \langle w, u_k \rangle u_k = \sum_{k: \mu_k = \mu} \langle w, u_k \rangle u_k$$

by orthogonality of eigenvectors belonging to different eigenspaces. In particular, the set $\{u_n \mid \mu_n = \mu\}$ forms an orthonormal basis for $\ker(T - \mu I)$.

6. Let $(v_n)_{n=1}^N \subset V$ be any other orthonormal sequence of eigenvectors with associated sequence of eigenvalues $(\nu_n)_{n=1}^N \subset \mathbb{R} \setminus \{0\}$ and assume that the sequence is complete in the sense that any other eigenvector $v \in V \setminus \ker T$ can be written as a finite linear combination of elements of $\{v_n\}_{n=1}^N$. Then $\lim_{n \rightarrow +\infty} \nu_n = 0$ if $N = +\infty$ by compactness of T and any $w \in \text{Im } T$ (or any $w \in \overline{V}$ if $T(V) \subset \overline{V}$ is dense) can be represented as the following limit in $(V, \|\cdot\|_{\langle \cdot, \cdot \rangle})$:

$$w = \lim_{n \rightarrow N} \sum_{k=1}^n \langle w, v_k \rangle v_k$$

Using (3.) and $\lim_{n \rightarrow +\infty} \nu_n = 0$, convergence is proven exactly in the same way as in the proof of Hilbert-Schmidt Theorem.

Theorem 1.18 (Expansion theorem)

Let L be a SL operator on $[a, b]$ with boundary conditions $R_1 u = R_2 u = 0$. Then there exists a countable sequence of eigenvalues $(\lambda_n)_{n=1}^\infty$ such that

$$|\lambda_{n+1}| \geq |\lambda_n| \quad \forall n \in \mathbb{N} \quad \text{and} \quad \lim_{n \rightarrow +\infty} |\lambda_n| = +\infty$$

as well as a sequence of associated eigenfunctions $(u_n)_{n=1}^\infty \subset C^2([a, b])$ such that $U = \{u_n\}_{n=1}^\infty$ forms an orthonormal set in $(C^0([a, b]), \langle \cdot, \cdot \rangle_{L^2})$ and such that any $w \in C^2([a, b])$ with $R_1 w =$

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$R_2 w = 0$ can be represented as the following Fourier series in $(C^0([a, b]), \|\cdot\|_{L^2})$:

$$w = \lim_{n \rightarrow +\infty} \sum_{k=1}^n \langle w, u_k \rangle u_k$$

In fact, we also have absolute and uniform convergence of the series on $[a, b]$.

Proof. This is a consequence of Hilbert-Schmidt Theorem. For the absolute and uniform convergence, see [Wal98] Section 28, p.297, or the proof of Theorem 3.10 which is essentially the same. $\square$

Theorem 1.19 (Expansion theorem for L^2 functions)

Let L be a SL operator on $[a, b]$ with boundary conditions $R_1 u = R_2 u = 0$. Then there exists a countable sequence of eigenvalues $(\lambda_n)_{n=1}^\infty$ such that

$$|\lambda_{n+1}| \geq |\lambda_n| \quad \forall n \in \mathbb{N} \quad \text{and} \quad \lim_{n \rightarrow +\infty} |\lambda_n| = +\infty$$

as well as a sequence of associated eigenfunctions $(u_n)_{n=1}^\infty \subset C^2([a, b])$ such that $U = \{u_n\}_{n=1}^\infty$ forms an orthonormal set in $(L^2([a, b]), \langle \cdot, \cdot \rangle_{L^2})$ and such that any $w \in L^2([a, b])$ can be represented as the following Fourier series in $(L^2([a, b]), \|\cdot\|_{L^2})$:

$$w = \lim_{n \rightarrow +\infty} \sum_{k=1}^n \langle w, u_k \rangle u_k$$

Proof. See the second part of Theorem 1.15. $\square$

2 The singular Sturm-Liouville problem

Let $p(s) \in C^1([0, +\infty))$, $p(s) > 0 \forall s \in [0, +\infty)$, and $q(s) \in C^0([0, +\infty))$. Let $u^{(1)}(s)$, resp. $u^{(2)}(s)$, be solutions to the homogeneous linear differential equation

$$L(u) = \frac{d}{ds} \left(p(s) \frac{du}{ds} \right) - q(s)u(s) = 0, \quad s \geq 0$$

with initial conditions

$$u(0) = 1, \quad \left[p(s) \frac{du}{ds} \right]_{s=0} = 0 \quad (2.1)$$

resp.

$$u(0) = 0, \quad \left[p(s) \frac{du}{ds} \right]_{s=0} = 1 \quad (2.2)$$

Then $\{u^{(1)}, u^{(2)}\}$ forms a fundamental system for $Lu = 0$.

We now introduce a parameter λ and consider the differential equation

$$L(u) + \lambda u = 0, \quad s \geq 0 \quad (2.3)$$

The solutions $u^{(1)}(s, \lambda)$, resp. $u^{(2)}(s, \lambda)$, of equation (2.3) satisfying the boundary conditions (1.1), resp. (1.2), are obtained from a complex Sturm-Liouville problem with $q(s) - \lambda$ instead of $q(s)$.

Let $h \in [0, \pi)$ be fixed for the rest of this work. We form

$$\phi(s, \lambda) = -\sin(h) \cdot u^{(1)}(s, \lambda) + \cos(h) \cdot u^{(2)}(s, \lambda). \quad (2.4)$$

Then $\phi(s, \lambda)$ is nothing other than the solution of (2.3) satisfying the initial conditions

$$u(0) = -\sin(h), \quad \left[p(s) \frac{du}{ds} \right]_{s=0} = \cos(h)$$

In other words, $\phi(s, \lambda)$ is the only solution (up to a scalar factor) of (2.3) satisfying the boundary condition

$$R_1 u = \left[\cos(h) \cdot u(s) + \sin(h) \cdot p(s) \frac{du}{ds} \right]_{s=0} = 0 \quad (2.5)$$

Chapter 2. The singular Sturm-Liouville problem

We examine (2.3) first for a fixed $\lambda \in \mathbb{C} \setminus \mathbb{R}$.

We form the two particular solutions

$$\phi(s, \lambda) = -\sin(h) \cdot u^{(1)}(s, \lambda) + \cos(h) \cdot u^{(2)}(s, \lambda) =: \eta(s, \lambda) =: \eta(s) \quad (2.6)$$

$$\cos(h) \cdot u^{(1)}(s, \lambda) + \sin(h) \cdot u^{(2)}(s, \lambda) =: \theta(s, \lambda) =: \theta(s) \quad (2.7)$$

Remark 2.1

The pairs $(\eta(s, \lambda), \eta_s(s, \lambda))$, $(\theta(s, \lambda), \theta_s(s, \lambda))$ are continuous on $[0, +\infty) \times \mathbb{C}$ and for any fixed $s_0 \geq 0$, they are holomorphic with respect to the parameter λ (see [Wal98] Chapter 3, Section 13 or [Lev55] Chapter 1, Section 8).

Notation

If $c \in \mathbb{C}$, then we write c_1 the real part of c and c_2 the imaginary part of c .

Furthermore, we define

$$(uv) := p(s) \left(u(s) \frac{dv}{ds}(s) - v(s) \frac{du}{ds}(s) \right)$$

and write $(uv)_a$ when this expression is evaluated at $s = a$.

Proposition 2.2

The map $(\cdot, \cdot)$ satisfies the following calculation rules :

1. $(uv) + (vu) = 0$, $(uu) = 0$.
2. $(\cdot, \cdot)$ is bilinear.
3. The real and imaginary parts of (uv) are $(u_1 v_1) - (u_2 v_2)$ and $(u_1 v_2) + (u_2 v_1)$ respectively.
4. $(u\bar{u}) = 2i(u_2, u_1)$.
5. If $u, v \in C^2([0, a], \mathbb{C})$, then Green's identity holds :

$$\int_0^s (uL(v) - vL(u)) dt = (uv)_s - (uv)_0, \quad s \in [0, a]$$

In particular, for any $\lambda \in \mathbb{C}$:

(a) If u, v both solve $Lu + \lambda u = 0$ on $[0, a]$, then (uv) is constant on $[0, a]$.

(b) If u solves $L(u) + \lambda u = 0$ on $[0, a]$, then

$$\lambda_2 \int_0^s |u|^2 dt = (u_2 u_1)_s - (u_2 u_1)_0, \quad s \in [0, a] \quad (2.8)$$

6. If $u, v \in C^2([0, a], \mathbb{C})$ satisfy

$$\begin{cases} \alpha_1 \cdot u(s) + \alpha_2 \cdot p(s)u'(s) = 0 \\ \alpha_1 \cdot v(s) + \alpha_2 \cdot p(s)v'(s) = 0 \end{cases}$$

for some $s \in [0, a]$ and $(\alpha_1, \alpha_2) \in \mathbb{C}^2 \setminus \{0\}$, then $(uv)_s = 0$.

It follows from (2.1) and (2.2) that

$$(\theta\eta)_0 = 1$$

and it follows from Green's identity that

$$(\theta\eta) \equiv 1$$

as well.

For $l \in \mathbb{C}$, we define

$$\beta^l(s) := \beta^l(s, \lambda) := \theta(s, \lambda) + l \cdot \eta(s, \lambda)$$

Then it also holds that

$$(\beta^l\eta) \equiv 1 \tag{2.9}$$

2.1 Weyl's limit-point and limit-circle theorem

In this Section 2.1, $\lambda \in \mathbb{C} \setminus \mathbb{R}$ is fixed. Let $a > 0$. We consider the general Sturm-Liouville problem $L(u) + \lambda u = 0$ on $[0, a]$ and then observe that by passing to the limit, we find a square-integrable solution on $[0, +\infty)$. We will follow the original proof of Weyl that appears in [Wey10].

Theorem 2.3 (Existence of a real boundary condition for β^l)

There exists $j \in [0, \pi)$ such that β^j solves $Lu + \lambda u = 0$ with real boundary condition

$$R_2 u := \left[\cos(j) \cdot u(s) + \sin(j) \cdot p(s) \frac{du}{ds} \right]_{s=a} = 0 \tag{2.10}$$

if and only if l lies on the circle C_a described by its diametrically opposed points

$$l^* = \frac{i(\theta\eta_1)_a}{(\eta_2\eta_1)_a}$$

and

$$l_* = \frac{(\theta\eta_2)_a}{(\eta_2\eta_1)_a}$$

If $\lambda_2 > 0$, they are respectively the highest and lowest points of C_a and if $\lambda_2 < 0$, they are respectively the lowest and highest points C_a .

Moreover, for $l \in C_a$, the associated $j \in [0, \pi)$ is unique and the map $l \in C_a \mapsto j(l) \in [0, \pi)$ is a

bijection.

Proof. Condition (2.10) breaks down into

$$\cos(j) \cdot \beta_1^l(a) + \sin(j) \cdot p(s) \frac{d\beta_1^l}{ds}(a) = 0 \quad (2.10.1)$$

$$\cos(j) \cdot \beta_2^l(a) + \sin(j) \cdot p(s) \frac{d\beta_2^l}{ds}(a) = 0 \quad (2.10.2)$$

Both equations have two unknowns, namely l_1 and l_2 . Therefore, they respectively describe either $\mathbb{C}$, $\emptyset$, or a straight line in $\mathbb{C}$.

We show that (2.10.1) and (2.10.2) are two straight lines forming a right angle. Observe first that we can find a point l^* of (2.10.1) such that

$$\beta_1^{l^*}(a) = 0, \quad \left[p(s) \frac{d\beta_1^{l^*}}{ds} \right]_{s=a} = 0$$

Since

$$\begin{aligned} \beta_1^{l^*}(a) &= \frac{1}{(\eta_2 \eta_1)_a} [\eta_2(a)(\beta_1^{l^*} \eta_1)_a - \eta_1(a)(\beta_1^{l^*} \eta_2)_a] \\ \left[p(s) \frac{d\beta_1^{l^*}}{ds} \right]_{s=a} &= \frac{1}{(\eta_2 \eta_1)_a} \left[\frac{d\eta_2}{ds}(a)(\beta_1^{l^*} \eta_1)_a - \frac{d\eta_1}{ds}(a)(\beta_1^{l^*} \eta_2)_a \right] \end{aligned}$$

and

$$(\eta_2 \eta_1)_a = (\eta_2 \eta_1)_a - (\eta_2 \eta_1)_0 = \lambda_2 \int_0^a |\eta|^2 ds \neq 0$$

it is enough to find l^* such that

$$(\beta_1^{l^*} \eta_1)_a = (\beta_1^{l^*} \eta_2)_a = 0 \quad (2.11)$$

By (2.9), $(\beta_1^{l^*} \eta_2)_a = (\beta_2^{l^*} \eta_1)_a$, so it suffices to find l^* such that

$$0 = (\beta_1^{l^*} \eta_1)_a = (\theta \eta_1)_a + l^* (\eta_2 \eta_1)_a$$

A point l_* of (2.10.2) is found similarly by looking for l_* such that

$$\beta_2^{l_*}(a) = 0, \quad \left[p(s) \frac{d\beta_2^{l_*}}{ds} \right]_{s=a} = 0$$

or sufficiently enough such that

$$(\beta_2^{l_*} \eta_1)_a = (\beta_2^{l_*} \eta_2)_a = 0 \quad (2.12)$$

We consider next the affine transformation

$$\begin{aligned} A_j : \mathbb{C} &\rightarrow \mathbb{C} \\ l &\mapsto \cos(j) \cdot \beta^l(a) + \sin(j) \cdot p(a) \frac{d\beta^l}{ds}(a) \\ &= (\cos(j) \cdot \theta(a) + \sin(j) \cdot p(a)\theta'(a)) + \\ &\quad l \cdot (\cos(j) \cdot \eta(a) + \sin(j) \cdot p(a)\eta'(a)) \end{aligned}$$

This transformation maps $\{l \in (2.10.1)\}$ into $i\mathbb{R}$ and $\{l \in (2.10.2)\}$ into $\mathbb{R}$. Furthermore, it is non constant. Otherwise, we would have $A_j \equiv 0$, i.e.

$$\begin{cases} \cos(j) \cdot \theta(a) + \sin(j) \cdot p(a)\theta'(a) = 0 \\ \cos(j) \cdot \eta(a) + \sin(j) \cdot p(a)\eta'(a) = 0 \end{cases}$$

which implies $(\eta\theta)_a = 0$: a contradiction.

In particular, we deduce that A_j is bijective. Thus, there is a unique point $l = l(j)$ such that $A_j(l) = 0$, i.e. a point satisfying (2.10). Therefore, (2.10.1) and (2.10.2) are two straight lines intersecting at a single point l . Since A_j preserves lines and angles, we deduce

$$A_j(\{l \in (2.10.1)\}) = i\mathbb{R}, \quad A_j(\{l \in (2.10.2)\}) = \mathbb{R}$$

and (2.10.1) intersects (2.10.2) at a right angle.

Finally, we let j vary over $[0, \pi)$. Since

$$\begin{aligned} l(j) &= \frac{-\cos(j) \cdot \theta(a) - \sin(j) \cdot p(a)\theta'(a)}{\cos(j) \cdot \eta(a) + \sin(j) \cdot p(a)\eta'(a)} \\ &= \frac{Az + B}{Cz + D} \end{aligned}$$

with $z = \cot(j)$ varying over $\mathbb{R}$, the set of admissible j must be a circle C_a . Since every point of this circle is obtained by the intersection at a right angle of a line coming from l^* and a line coming from l_* , we deduce that l^* and l_* are diametrically opposed points of C_a that differ by a pure imaginary number

$$l^* - l_* = \frac{i}{(\eta_2\eta_1)_a} \in i\mathbb{R}$$

which belongs to $i\mathbb{R}_{>0}$ if $\lambda_2 > 0$ and to $i\mathbb{R}_{<0}$ if $\lambda_2 < 0$. □

Remark 2.4 (On the diameter of C_a)

C_a has diameter

$$2r_a = |l^* - l_*| = \frac{1}{|(\eta_2\eta_1)_a|} = (|\lambda_2| \int_0^a |\eta|^2 ds)^{-1} \quad (2.13)$$

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Theorem 2.5 (Weyl's limit-point, limit-circle theorem)

The circle C_a lies in the half-plane $\{l \in \mathbb{C} : \lambda_2^{-1} l_2 > 0\}$ and contains every other circle C_b with $b > a$. Therefore, C_a either shrinks to a limit-circle or to a limit-point as $a \rightarrow +\infty$.

Proof. Initial conditions for θ, η yields $(\beta^l \overline{\beta^l})_0 = (\bar{l} - l)$ and by property (4) of $(\cdot, \cdot)$:

$$(\beta_2^l \beta_1^l)_0 = -\frac{i}{2} (\beta^l \overline{\beta^l})_0 = -\frac{i}{2} (\bar{l} - l) = -l_2 \quad (2.14)$$

If $l \in C_a$, conditions (2.10.1) and (2.10.2) are satisfied for some $j \in [0, \pi)$ and those imply $(\beta_2^l \beta_1^l)_a = 0$. Thus,

$$\lambda_2^{-1} l_2 = \lambda_2^{-1} \left((\beta_2^l \beta_1^l)_a - (\beta_2^l \beta_1^l)_0 \right) = \int_0^a |\beta^l|^2 ds > 0$$

proving the first assertion.

Thus, we have proved that the equation

$$\int_0^a |\beta^l|^2 ds = \lambda_2^{-1} l_2, \quad l \in \mathbb{C} \quad (2.15)$$

is satisfied by all $l \in C_a$. Since

$$\int_0^a |\beta^l|^2 ds = |l|^2 \int_0^a |\eta|^2 ds + l \int_0^a \bar{\theta} \eta ds + \bar{l} \int_0^a \theta \bar{\eta} ds + \int_0^a |\theta|^2 ds, \quad \forall l \in \mathbb{C} \quad \forall a > 0 \quad (2.16)$$

we deduce by completing the square that (2.15) is precisely the cartesian equation describing C_a . Therefore, relaxing the equality constraint for an inequality

$$\int_0^a |\beta^l|^2 ds < \lambda_2^{-1} l_2, \quad l \in \mathbb{C}$$

should either give the interior of the circle or the exterior of the circle.

In fact, it gives the interior of the circle. Indeed, the point $l := 0 \notin C_a$ satisfies

$$\int_0^a |\beta^l|^2 ds = \int_0^a |\theta^l|^2 ds > 0 = l_2 = \lambda_2^{-1} l_2$$

Now let $b > a$ and let $l \in C_b$. Then l lies in the interior of C_a by monotonicity of the integral :

$$\lambda_2^{-1} l_2 = \int_0^b |\beta^l|^2 ds > \int_0^a |\beta^l|^2 ds$$

□

Remark 2.6 (On the holomorphicity of l with respect to λ)

Let $j \in [0, \pi)$ be fixed. For $a > 0$ and $\lambda \in \mathbb{C} \setminus \mathbb{R}$, consider the unique $l_a(\lambda) \in C_{a,\lambda}$ given by

$$l_a(\lambda) = \frac{-\cos(j) \cdot \theta(a, \lambda) - \sin(j) \cdot p(a) \theta'(a, \lambda)}{\cos(j) \cdot \eta(a, \lambda) + \sin(j) \cdot p(a) \eta'(a, \lambda)}$$

2.1. Weyl's limit-point and limit-circle theorem

For a fixed $a > 0$, $l_a(\lambda)$ is holomorphic on $\mathbb{C} \setminus \mathbb{R}$ as it is a well-defined quotient of holomorphic functions. Moreover, the family of holomorphic functions $(l_a(\lambda))_{a \geq 1}$ is uniformly bounded on every compact subset of $\mathbb{C} \setminus \mathbb{R}$. By Montel's Theorem (which is a corollary of Arzela-Ascoli Theorem), this family is normal so that for every unbounded and increasing sequences $(a_n)_{n=1}^\infty$ in $\mathbb{R}_{\geq 1}$, there exists a subsequence $(a_{n_k})_{k=1}^\infty$ for which $(l_{a_{n_k}}(\lambda))_{k=1}^\infty$ converges uniformly on every compact subset of $\mathbb{C} \setminus \mathbb{R}$ to some holomorphic function $m(\lambda)$ defined on $\mathbb{C} \setminus \mathbb{R}$. In the limit-point case, the limit function is unique in the sense that it does not depend on j , the sequence $(a_n)_{n=1}^\infty$ and its subsequence. In the limit-circle case, this is not the case.

Observe also that $l_a(\bar{\lambda}) = \overline{l_a(\lambda)}$ and this property is preserved at the limit.

Notation

For a circle C_a , we denote D_a the compact disc with boundary C_a .

In the limit-circle case, we denote C_∞ the limit-circle and D_∞ the limit-disc with boundary circle C_∞ .

In the limit-point case, we denote l_∞ the limit-point.

Remark 2.7 (Characterization of C_a and D_a)

In particular, we have shown

$$C_a = \{l \in \mathbb{C} \mid \int_0^a |\beta^l|^2 ds = \lambda_2^{-1} l_2\} \quad (2.17)$$

$$D_a = \{l \in \mathbb{C} \mid \int_0^a |\beta^l|^2 ds \leq \lambda_2^{-1} l_2\} \quad (2.18)$$

Furthermore, the inclusion $D_b \subset D_a$ for $b > a$ yields

$$l_\infty \in \bigcap_{a>0} D_a = \{l \in \mathbb{C} \mid \int_0^{+\infty} |\beta^l|^2 ds \leq \lambda_2^{-1} l_2\}$$

$$D_\infty \subset \bigcap_{a>0} D_a = \{l \in \mathbb{C} \mid \int_0^{+\infty} |\beta^l|^2 ds \leq \lambda_2^{-1} l_2\}$$

in the limit-point case and limit-circle case respectively.

Corollary 2.8 (Weyl alternative)

Let $\lambda \in \mathbb{C} \setminus \mathbb{R}$ be fixed. For the equation $L(u) + \lambda u = 0$ on $[0, +\infty)$, exactly one of the following situations happens :

1. *Limit-circle case* : Every solution to $L(u) + \lambda u = 0$ on $[0, +\infty)$ is square-integrable.
2. *Limit-point case* : There is only one (up to a scalar factor) square-integrable solution to $L(u) + \lambda u = 0$ on $[0, +\infty)$.

Proof. Let l be the limit-point or l be in the limit-circle. Then $\{\eta, \beta^l\}$ forms a fundamental

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system of solutions for $L(u) + \lambda u = 0$ on $[0, +\infty)$.

In the limit-circle case (i.e. $r_a \rightarrow r_\infty > 0$), $\beta^l \in L^2([0, +\infty), \mathbb{C})$ and $\eta \in L^2([0, +\infty), \mathbb{C})$.

In the limit-point case (i.e. $r_a \rightarrow 0$), $\beta^l \in L^2([0, +\infty), \mathbb{C})$ but $\eta \notin L^2([0, +\infty), \mathbb{C})$. $\square$

Remark 2.9

We will show in Section 2.3 that the alternative generalizes to real values of λ and that it is independent of λ , i.e. if 1. (resp. 2.) holds for one particular value of $\lambda^* \in \mathbb{C}$, then it holds for any other $\lambda \in \mathbb{C}$.

Theorem 2.10 (Characterization of the limit-point and limit-circle)

We have

$$l_\infty = \{l \in \mathbb{C} \mid \int_0^{+\infty} |\beta^l|^2 ds = \lambda_2^{-1} l_2\} \quad (2.19)$$

$$C_\infty = \{l \in \mathbb{C} \mid \int_0^{+\infty} |\beta^l|^2 ds = \lambda_2^{-1} l_2\} \quad (2.20)$$

$$D_\infty = \{l \in \mathbb{C} \mid \int_0^{+\infty} |\beta^l|^2 ds \leq \lambda_2^{-1} l_2\} \quad (2.21)$$

in the limit-point case and limit-circle case respectively.

Proof.

We prove (2.20) first. The equation for (2.21) follows as in the proof of Theorem 2.5.

Let $l \in C_\infty$ and let $(l_n)_{n=1}^\infty \subset \mathbb{C}$ be a convergent sequence such that $l_n \in C_n \forall n \in \mathbb{N}$ and $\lim_{n \rightarrow +\infty} l_n = l$. For all $n \in \mathbb{N}$, it holds that

$$\int_0^n |\beta^{l_n}|^2 ds = \lambda_2^{-1} (l_n)_2$$

Since η, θ are $L^2([0, +\infty), \mathbb{C})$ functions, the left-hand side converges to

$$\lim_{n \rightarrow +\infty} \int_0^n |\beta^{l_n}|^2 ds = \int_0^{+\infty} |\beta^l|^2 ds$$

by (2.16) and the right-hand side converges to $\lambda_2^{-1} l_2$.

Next, we turn to (2.19). We already know that

$$\int_0^{+\infty} |\beta^{l_\infty}|^2 ds \leq \lambda_2^{-1} (l_\infty)_2$$

Let l_a denote the center of C_a . Notice that

$$\int_0^a |\beta^l|^2 ds - \lambda_2^{-1} l_2 = 0, \quad \int_0^a |\eta|^2 ds (|l - l_a|^2 - r_a^2) = 0$$

are both cartesian equation of the circle C_a with the same coefficient in front of l_1^2 and l_2^2 . Therefore,

$$\begin{aligned} \int_0^a |\beta^l|^2 ds - \lambda_2^{-1} l_2 &= \int_0^a |\eta|^2 ds (|l - l_a|^2 - r_a^2), \quad \forall l \in \mathbb{C} \quad \forall a > 0 \\ &\geq -r_a^2 \int_0^a |\eta|^2 ds = -\frac{1}{2} r_a \end{aligned} \quad (2.22)$$

Letting $a \rightarrow +\infty$, we get

$$\int_0^{+\infty} |\beta^l|^2 ds - \lambda_2^{-1} l_2 \geq -\frac{1}{2} r_\infty \quad \forall l \in \mathbb{C}$$

which gives the wished inequality for $l = l_\infty$ in the limit point case $r_\infty = 0$. $\square$

Corollary 2.11

If l is the limit point or $l \in C_\infty$, $(a_n)_{n=1} \subset \mathbb{R}_{>0}$ is an increasing and unbounded sequence, and $(l_n)_{n=1}^\infty \subset \mathbb{C}$ is a convergent sequence such that $l_n \in C_{a_n} \quad \forall n \in \mathbb{N}$ and $\lim_{n \rightarrow +\infty} l_n = l$, then

$$\lim_{n \rightarrow +\infty} \int_0^{a_n} |\beta^{l_n}|^2 ds = \int_0^{+\infty} |\beta^l|^2 ds$$

In particular,

$$\lim_{n \rightarrow +\infty} \int_0^{+\infty} |\beta^{l_n} \cdot \chi_{[0, a_n]} - \beta^l|^2 ds = 0$$

Proof. The first assertion follows from equations (2.17), (2.19) and (2.20).

The second assertion is a consequence of the following Lemma from measure theory : $\square$

Lemma 2.12

Let $(f_n)_{n=1}^\infty \subset L^p(\mathbb{R})$, $1 \leq p < +\infty$, and assume there exists a measurable f such that

1. $\lim_{n \rightarrow +\infty} f_n = f$ almost everywhere on $\mathbb{R}$.
2. $\lim_{n \rightarrow +\infty} \int_{\mathbb{R}} |f_n|^p = \int_{\mathbb{R}} |f|^p$.

Then

$$\lim_{n \rightarrow +\infty} \int_{\mathbb{R}} |f_n - f|^p = 0$$

2.2 The Green function for $Lu + iu = 0$ on $[0, +\infty)$

Theorem 2.13

Let $\lambda \in \mathbb{C} \setminus \mathbb{R}$ be fixed. For $a > 0$, $l \in \mathbb{C}$, we consider the unscaled Green function on $[0, a]^2$:

$$G^l(s, t, \lambda) := \begin{cases} \eta(s, \lambda) \beta^l(t, \lambda) & 0 \leq s \leq t \leq a \\ \beta^l(s, \lambda) \eta(t, \lambda) & a \geq s > t \geq 0 \end{cases}$$

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For $g \in C^0([0, a], \mathbb{C})$,

$$u(s) = \int_0^a G^l(s, t, \lambda) g(t) dt$$

is of class $C^2([0, a], \mathbb{C})$ and solves

$$Lu + \lambda u = (\eta\beta^l)_0 g = -g$$

on $[0, a]$ with boundary condition (2.5).

Moreover, if $l \in C_{a,\lambda}$, then $u(s)$ uniquely solves $Lu + \lambda u = -g$ on $[0, a]$ with boundary conditions (2.5) and (2.10) (where $j = j(l) \in [0, \pi)$ is uniquely given by Theorem 2.3).

Proof. Proceed as in the proof of Theorem 1.7. □

Notation

For the rest of this Section 2.2, the parameter $\lambda = i$ is chosen and for simplicity, we will write $G^l(s, t) := G^l(s, t, i)$.

Theorem 2.14

Let $G_1^l(s, t)$ and $G_2^l(s, t)$ denote respectively the real and imaginary part of $G^l(s, t)$.

Let $a > 0$ and $l \in D_a$. For all $v \in C^0([0, a])$ with $\int_0^a v^2 ds \leq 1$, we have

$$-\frac{1}{2} \leq \int_0^a \int_0^a G_1^l(s, t) v(s) v(t) ds dt \leq \frac{1}{2} \quad (2.23)$$

$$0 \leq \int_0^a \int_0^a G_2^l(s, t) v(s) v(t) ds dt \leq 1 \quad (2.24)$$

Remark 2.15

Let $T : V \rightarrow V$ be a linear, compact, self-adjoint operator from an inner product space $(V, \langle \cdot, \cdot \rangle)$ into itself. Then it follows from Hilbert-Schmidt Theorem that

$$\begin{aligned} \|T\| &:= \sup\{\|Tv\| : \|v\| \leq 1\} \\ &= \sup\{|\langle Tv, v \rangle| : \|v\| \leq 1\} \\ &= \sup\{|\lambda| : \lambda \text{ is an eigenvalue of } T\} \end{aligned}$$

and, assuming that $T(V)$ is dense in $(V, \langle \cdot, \cdot \rangle_V)$, it also follows that

$$\langle Tv, v \rangle \geq 0 \quad \forall v \in V \iff \{\text{eigenvalues of } T\} \subset [0, +\infty)$$

Let $l \in C_a$ first. We look at $G_i^l(s, t)$, $i = 1, 2$, as a symmetric kernel on $I = [0, a]$ and define

$$\begin{aligned} T_{a,i} : (C^0([0, a]), \langle \cdot, \cdot \rangle_{L^2}) &\rightarrow (C^0([0, a]), \langle \cdot, \cdot \rangle_{L^2}) \\ f &\mapsto (T_{a,i}f)(s) := \int_0^a G_i^l(s, t) f(t) dt \end{aligned}$$

to be the associated bounded, compact, self-adjoint integral operator. It can be proven, quite similarly to Theorem 3.3, that $T_{a,2}(C^0([0, a]))$ is dense in $(C^0([0, a]), \langle \cdot, \cdot \rangle_{L^2})$. Then for $l \in C_a$, the theorem is equivalent to

$$\begin{aligned} \{\text{eigenvalues of } T_{a,1}\} &\subset [-1/2, 1/2] \\ \{\text{eigenvalues of } T_{a,2}\} &\subset [0, 1] \end{aligned}$$

Proof. We will only prove (2.23). The other one is proved in a similar fashion.

First assume that $l \in C_a$. Suppose for a contradiction that $\nu \in \mathbb{R} \setminus [-\frac{1}{2}, \frac{1}{2}]$ is an eigenvalue of $T_{a,1}$ with eigenfunction ψ . We set $\phi := \nu^{-1} T_{a,2}(\psi)$. Then

$$\psi + i\phi = \nu^{-1} \int_0^a G^l(s, t) \psi(t) dt$$

so we deduce

$$L(\psi + i\phi) + i(\psi + i\phi) = -\nu^{-1}\psi$$

By separating the real and imaginary part, this amounts to

$$\begin{cases} L(\psi) &= -\nu^{-1}\psi + \phi \\ L(\phi) &= -\psi \end{cases}$$

so that

$$LL(\psi) + \nu^{-1}L(\psi) + \psi = 0$$

This can be rewritten as

$$\left(L - \frac{\nu^{-1} + i\sqrt{4 - \nu^{-2}}}{2}I\right) \circ \left(L - \frac{\nu^{-1} - i\sqrt{4 - \nu^{-2}}}{2}I\right)(\psi) = 0$$

Thus, there exists a non identically zero $u \in C^2([0, a], \mathbb{C})$ such that either

$$L(u) = \left(\frac{\nu^{-1} + i\sqrt{4 - \nu^{-2}}}{2}\right)u \quad \text{or} \quad L(u) = \left(\frac{\nu^{-1} - i\sqrt{4 - \nu^{-2}}}{2}\right)u$$

holds. By Green's formula,

$$0 = \int_0^a (uL(\bar{u}) - \bar{u}L(u)) ds = 2\left(\frac{\nu^{-1} \pm i\sqrt{4 - \nu^{-2}}}{2}\right) \int_0^a |u|^2 ds$$

implying $u \equiv 0$, a contradiction.

Now, assume that $l \in D_a$. Then we can find $l_1, l_2 \in C_a$ and $t \in [0, 1]$ such that

$$l = t \cdot l_1 + (1 - t) \cdot l_2$$

which yields

$$\beta^l = t \cdot \beta^{l_1} + (1 - t) \cdot \beta^{l_2}$$

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and

$$G^l(s, t) = t \cdot G^{l_1}(s, t) + (1 - t) \cdot G^{l_2}(s, t)$$

Since both G^{l_1} and G^{l_2} satisfy condition (2.23), it follows that G^l satisfy it as well. $\square$

Corollary 2.16

Let l be the limit point or $l \in D_\infty$. Then for all $0 < a \leq +\infty$, for all $v \in L^2([0, a])$ with $\|v\|_{L^2}^2 \leq 1$, we have

$$-\frac{1}{2} \leq \int_0^a \int_0^a G_1^l(s, t) v(s) v(t) ds dt \leq \frac{1}{2} \quad (2.25)$$

$$0 \leq \int_0^a \int_0^a G_2^l(s, t) v(s) v(t) ds dt \leq 1 \quad (2.26)$$

In particular, for all $0 < a \leq +\infty$, for all $v \in C^0([0, a])$ with $\|v\|^2 = \int_0^a v^2 ds \leq 1$, we have

$$\int_0^a \left(\int_0^a G_1^l(s, t) v(t) dt \right)^2 ds \leq \frac{1}{4} \quad (2.27)$$

$$\int_0^a \left(\int_0^a G_2^l(s, t) v(t) dt \right)^2 ds \leq 1 \quad (2.28)$$

Notation

Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. We write

$$C_{L^2}^k([0, +\infty), \mathbb{F}) = \{f \in C^k([0, +\infty), \mathbb{F}) \mid \int_0^{+\infty} |f(s)|^2 ds < +\infty\}$$

Theorem 2.17 (The Green function for $Lu + iu = 0$ on $[0, +\infty)$)

For the limit-point or points in D_∞ , we consider the unscaled Green function on $[0, +\infty)^2$:

$$G^l(s, t) = \begin{cases} \eta(s, i) \beta^l(t, i) & 0 \leq s \leq t \\ \beta^l(s, i) \eta(t, i) & s > t \geq 0 \end{cases}$$

For $g \in C_{L^2}^0([0, +\infty), \mathbb{C})$,

$$u(s) = \int_0^{+\infty} G^l(s, t) g(t) dt$$

is of class $C_{L^2}^2([0, +\infty), \mathbb{C})$ and solves

$$Lu + iu = (\eta \beta^l)_0 g = -g$$

on $[0, +\infty)$ with boundary condition (2.5).

Proof. The solution u is in $L^2([0, +\infty), \mathbb{C})$ because of (2.27) and (2.28). Moreover, observe that

$$u(s) = \beta^l(s) \int_0^s \eta(t) g(t) dt + \eta(s) \int_s^{+\infty} \beta^l(t) g(t) dt$$

where the rightmost integral is well-defined since $\beta^l, g \in L^2([0, +\infty), \mathbb{C})$ and is continuously differentiable. Indeed, for any $s_0 \in \mathbb{R}$:

$$\begin{aligned} \lim_{s \rightarrow s_0} (s - s_0)^{-1} \left(\int_s^{+\infty} \beta^l(t) g(t) dt - \int_{s_0}^{+\infty} \beta^l(t) g(t) dt \right) &= \lim_{s \rightarrow s_0} (s - s_0)^{-1} \left(\int_s^{s_0} \beta^l(t) g(t) dt \right) \\ &= -\beta^l(s_0) g(s_0) \end{aligned}$$

Then, we proceed as in the proof of Theorem 1.7 to show that $u \in C^2([0, +\infty), \mathbb{C})$, $Lu + iu = -g$ and $R_1 u = 0$. $\square$

Corollary 2.18 (Weyl alternative for the inhomogeneous equation $Lu + iu = -g$)

Let $g \in C_{L^2}^0([0, +\infty), \mathbb{C})$ be fixed.

In the limit-circle case, any solution to $Lu + iu = -g$ on $[0, +\infty)$ is square-integrable.

In the limit-point case, there is exactly one square-integrable solution (up to the addition of $c \cdot \beta^{l_\infty}$ where l_∞ is the limit-point and $c \in \mathbb{C}$ is a constant).

Proof. Let v be any solution to $Lu + iu = -g$ on $[0, +\infty)$ and let u be the square-integrable solution obtained by means of the Green function. Then $w := u - v$ solves $Lu + iu = 0$ on $[0, +\infty)$. In the limit-circle case, w is square-integrable and so is v .

In the limit-point case, w is square-integrable if and only if $w = c \cdot \beta^{l_\infty}$ for some $c \in \mathbb{C}$. This implies that v is square-integrable if and only if $v = u + c \cdot \beta^{l_\infty}$ for some $c \in \mathbb{C}$. $\square$

Proposition 2.19 (Smoothness of $\|G_i^l(s, \cdot)\|_{L^2}$)

Let $l \in \mathbb{C}$ be such that $\beta^l \in L^2([0, +\infty), \mathbb{C})$. Let $G_1^l(s, t)$ and $G_2^l(s, t)$ denote respectively the real and imaginary part of $G^l(s, t)$. Then

$$s \mapsto \int_0^{+\infty} G_1^l(s, t)^2 dt, \quad s \mapsto \int_0^{+\infty} G_2^l(s, t)^2 dt$$

exist and are twice continuously differentiable on $[0, +\infty)$.

Proof. It suffices to prove that

$$\int_0^{+\infty} |G^l(s, t)|^2 dt, \quad \int_0^{+\infty} G^l(s, t)^2 dt, \quad \int_0^{+\infty} \overline{G^l(s, t)}^2 dt$$

exist and are twice continuously differentiable on $[0, +\infty)$. Observe that

$$\int_0^{+\infty} G^l(s, t)^2 dt = \beta^l(s)^2 \int_0^s \eta(t)^2 dt + \eta(s)^2 \int_s^{+\infty} \beta^l(t)^2 dt$$

Except for the rightmost integral, every term is well-defined and of class $C^2([0, +\infty), \mathbb{C})$. In fact, this is also the case for the rightmost integral : it is well-defined since $\beta^l \in L^2([0, +\infty), \mathbb{C})$

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and is twice continuously differentiable because for any $s_0 \in [0, +\infty)$:

$$\lim_{s \rightarrow s_0} (s - s_0)^{-1} \left(\int_s^{+\infty} \beta^l(t)^2 dt - \int_{s_0}^{+\infty} \beta^l(t)^2 dt \right) = (s - s_0)^{-1} \lim_{s \rightarrow s_0} \left(\int_s^{s_0} \beta^l(t)^2 dt \right) = -\beta^l(s_0)^2$$

We proceed similarly for

$$\begin{aligned} \int_0^{+\infty} \overline{G^l(s, t)}^2 dt &= \overline{\beta^l(s)}^2 \int_0^s \overline{\eta(t)}^2 dt + \overline{\eta(s)}^2 \int_s^{+\infty} \overline{\beta^l(t)}^2 dt \\ \int_0^{+\infty} |G^l(s, t)|^2 dt &= |\beta^l(s)|^2 \int_0^s |\eta(t)|^2 dt + |\eta(s)|^2 \int_s^{+\infty} |\beta^l(t)|^2 dt \end{aligned}$$

□

Notation

For the rest of this section, we choose and fix $l = l_\infty$ or $l \in D_\infty$ depending on whether L is in the limit-point or in the limit-circle case for the parameter $\lambda = i$.

Proposition 2.20 (Basic properties of T_i)

For $i = 1, 2$, we define the following linear operator :

$$\begin{aligned} T_i : (C_{L^2}^0([0, +\infty)), \langle \cdot, \cdot \rangle_{L^2}) &\rightarrow (C_{L^2}^0([0, +\infty)), \langle \cdot, \cdot \rangle_{L^2}) \\ f(t) &\mapsto (T_i f)(s) := \int_0^{+\infty} G_i^l(s, t) f(t) dt \end{aligned}$$

Then T_i is bounded, self-adjoint and it maps $C_{L^2}^0([0, +\infty))$ to a subspace of $C_{L^2}^2([0, +\infty))$.

Proof. The boundedness follows from (2.27) and (2.28), the self-adjointness follows from the symmetry $G_i^l(s, t) = G_i^l(t, s)$ of the kernel and the assertion about the image of $C_{L^2}^0([0, +\infty))$ under T_i follows from Theorem 2.17. □

Definition 2.1 (Hilbert-Schmidt integral operator)

Let $I \subset \mathbb{R}$ be a closed interval and let $k(s, t) \in L^2(I \times I, \mathbb{C})$ be a square-integrable kernel. The integral operator

$$\begin{aligned} K : (L^2(I, \mathbb{C}), \langle \cdot, \cdot \rangle_L^2) &\rightarrow (L^2(I, \mathbb{C}), \langle \cdot, \cdot \rangle_L^2) \\ f(t) &\mapsto (Kf)(s) := \int_I K(s, t) f(t) dt \end{aligned}$$

is called a *Hilbert-Schmidt integral operator*.

By Cauchy-Schwarz inequality, such an operator is bounded. Moreover, if $k(s, t) = \overline{k(t, s)}$, then it is also self-adjoint.

Proposition 2.21 (Compactness of Hilbert-Schmidt integral operators)

A Hilbert-Schmidt integral operator K is compact.

Proof. Let $\{\phi_n(s)\}_{n=1}^\infty$ be an orthonormal basis for $(L^2(I, \mathbb{C}), \langle \cdot, \cdot \rangle_{L^2})$. Then $\{\phi_n(s)\phi_m(t)\}_{n,m=1}^\infty$ is an orthonormal basis for $(L^2(I \times I, \mathbb{C}), \langle \cdot, \cdot \rangle_{L^2})$ and we can write

$$k(s, t) = \sum_{n,m=1}^\infty a_{n,m} \phi_n(s) \phi_m(t)$$

for some $a_{n,m} \in \mathbb{C}$, where the series converges in $(L^2(I \times I, \mathbb{C}), \|\cdot\|_{L^2})$. For $n \in \mathbb{N}$, we define

$$k_n(s, t) = \sum_{i=1}^n \sum_{j=1}^\infty a_{i,j} \phi_i(s) \phi_j(t)$$

$$A_n : (L^2(I, \mathbb{C}), \langle \cdot, \cdot \rangle_{L^2}) \rightarrow (L^2(I, \mathbb{C}), \langle \cdot, \cdot \rangle_{L^2})$$

$$f(t) \mapsto (A_n f)(s) := \int_0^{+\infty} k_n(s, t) f(t) dt$$

The image of A_n is finite-dimensional (it is a subspace of $\text{span}(\phi_1, \dots, \phi_n)$), so A_n is compact. Furthermore, $A_n \rightarrow T$ in operator norm as $n \rightarrow +\infty$. Indeed, for $f \in L^2(I, \mathbb{C})$, $\|f\|_{L^2} \leq 1$, we have

$$\begin{aligned} \|(A_n - T)f\|_{L^2}^2 &= \int_0^{+\infty} \left(\int_0^{+\infty} (k_n(s, t) - k(s, t)) f(t) dt \right)^2 ds \\ &\leq \int_0^{+\infty} \left(\int_0^{+\infty} (k_n(s, t) - k(s, t))^2 dt \right) \left(\int_0^{+\infty} f(t)^2 dt \right) ds \quad (\text{Cauchy-Schwarz}) \\ &\leq \int_0^{+\infty} \int_0^{+\infty} (k_n(s, t) - k(s, t))^2 dt ds \\ &= \int_0^{+\infty} \int_0^{+\infty} \left(\sum_{i=n+1}^\infty \sum_{j=1}^\infty a_{i,j} \phi_i(s) \phi_j(t) \right)^2 dt ds \\ &= \sum_{i=n+1}^\infty \sum_{j=1}^\infty a_{i,j}^2 \quad (\text{Orthonormality}) \end{aligned}$$

Since $\|k(s, t)\|_{L^2}^2 = \sum_{n,m=1}^\infty a_{n,m}^2 < +\infty$, we deduce that $\|(A_n - T)f\|_{L^2}^2 \rightarrow 0$ as $n \rightarrow +\infty$ independently of $f \in L^2(I, \mathbb{C})$, $\|f\|_{L^2} \leq 1$.

Thus, K is compact as $(L^2(I, \mathbb{C}), \langle \cdot, \cdot \rangle_{L^2})$ is a Hilbert space and K is the limit of a sequence of compact operators from $L^2(I, \mathbb{C})$ into itself. $\square$

Corollary 2.22 (L^2 -compactness of T_i in limit-circle case)

Let L be in the limit-circle case. If we look at T_i as an operator from $(L^2([0, +\infty)), \langle \cdot, \cdot \rangle_{L^2})$ to $(L^2([0, +\infty)), \langle \cdot, \cdot \rangle_{L^2})$, then T_i is a self-adjoint Hilbert-Schmidt integral operator.

Proof. We fix $i \in \{1, 2\}$. In the limit-circle case, $G_i^l(s, t) \in L^2([0, +\infty) \times [0, +\infty))$, so T_i is a Hilbert-Schmidt integral operator, hence bounded and compact (see Definition 2.1 and Proposition

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2.21). It is self-adjoint because of the symmetry $G_i^l(s, t) = G_i^l(t, s)$ of the kernel. $\square$

Proposition 2.23 (Compactness of T_i in limit-circle case)

In the limit-circle case, $T_i : (C_{L^2}^0([0, +\infty)), \langle \cdot, \cdot \rangle_{L^2}) \rightarrow (C_{L^2}^0([0, +\infty)), \langle \cdot, \cdot \rangle_{L^2})$ is compact.

Remark 2.24

In that sense, the limit-circle case is the regular case as Hilbert-Schmidt theorem will apply.

Proof. We fix $i \in \{1, 2\}$. Let $(f_n)_{n=1}^\infty \subset (C_{L^2}^0([0, +\infty)), \langle \cdot, \cdot \rangle_{L^2})$ be uniformly bounded with respect to the L^2 -norm. By passing to a subsequence, we can assume without loss of generality that $(T_i f_n)_{n=1}^\infty$ converges in $(L^2([0, +\infty)), \langle \cdot, \cdot \rangle_{L^2})$ (Proposition 2.22). It suffices to show now that there exists a subsequence of $(T_i f_n)_{n=1}^\infty$ which converges uniformly on every compact subsets of $[0, +\infty)$.

First, we show that for any fixed $M > 0$, $(T_i f_n)_{n=1}^\infty$ is uniformly bounded and uniformly equicontinuous on $[0, M]$. Indeed, it is uniformly bounded since T_i is a bounded operator:

$$|(T_i f_n)(s)| \leq \|T_i\| \cdot \|f_n\|_{L^2} \quad \forall s \geq 0$$

Next, we prove uniform equicontinuity. Let $\varepsilon > 0$ be fixed. Since $G_i^l(s, t)$ is uniformly continuous on $[0, M] \times [0, M]$ and η is uniformly continuous on $[0, M]$, there exists $\delta > 0$ such that for all $s, s', t, t' \in [0, M]$:

$$|s - s'| + |t - t'| \leq \delta \Rightarrow |G_i^l(s, t) - G_i^l(s', t')| \leq \frac{\varepsilon}{2M^{1/2} \cdot (\sup_n \|f_n\|_{L^2} + 1)} \quad (\star)$$

and

$$|s - s'| \leq \delta \Rightarrow |\eta(s) - \eta(s')| \leq \frac{\varepsilon}{2\|\beta^l\|_{L^2} \cdot (\sup_n \|f_n\|_{L^2} + 1)} \quad (\star\star)$$

Thus, for all $s, s' \in [0, M]$ such that $|s - s'| \leq \delta$, we have

$$\begin{aligned} |T_i f_n(s) - T_i f_n(s')| &= \left| \int_0^{+\infty} (G_i^l(s, t) - G_i^l(s', t)) f(t) dt \right| \\ &\leq \left| \int_0^M \dots dt \right| + \left| \int_M^{+\infty} \dots dt \right| \end{aligned}$$

The first term is bounded using Cauchy-Schwarz inequality and then our first bound $(\star)$

$$\begin{aligned} \left| \int_0^M \dots dt \right| &\leq \left(\int_0^M (G_i^l(s, t) - G_i^l(s', t))^2 dt \right)^{1/2} \cdot \left(\int_0^M f(t)^2 dt \right)^{1/2} \\ &\leq \varepsilon/2 \end{aligned}$$

For the second term, observe that

$$|G_i^l(s, t) - G_i^l(s', t)| \leq |G^l(s, t) - G^l(s', t)|$$

and

$$\int_M^{+\infty} (G^l(s, t) - G^l(s', t)) f(t) dt = (\eta(s) - \eta(s')) \int_M^{+\infty} \beta^l(t) f(t) dt$$

Thus, by Cauchy-Schwarz inequality and our second bound ($\star\star$)

$$\begin{aligned} \left| \int_M^{+\infty} \dots dt \right| &\leq |\eta(s) - \eta(s')| \cdot \int_M^{+\infty} |\beta^l(t) f(t)| dt \\ &\leq |\eta(s) - \eta(s')| \cdot \left(\int_M^{+\infty} |\beta^l(t)|^2 dt \right)^{1/2} \cdot \left(\int_M^{+\infty} f(t)^2 dt \right)^{1/2} \\ &\leq \varepsilon/2 \end{aligned}$$

This proves uniform equicontinuity.

By Arzela-Ascoli theorem, we can find a sequence of subsequences $(f_{n,m})_{n=1}^\infty$, $m = 0, 1, 2, 3, \dots$, with the following properties :

1. $(f_{n,0})_{n=1}^\infty = (f_n)_{n=1}^\infty$.
2. For all $m \geq 1$, $(f_{n,m})_{n=1}^\infty$ is a subsequence of $(f_{n,m-1})_{n=1}^\infty$. In particular, it is a subsequence of $(f_{n,k})_{n=1}^\infty$ for all $0 \leq k < m$.
3. For all $m \geq 1$, $(f_{n,m})_{n=1}^\infty$ converges uniformly on $[0, m]$.

We conclude by passing to the diagonal subsequence $(f_{n,n})_{n=1}^\infty$ of $(f_n)_{n=1}^\infty$. □

2.3 Weyl alternative for arbitrary λ

To show that a SL operator L is in the limit-point case or in the limit-circle case, it suffices to prove Weyl alternative for an arbitrary parameter $\lambda \in \mathbb{C}$. Indeed, we have the following theorem :

Theorem 2.25 (Weyl alternative for arbitrary λ)

Let L be a SL operator on $[0, +\infty)$. For any $\lambda \in \mathbb{C}$ and any bounded $r(s) \in C^0([0, +\infty))$, exactly one of the following situations happens :

1. *Limit-circle case* : Every solution to $L(u) + (\lambda - r)u = 0$ on $[0, +\infty)$ is square-integrable.
2. *Limit-point case* : There is at most one (up to a scalar factor) square-integrable solution to $L(u) + (\lambda - r)u = 0$ on $[0, +\infty)$. Furthermore, if $\lambda \in \mathbb{C} \setminus \mathbb{R}$, then there is exactly one (up to a scalar factor) square-integrable solution.

Moreover, if 1. (resp. 2.) holds for one particular pair of $\lambda^* \in \mathbb{C}$ and bounded $r^* \in C^0([0, +\infty))$, then it holds for any other pair of $\lambda \in \mathbb{C}$ and bounded $r \in C^0([0, +\infty))$.

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Proposition 2.26

Let $\lambda \in \mathbb{C}$ and let $\{u_\lambda(s), v_\lambda(s)\}$ be a fundamental system of solutions for $Lu + \lambda u = 0$ on $[0, +\infty)$ satisfying $(u_\lambda v_\lambda) \equiv 1$.

Let $r(x) \in C^0([0, +\infty))$ be any continuous function and let $\varphi \in C^2([0, +\infty), \mathbb{C})$ be a solution to $Lu + (\lambda - r)u = 0$ on $[0, +\infty)$. Then

$$\psi(s) := \varphi(s) - \int_0^s (u_\lambda(s)v_\lambda(t) - v_\lambda(s)u_\lambda(t)) r(t)\varphi(t) dt$$

is of class $C^2([0, +\infty), \mathbb{C})$ and solves $Lu + \lambda u = 0$ on $[0, +\infty)$.

In particular, the subspace

$$\{\varphi \in C^2([0, +\infty), \mathbb{C}) \mid \varphi(s) = c \cdot u_\lambda(s) + d \cdot v_\lambda(s) + \int_0^s (u_\lambda(s)v_\lambda(t) - v_\lambda(s)u_\lambda(t)) r(t)\varphi(t) dt, c, d \in \mathbb{C}\}$$

forms the set of solutions to $Lu + (\lambda - r)u = 0$ on $[0, +\infty)$.

Proof. We have

$$\begin{aligned} \psi(s) &= \varphi(s) - \int_0^s (u_\lambda(s)v_\lambda(t) - v_\lambda(s)u_\lambda(t)) r(t)\varphi(t) dt \in C^1([0, +\infty), \mathbb{C}) \\ \psi'(s) &= \varphi'(s) - \int_0^s (u'_\lambda(s)v_\lambda(t) - v'_\lambda(s)u_\lambda(t)) r(t)\varphi(t) dt \in C^1([0, +\infty), \mathbb{C}) \\ (p(s)\psi'(s))' &= (p(s)\varphi'(s))' - \int_0^s [(p(s)u'_\lambda(s))'v_\lambda(t) - (p(s)v'_\lambda(s))'u_\lambda(t)] r(t)\varphi(t) dt \\ &\quad - p(s) \cdot (u'_\lambda(s)v_\lambda(s) - v'_\lambda(s)u_\lambda(s)) r(s)\varphi(s) \in C^0([0, +\infty), \mathbb{C}) \end{aligned}$$

so that

$$\begin{aligned} L(\psi)(s) &= L(\varphi)(s) - \int_0^s (L(u_\lambda)(s)v_\lambda(t) - L(v_\lambda)(s)u_\lambda(t)) r(t)\varphi(t) dt - r(s)\varphi(s) \\ &= -\lambda\varphi(s) + \lambda \int_0^s (u_\lambda(s)v_\lambda(t) - v_\lambda(s)u_\lambda(t)) r(t)\varphi(t) dt \\ &= -\lambda\psi(s) \end{aligned}$$

□

Lemma 2.27 (Grönwall's inequality)

Let $\alpha, \beta, \phi \in C^0([0, +\infty))$ with $\beta \geq 0$. Assume that

$$\phi(r) \leq \alpha(r) + \int_0^r \beta(s)\phi(s) ds \quad \forall r \in [0, +\infty)$$

Then

$$\phi(r) \leq \alpha(r) + \int_0^r \alpha(s)\beta(s) \exp\left(\int_s^r \beta(t) dt\right) ds \quad \forall r \in [0, +\infty)$$

Furthermore, if α is non-decreasing, then

$$\phi(r) \leq \alpha(r) \exp \left(\int_0^r \beta(t) dt \right) \forall r \in [0, +\infty)$$

Theorem 2.28

Assume that $\lambda \in \mathbb{C}$ is such that every solution to $Lu + \lambda u = 0$ on $[0, +\infty)$ is square-integrable. Then for every bounded $r(s) \in C^0([0, +\infty))$, every solution to $Lu + (\lambda - r)u = 0$ on $[0, +\infty)$ is square-integrable.

Proof. Let $\varphi \in C^2([0, +\infty), \mathbb{C})$ be a solution to $Lu + (\lambda - r)u = 0$ on $[0, +\infty)$. We write

$$\varphi(s) = c \cdot u_\lambda(s) + d \cdot v_\lambda(s) + \int_0^s (u_\lambda(s)v_\lambda(t) - v_\lambda(s)u_\lambda(t)) r(t) \varphi(t) dt$$

for some $c, d \in \mathbb{C}$ as in Proposition 2.26. By applying the triangle inequality, we get

$$|\varphi(s)| \leq \left[|c| \cdot |u_\lambda(s)| + |d| \cdot |v_\lambda(s)| + M_0 \int_0^s (|u_\lambda(s)| \cdot |v_\lambda(t)| + |v_\lambda(s)| \cdot |u_\lambda(t)|) |\varphi(t)| dt \right]$$

where $M_0 = \|r\|_\infty$.

Since for $a, b \in \mathbb{R}$: $(a + b)^2 \leq 2(a^2 + b^2)$, we have

$$\begin{aligned} |\varphi(s)|^2 &\leq 4 \left[|c|^2 \cdot |u_\lambda(s)|^2 + |d|^2 \cdot |v_\lambda(s)|^2 \right. \\ &\quad \left. + M_0^2 \left(\int_0^s (|u_\lambda(s)| \cdot |v_\lambda(t)| + |v_\lambda(s)| \cdot |u_\lambda(t)|) |\varphi(t)|^2 dt \right)^2 \right] \end{aligned}$$

Applying Cauchy-Schwarz inequality and then $(a + b)^2 \leq 2(a^2 + b^2)$ again, we find

$$\begin{aligned} |\varphi(s)|^2 &\leq 8 \left[|c|^2 \cdot |u_\lambda(s)|^2 + |d|^2 \cdot |v_\lambda(s)|^2 \right. \\ &\quad \left. + M_0^2 \int_0^s (|u_\lambda(s)|^2 \cdot |v_\lambda(t)|^2 + |v_\lambda(s)|^2 \cdot |u_\lambda(t)|^2) dt \cdot \int_0^s |\varphi(t)|^2 dt \right] \end{aligned}$$

Let $M_1 = 8 \cdot \max\{|c|^2, |d|^2, M_0^2 \cdot \|u_\lambda\|_{L_2}^2, M_0^2 \cdot \|v_\lambda\|_{L_2}^2\}$. Then

$$|\varphi(s)|^2 \leq M_1 \left[|u_\lambda(s)|^2 + |v_\lambda(s)|^2 + (|u_\lambda(s)|^2 + |v_\lambda(s)|^2) \cdot \int_0^s |\varphi(t)|^2 dt \right]$$

Integrating both sides with respect to s , we then get

$$\begin{aligned} \int_0^r |\varphi(s)|^2 ds &\leq M_1 \left[\int_0^r (|u_\lambda(s)|^2 + |v_\lambda(s)|^2) ds + \int_0^r (|u_\lambda(s)|^2 + |v_\lambda(s)|^2) \left(\int_0^s |\varphi(t)|^2 dt \right) ds \right] \\ &\leq M_2 + M_1 \cdot \int_0^r (|u_\lambda(s)|^2 + |v_\lambda(s)|^2) \left(\int_0^s |\varphi(t)|^2 dt \right) ds \\ &= M_2 + M_1 \cdot \int_0^r (|u_\lambda(t)|^2 + |v_\lambda(t)|^2) \left(\int_0^t |\varphi(s)|^2 ds \right) dt \end{aligned}$$

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where $M_2 = M_1 \cdot (||u_\lambda||_{L_2}^2 + ||v_\lambda||_{L_2}^2)$. We finally apply Grönwall's inequality to

$$\alpha(r) \equiv M_2, \beta(r) = M_1 \cdot (|u_\lambda(r)|^2 + |v_\lambda(r)|^2), \phi(r) = \int_0^r |\varphi(s)|^2 ds$$

and deduce that

$$\int_0^r |\varphi(s)|^2 ds \leq M_2 \exp \left(M_1 \int_0^r |u_\lambda(t)|^2 + |v_\lambda(t)|^2 dt \right), \forall r \in [0, +\infty)$$

□

Proof of Theorem 2.25.

Let $L, \lambda \in \mathbb{C}, r \in C^0([0, +\infty))$ be as in the theorem. We consider the auxiliary real SL operator L^* defined with $p^*(s) = p(s)$ and $q^*(s) = q(s) + r(s)$ and apply Corollary 2.8 to get that the alternative holds when λ is non-real.

The fact that it holds for real λ and the fact that the alternative is independent from the pair (λ, r) follow from Theorem 2.28. □

Next, we show with a simple example that we cannot always expect $Lu + \lambda u = 0$ to have non-trivial square-integrable solutions when $\lambda \in \mathbb{R}$.

Proposition 2.29

For any $\lambda \geq 0$, the differential equation

$$\frac{d^2 u}{ds^2}(s) + \lambda u(s) = 0, s \geq 0$$

has no non-trivial square-integrable solution.

In particular, the SL operator $L = d^2/ds^2$ is in the limit-point case.

Proof. If $\lambda = 0$, any such solution is of the form

$$u(s) = a + b \cdot s$$

for some $a, b \in \mathbb{C}$.

If $\lambda > 0$, any such solution is of the form

$$u(s) = a \cdot \cos(\sqrt{\lambda}s) + b \cdot \sin(\sqrt{\lambda}s)$$

for some $a, b \in \mathbb{C}$.

Any such solution is not square-integrable on $[0, +\infty)$ unless $a = b = 0$. □

Finally, we conclude with some simple sufficient criterions for L to be in the limit-point case.

Proposition 2.30

Let L be a SL operator on $[0, +\infty)$ such that $q(s) \in C^0([0, +\infty))$ is bounded. Then L is in the limit-point case.

Proof. Observe first that the differential equation

$$\frac{d}{ds} \left(p(s) \frac{du}{ds}(s) \right) = 0, \quad s \geq 0$$

has set of solutions

$$\{u(s) \in C^2([0, +\infty)) \mid u(s) = a + b \cdot \int_0^s \frac{1}{p(t)} dt, \quad a, b \in \mathbb{C}\}$$

Since $p(s) > 0$ on $[0, +\infty)$, any such solution is not square-integrable unless $a = b = 0$.

If L was in the limit-circle case, then for $\lambda = 0$ and $r(s) = q(s)$, every solution to

$$\frac{d}{ds} \left(p(s) \frac{du}{ds}(s) \right) = Lu + (\lambda + r(s))u = 0$$

would be square-integrable by Theorem 2.25, which is a contradiction. $\square$

Theorem 2.31

Let L be a SL operator on $[0, +\infty)$ and assume there exists a positive function $M \in C^1([0, +\infty))$ and two constants $k_1, k_2 > 0$ such that for all large s :

$$\begin{aligned} q(s) &\geq -k_1 M(s), \quad \int_1^{+\infty} (p(t) M(t))^{-1/2} dt = +\infty \\ |p(s)^{1/2} M'(s) M(s)^{-3/2}| &\leq k_2 \end{aligned}$$

Then L is in the limit-point case.

Proof. We show that $Lu = 0$ cannot have two linearly independent solutions $\varphi, \psi \in L^2([0, +\infty))$.

First, assume that $\varphi \in L^2([0, +\infty))$ solves $Lu = 0$ on $[0, +\infty)$. Then

$$\int_1^x \frac{(p(t)\varphi'(t))'\varphi(t)}{M(t)} dt = \int_1^x \frac{q(t)}{M(t)} \varphi(t)^2 dt \geq -k_1 \int_1^x \varphi(t)^2 dt, \quad x \geq 1$$

using $L\varphi = 0$ and

$$\int_0^x \frac{(p(t)\varphi'(t))'\varphi(t)}{M(t)} = \left[\frac{p(t)\varphi'(t)\varphi(t)}{M(t)} \right]_1^{t=x} - \int_1^x p(t) \frac{(\varphi'(t))^2}{M(t)} dt + \int_1^x p(t) \frac{\varphi'(t)\varphi(t)M'(t)}{M(t)^2} dt, \quad x \geq 1$$

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using integration by parts. Thus, for $c_1 := k_1 \|\varphi\|_{L^2}^2 - p(1)\varphi'(1)\varphi(1)M(1)^{-1}$, we have

$$-p(x)\frac{\varphi'(x)\varphi(x)}{M(x)} + \int_1^x p(t)\frac{(\varphi'(t))^2}{M(t)}dt - \int_1^x p(t)\frac{\varphi'(t)\varphi(t)M'(t)}{M(t)^2}dt \leq c_1, \quad x \geq 1$$

We define

$$H(x) = \int_1^x p(t)\frac{(\varphi'(t))^2}{M(t)}dt, \quad x \geq 1$$

By Cauchy-Schwarz inequality, we have

$$\left| \int_1^x p(t)\frac{\varphi'(t)\varphi(t)M'(t)}{M(t)^2}dt \right| \leq \int_1^x k_2 p(t)^{1/2} \frac{|\varphi'(t)|}{M(t)^{1/2}} \cdot |\varphi(t)| dt \leq k_2 H(x)^{1/2} \left(\int_1^x \varphi(t)^2 dt \right)^{1/2}, \quad x \geq 1$$

so that for $c_2 := k_2 \|\varphi\|_{L^2}$, $H(x)$ satisfies

$$-p(x)\frac{\varphi'(x)\varphi(x)}{M(x)} + H(x) - c_2 H(x)^{1/2} \leq c_1, \quad x \geq 1$$

If $H(x) \rightarrow +\infty$ as $x \rightarrow +\infty$, then

$$p(x)\frac{\varphi'(x)\varphi(x)}{M(x)} \geq \frac{1}{2}H(x)$$

for all x large enough. In particular, $\varphi(x)$ and $\varphi'(x)$ have the same sign for all x large enough (say positive sign without loss of generality). Thus, for x large enough, $\varphi(x)$ is a positive and increasing function. In particular, it is bounded below by a positive constant $c > 0$, which contradicts its square-integrability.

Therefore,

$$H(x) = \int_1^x p(t)\frac{(\varphi'(t))^2}{M(t)}dt, \quad x \geq 1$$

is bounded. Finally, if $\varphi, \psi \in L^2([0, +\infty))$ are two linearly independent solutions to $Lu = 0$ such that $(\varphi\psi) \equiv 1$, then

$$p(x)^{1/2} \cdot \left(\varphi(x)\frac{\psi'(x)}{M(x)^{1/2}} - \psi(x)\frac{\varphi'(x)}{M(x)^{1/2}} \right) = \frac{1}{(p(x)M(x))^{1/2}}, \quad x > 0$$

The left-hand side is square-integrable on $[1, +\infty)$ by what we have just proved but the right-hand side is not, which is a contradiction. $\square$

Corollary 2.32

Let L be a SL operator on $[0, +\infty)$ such that

$$q(s) \geq -k \quad \forall s \geq 0, \quad \int_1^{+\infty} p(t)^{-1/2} dt = +\infty$$

for some $k > 0$. Then L is in the limit-point case.

Proof. This follows from Theorem 2.31 with $M(s) \equiv 1$. □

Corollary 2.33

Let L be a SL operator on $[0, +\infty)$ such that $p(s)$ is bounded and for all large s :

$$q(s) \geq -ks^2$$

for some $k > 0$. Then L is in the limit-point case.

Proof. This follows from Theorem 2.31 with $M(s) = s^2$. □

3 Eigenfunctions and the expansion theorem in the limit-circle case

For the remaining of this Chapter 3, we assume that L is a SL operator with boundary condition (2.5) in the limit-circle case. Furthermore, we choose $\lambda = i$ and fix some $l \in C_\infty$, where C_∞ denotes the limit-circle for the parameter $\lambda = i$.

For the non-degenerate SL problem on $[a, b]$, the eigenfunctions of the SL operator L are exactly the eigenvectors of a bounded, compact and self-adjoint integral operator. This correspondance is made possible because the solution to the problem $Lu = g$, $R_1 u = R_2 u = 0$, is unique and obtained using the Green function as a symmetric kernel. In the limit-circle case, the kernel G_2^l will allow to obtain such a transition.

First, we have to strengthen our boundary condition at $s = +\infty$. Indeed, our former boundary condition at $s = +\infty$, namely $u \in L^2([0, +\infty), \mathbb{C})$, is useless as any solution to $Lu + iu = -g$ will be square-integrable.

Furthermore, we also need to determine the image of

$$T_2 : (C_{L^2}^0([0, +\infty)), \langle \cdot, \cdot \rangle_{L^2}) \rightarrow (C_{L^2}^0([0, +\infty)), \langle \cdot, \cdot \rangle_{L^2})$$

$$f(t) \mapsto (T_2 f)(s) := \int_0^{+\infty} G_2^l(s, t) f(t) dt$$

if we want to use Hilbert-Schmidt theorem.

3.1 The boundary condition at infinity

Theorem 3.1

Let $g(s) \in C_{L^2}^0([0, +\infty), \mathbb{C})$.

We consider the following Sturm-Liouville problem on the real half-line :

$$L(u) + iu = \frac{d}{ds} \left(p(s) \frac{du}{ds}(s) \right) - q(s)u(s) + iu(s) = -g(s), \quad s \geq 0$$

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with boundary condition (2.5) at $s = 0$ and the following boundary conditions at $s = +\infty$

$$\lim_{s \rightarrow +\infty} (\beta_1^l u)_s = \lim_{s \rightarrow +\infty} (\beta_2^l u)_s = 0 \quad (3.1)$$

Then this problem has a unique solution given by

$$u(s) = \int_0^{+\infty} G^l(s, t) g(t) dt$$

Proof. We have to verify that $u(s)$ satisfies (3.1) and that the solution is unique.

Since

$$\begin{aligned} u(s) &= \int_0^\infty G^l(s, t) g(t) dt = \int_0^s G^l(s, t) g(t) dt + \int_s^\infty G^l(s, t) g(t) dt \\ &= \beta^l(s) \int_0^s \eta(t) g(t) dt + \eta(s) \int_s^\infty \beta^l(t) g(t) dt \\ \frac{du}{ds}(s) &= \frac{d\beta^l}{ds}(s) \int_0^s \eta(t) g(t) dt + \frac{d\eta}{ds}(s) \int_s^\infty \beta^l(t) g(t) dt \end{aligned}$$

we have

$$(\beta^l u)_s = \underbrace{(\beta^l \beta^l)_s \int_0^s \eta(t) g(t) dt}_{(A)} + \underbrace{(\beta^l \eta)_s \int_s^\infty \beta^l(t) g(t) dt}_{(B)}$$

The (A) term is zero for all $s \geq 0$ because $(\beta^l \beta^l) \equiv 0$ and the (B) term goes to zero as $s \rightarrow +\infty$ because $(\beta^l \eta) \equiv 1$ and $\beta^l, g \in L^2([0, +\infty), \mathbb{C})$.

Similarly, we have

$$(\overline{\beta^l} u)_s = \underbrace{(\overline{\beta^l} \beta^l)_s \int_0^s \eta(t) g(t) dt}_{(A)} + \underbrace{(\overline{\beta^l} \eta)_s \int_s^\infty \beta^l(t) g(t) dt}_{(B)}$$

The (A) term goes to zero as $s \rightarrow +\infty$. Indeed, since $l \in C_\infty$, we have

$$\int_0^{+\infty} |\beta^l|^2 ds = l_2 = -(\beta_2^l \beta_1^l)_0$$

by (2.14), (2.20) and

$$(\overline{\beta^l} \beta^l)_s = 2i(\beta_2 \beta_1)_s = 2i \left[\int_0^s |\beta^l|^2 ds + (\beta_2^l \beta_1^l)_0 \right] \rightarrow 0 \text{ as } s \rightarrow +\infty$$

The (B) term goes to zero as $s \rightarrow +\infty$ because

$$(\overline{\beta^l} \eta)_s = \int_0^s \left(\overline{\beta^l} L(\eta) - \eta L(\overline{\beta^l}) \right) dt + (\overline{\beta^l} \eta)_0 = 2i \int_0^s \overline{\beta^l} \eta dt + (\overline{\beta^l} \eta)_0$$

is bounded on $[0, +\infty)$ and $\beta^l, g \in L^2([0, +\infty))$.

Thus,

$$\lim_{s \rightarrow +\infty} (\beta_1^l u)_s = \lim_{s \rightarrow +\infty} (\beta_2^l u)_s = 0$$

Finally, assume that u, v are two solutions to the above problem. Then $w := u - v$ satisfies $Lw + iw = 0$ with boundary conditions (2.5) and (3.1). We write

$$w = c \cdot \eta + d \cdot \beta^l, \quad c, d \in \mathbb{C}$$

Observe that

$$(\beta^l w) = c(\beta^l \eta) \equiv c$$

so c must be zero by condition (3.1).

Furthermore, $d \cdot \beta^l$ satisfies condition (2.5) if and only if $d = 0$. □

Remark 3.2

For $u, v \in C^2([0, +\infty), \mathbb{C})$, we write

$$(uv)_\infty := \lim_{s \rightarrow +\infty} (uv)_s$$

if the limit exists.

If $u, L(u), v, L(v) \in L^2([0, +\infty), \mathbb{C})$, the limit $(uv)_\infty$ exists and is equal to

$$(uv)_\infty = \int_0^{+\infty} (uL(v) - vL(u)) dt + (uv)_0$$

Theorem 3.3 (On the image of T_2)

Let $f : [0, +\infty) \rightarrow \mathbb{R}$ be fixed. Then there exists $g(s) \in C_{L^2}^0([0, +\infty))$ such that

$$f(s) = \int_0^\infty G_2^l(s, t) g(t) dt, \quad s \geq 0$$

if and only if $f, L(f) \in C_{L^2}^2([0, +\infty))$, $LL(f) \in C_{L^2}^0([0, +\infty))$ and $f, L(f)$ both satisfy the boundary condition (2.5) at $s = 0$ and the boundary condition (3.1) at $s = +\infty$.

Proof.

$\Rightarrow$: Assume first that

$$f(s) = \int_0^\infty G_2^l(s, t) g(t) dt, \quad s \geq 0$$

and define $u(s) := f^* + if$, where

$$f^*(s) := \int_0^\infty G_1^l(s, t) g(t) dt, \quad s \geq 0$$

Then u is of class $C_{L^2}^2([0, +\infty), \mathbb{C})$ and uniquely solves $Lu + iu = -g$ on $[0, +\infty)$ with boundary

conditions (2.5) and (3.1). Separating the real and imaginary parts yields

$$\begin{cases} L(f^*) - f &= -g \\ L(f) + f^* &= 0 \end{cases}$$

Thus, f , $L(f) = -f^*$ and $LL(f) = g - f$ are all continuous, square-integrable functions.

The boundary conditions (2.5) and (3.1) are satisfied by both the real part $f^* = -L(f)$ and the imaginary part f of u .

$\Leftarrow$: Set $f^* := -L(f)$, $g := f - L(f^*)$ and define $u := f^* + if$. Then u solves $Lu + iu = -g$ on $[0, +\infty)$ with boundary condition (2.5) and (3.1). By unicity,

$$u(s) = \int_0^\infty G^l(s, t) g(t) dt$$

so that

$$f(s) = \int_0^\infty G_2^l(s, t) g(t) dt$$

□

Proposition 3.4

T_2 has trivial kernel.

Proof. Assume that $u \in C_{L^2}^0([0, +\infty))$ is such that $T_2 u = 0$. Define

$$f(s) := 0 = \int_0^\infty G_2^l(s, t) u(t) dt, \quad f^*(s) := \int_0^\infty G_1^l(s, t) u(t) dt$$

Then $v := f^* + if = f^*$ solves $Lv + iv = -u$ on $[0, +\infty)$. Since u, v, Lv are real-valued, we must have $v \equiv 0$. Hence, $u = -Lv - iv \equiv 0$ as well. □

3.2 Eigenfunctions and the expansion theorem

Definition 3.1 (Eigenfunction)

Let $u \in C^2([0, +\infty))$ be non-trivial. We call u an *eigenfunction* for the SL operator L on $[0, +\infty)$ with boundary conditions (2.5) and (3.1) if $Lu + \lambda u = 0$ for some $\lambda \in \mathbb{R}$ and u satisfies (2.5) and (3.1). In that case, we call λ an *eigenvalue* of L and the set of eigenvalues is called the *spectrum* of L .

Remark 3.5 (About multiplicity of eigenvalues)

In other words, u is an eigenfunction for L if and only if it is equal to $u = c \cdot \phi(s, \lambda)$ for some $\lambda \in \mathbb{R}$, $c \in \mathbb{C}$, and if it satisfies (3.1). In particular, each eigenvalue of L has multiplicity 1.

Remark 3.6

If $\phi(s, \lambda)$ is an eigenfunction, then $\phi(s, \lambda)$ solves

$$Lu + iu = (i - \lambda)\phi(s, \lambda), \quad s \geq 0$$

with boundary conditions (2.5) and (3.1). Thus,

$$\phi(s, \lambda) = (\lambda - i) \int_0^{+\infty} G^l(s, t) \phi(t, \lambda) dt \quad (3.2)$$

Theorem 3.7

Let L be a SL operator on $[0, +\infty)$ with boundary conditions (2.5) and (3.1).

If $\lambda \in \mathbb{R}$ is an eigenvalue of L with associated eigenfunction u , then $\mu := (1 + \lambda^2)^{-1}$ is an eigenvalue of the linear operator

$$\begin{aligned} T_2 : C_{L^2}^0([0, +\infty)) &\rightarrow C_{L^2}^0([0, +\infty)) \\ f(t) &\mapsto (T_2 f)(s) := \int_0^{+\infty} G_2^l(s, t) f(t) dt \end{aligned}$$

with eigenvector u .

Conversely, if $\mu \in \mathbb{R}$ is an eigenvalue of T_2 with associated eigenvector $u \in C_{L^2}^0([0, +\infty))$, then $\mu \in (0, 1]$, $u \in C^2([0, +\infty))$, u satisfies (2.5), (3.1) and

$$u(s) = c^+ \cdot \phi(s, \lambda^+) + c^- \cdot \phi(s, \lambda^-)$$

where $c^+, c^- \in \mathbb{R}$, $\lambda^+ := \sqrt{\mu^{-1} - 1}$, $\lambda^- := -\sqrt{\mu^{-1} - 1}$.

Furthermore, c^+ is always zero if $\phi(s, \lambda^+)$ is not an eigenfunction of L and c^- is always zero if $\phi(s, \lambda^-)$ is not an eigenfunction of L .

In particular, a (possibly empty) basis for the eigenspace E_μ of T_2 is given by

$$\{\phi(s, \lambda^\epsilon) \mid \epsilon \in \{+, -\}, \phi(s, \lambda^\epsilon) \text{ is an eigenfunction of } L\}$$

Proof.

$\Rightarrow$: Assume that $u(s) = \phi(s, \lambda)$ is an eigenfunction of L . Then by (3.2) :

$$\phi(s, \lambda) = (\lambda - i) \int_0^{+\infty} G^l(s, t) \phi(t, \lambda) dt$$

which is equivalent to

$$\begin{cases} \phi(s, \lambda) &= \lambda \int_0^{+\infty} G_1^l(s, t) \phi(t, \lambda) dt + \int_0^{+\infty} G_2^l(s, t) \phi(t, \lambda) dt \\ 0 &= \lambda \int_0^{+\infty} G_2^l(s, t) \phi(t, \lambda) dt - \int_0^{+\infty} G_1^l(s, t) \phi(t, \lambda) dt \end{cases}$$

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This finally implies

$$\phi(s, \lambda) = (1 + \lambda^2) \int_0^{+\infty} G_2^l(s, t) \phi(t, -\lambda) dt = (1 + \lambda^2) T_2(\phi)$$

$\Leftarrow$: Assume that $u \in C_{L^2}^0([0, +\infty))$ is an eigenvector of T_2 with associated eigenvalue $\mu \in \mathbb{R}$. Proposition 3.4 and estimates (2.26) show that $\mu \in (0, 1]$. Hence, u satisfies

$$u(s) = \mu^{-1} \int_0^{+\infty} G_2^l(s, t) u(t) dt, \quad s \geq 0$$

and $u \in C^2([0, +\infty))$ by Proposition 2.20. We set

$$u^*(s) := \mu^{-1} \int_0^{+\infty} G_1^l(s, t) u(t) dt, \quad s \geq 0$$

Therefore,

$$u^* + iu = \mu^{-1} \int_0^{+\infty} G^l(s, t) u(t) dt$$

is the unique solution to $Lv + i\mu v = -\mu^{-1}u$ on $[0, +\infty)$ with boundary conditions (2.5) and (3.1). By separating the real and imaginary part, we get

$$\begin{cases} L(u^*) - u = -\mu^{-1}u \\ L(u) + u^* = 0 \end{cases}$$

which implies

$$LL(u) - (\mu^{-1} - 1)u = 0$$

First, we study the case $\mu \neq 1$.

This can be rewritten as

$$(L + \lambda^- I) \circ (L + \lambda^+ I)(u) = (L + \lambda^+ I) \circ (L + \lambda^- I)(u) = 0$$

We set

$$v := (L + \lambda^- I)(u)$$

This function satisfies $(L + \lambda^+ I)(v) = 0$ with boundary conditions (2.5) and (3.1). Therefore, there exists $c^+ \in \mathbb{R}$ such that

$$v = c^+ \cdot \phi(s, \lambda^+)$$

with $c^+ = 0$ if $\phi(s, \lambda^+)$ is not an eigenfunction of L .

Similarly, we set

$$w := (L + \lambda^+ I)(u)$$

This function satisfies $(L + \lambda^- I)(w) = 0$ with boundary conditions (2.5) and (3.1). Therefore,

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there exists $c^- \in \mathbb{R}$ such that

$$v = c^- \cdot \phi(s, \lambda^-)$$

with $c^- = 0$ if $\phi(s, \lambda^-)$ is not an eigenfunction of L .

Combining both v and w gives

$$u = \frac{v - w}{\lambda^+ - \lambda^-} = \frac{c^+ \cdot \phi(s, \lambda^+) + c^- \cdot \phi(s, \lambda^-)}{2\lambda^+}$$

If $\mu = 1$, we have

$$LL(u) = 0$$

If $\phi(s, 0)$ is not an eigenfunction of L , then $LL(u) = 0 \Rightarrow L(u) = 0 \Rightarrow u = 0$.

If $\phi(s, 0)$ is an eigenfunction of L , we prove that $LL(u) = 0 \Rightarrow L(u) = 0$. Then in that case, there exists $c \in \mathbb{R}$ such that $u = c \cdot \phi(s, 0)$.

Assume that $LL(u) = 0$. Then there exists $c \in \mathbb{R}$ such that $L(u) = c \cdot \phi(s, 0)$. We show that $c = 0$. Indeed, we have

$$L(u) + iu = c \cdot \phi(s, 0) + iu, \quad L(\phi) + i\phi = i\phi$$

on $[0, +\infty)$ with u and ϕ both satisfying boundary conditions (2.5) and (3.1). Therefore,

$$\begin{aligned} u(s) &= - \int_0^{+\infty} G^l(s, t)(c \cdot \phi(t, 0) + iu(t))dt \\ \phi(s, 0) &= -i \int_0^{+\infty} G^l(s, t)\phi(t, 0)dt \end{aligned}$$

and thus

$$u(s) = -i \int_0^{+\infty} G^l(s, t)u(t)dt - i \cdot c\phi(s, 0)$$

Finally,

$$\begin{aligned} \int_0^{+\infty} u(s)\phi(s, 0)ds &= -i \int_0^{+\infty} \int_0^{+\infty} G^l(s, t)u(t)\phi(s, 0)dt ds - i \cdot c \int_0^{+\infty} \phi(s, 0)^2 ds \\ &= \int_0^{+\infty} -i \left(\int_0^{+\infty} G^l(s, t)\phi(s, 0)ds \right) u(t)dt - i \cdot c \int_0^{+\infty} \phi(s, 0)^2 ds \\ &= \int_0^{+\infty} \phi(s, 0)u(t)dt - i \cdot c \int_0^{+\infty} \phi(s, 0)^2 ds \end{aligned}$$

where we used Fubini-Tonelli theorem. Thus, $c = 0$. □

Proposition 3.8 (On the orthogonality of eigenfunctions)

Any two eigenfunctions $\phi(s, \lambda)$, $\phi(s, \mu)$ with $\lambda \neq \mu$ are L^2 -orthogonal.

Proof. If $\mu \neq -\lambda$, we know this is true since T_2 is self-adjoint and $\phi(s, \lambda)$, $\phi(s, \mu)$ are eigenvectors of T_2 belonging to different eigenspaces.

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Still, we will prove the assertion in the general case $\mu \neq \lambda$. Using the symmetry of the kernel G and Fubini-Tonelli Theorem, we have

$$\begin{aligned}
 \langle \phi(s, \mu), \phi(s, \lambda) \rangle &= \int_0^{+\infty} \phi(s, \mu) \phi(s, \lambda) ds \\
 &= (\mu - i) \int_0^{+\infty} \left(\int_0^{+\infty} G^l(s, t) \phi(t, \mu) dt \right) \phi(s, \lambda) ds \\
 &= (\mu - i) \int_0^{+\infty} \left(\int_0^{+\infty} G^l(s, t) \phi(s, \lambda) ds \right) \phi(t, \mu) dt \\
 &= (\mu - i)(\lambda - i)^{-1} \int_0^{+\infty} \phi(t, \lambda) \phi(t, \mu) dt \\
 &= (\mu - i)(\lambda - i)^{-1} \langle \phi(s, \mu), \phi(s, \lambda) \rangle
 \end{aligned}$$

which implies $\langle \phi(s, \mu), \phi(s, \lambda) \rangle = 0$. □

Proposition 3.9

Assume that $u, v \in C_{L^2}^2([0, +\infty))$ both satisfy Theorem 3.3 hypotheses. Then

$$\int_0^{+\infty} uL(v) - vL(u) dt = 0$$

Proof. Since u, v both satisfy (2.5), we have $(uv)_0 = 0$. By Remark 3.2, it remains to show that $(uv)_\infty = 0$. By Theorem 3.1, we can write

$$\begin{aligned}
 u(s) &= \int_0^{+\infty} G^l(s, t) g_u(t) dt \\
 v(s) &= \int_0^{+\infty} G^l(s, t) g_v(t) dt
 \end{aligned}$$

where $-g_u = L(u) + iu$ and $-g_v = L(v) + iv$. Then

$$(uv)_s = (\beta^l \eta)_s \left(\int_0^s \eta(t) g_u(t) dt \right) \left(\int_s^\infty \beta^l(t) g_v(t) dt \right) - (\beta^l \eta)_s \left(\int_0^s \eta(t) g_v(t) dt \right) \left(\int_s^\infty \beta^l(t) g_u(t) dt \right)$$

Since $(\beta^l \eta) \equiv 1$ and $\eta, \beta^l, g_u, g_v \in L^2([0, +\infty), \mathbb{C})$, everything goes to zero as $s \rightarrow +\infty$. □

Theorem 3.10 (Expansion theorem)

Let L be a SL operator on $[0, +\infty)$ with boundary conditions (2.5) and (3.1).

Then there exists a countable sequence of eigenvalues $(\lambda_n)_{n=1}^\infty$ such that

$$|\lambda_{n+1}| \geq |\lambda_n| \quad \forall n \in \mathbb{N} \quad \text{and} \quad \lim_{n \rightarrow +\infty} |\lambda_n| = +\infty$$

as well as a sequence of associated eigenfunctions

$$(u_n)_{n=1}^\infty := \left(\frac{\phi(s, \lambda_n)}{\|\phi(s, \lambda_n)\|_{L^2}} \right)_{n=1}^\infty \subset C_{L^2}^2([0, +\infty))$$

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such that $U = \{u_n\}_{n=1}^{\infty}$ forms an orthonormal set in $(C_{L^2}^0([0, +\infty)), \langle \cdot, \cdot \rangle_{L^2})$ and such that any $f : [0, +\infty) \rightarrow \mathbb{R}$ satisfying Theorem 3.3 hypotheses can be represented as the following Fourier series in $(C_{L^2}^0([0, +\infty)), \|\cdot\|_{L^2})$:

$$f = \lim_{n \rightarrow +\infty} \sum_{k=1}^n \langle f, u_k \rangle u_k \quad (3.3)$$

In fact, we also have absolute and uniform convergence of the series on every compact subset of $[0, +\infty)$.

Proof. The first part of the theorem follows from Hilbert-Schmidt Theorem (Theorem 1.15 and Remark 1.17). For the absolute and uniform convergence on every compact subset of $[0, +\infty)$, it suffices to show that there exists $g \in C^0([0, +\infty))$ such that

$$\lim_{n \rightarrow +\infty} \sum_{k=1}^n \langle f, u_k \rangle u_k = g$$

absolutely and uniformly on every compact subset of $[0, +\infty)$. Indeed, if we have shown the existence of such a g , we then have for any $a > 0$:

$$\begin{aligned} \int_0^a \left| \sum_{k=1}^n \langle f, u_k \rangle u_k - f \right|^2 ds &\leq \int_0^a \left| \sum_{k=1}^n \langle f, u_k \rangle u_k - g \right|^2 ds \rightarrow 0 \text{ as } n \rightarrow +\infty \\ \int_0^a \left| \sum_{k=1}^n \langle f, u_k \rangle u_k - g \right|^2 ds &\leq a^2 \sup \left\{ \left| \sum_{k=1}^n \langle f, u_k \rangle u_k(s) - g(s) \right|^2 : s \in [0, a] \right\} \rightarrow 0 \text{ as } n \rightarrow +\infty \end{aligned}$$

and by completeness of $L^2([0, a])$, this implies $f = g$ on $[0, a]$.

To prove the existence of g , it suffices to show that

$$\lim_{i, j \rightarrow +\infty} \sum_{n=i}^j |\langle f, u_n \rangle u_n(s)| = 0$$

uniformly on every compact subset of $[0, +\infty)$. By Theorem 3.7 and Proposition 3.9, we have

$$\begin{aligned} u_n(s) &= (1 + \lambda_n^2)(T_2 u_n)(s) = (1 + \lambda_n^2) \left\langle G_2^l(s, \cdot), u_n(\cdot) \right\rangle \\ (1 + \lambda_n^2) |\langle f, u_n \rangle| &\leq 2\lambda_n^2 |\langle f, u_n \rangle| = 2 |\langle f, LLu_n \rangle| = 2 |\langle LLf, u_n \rangle| \text{ (for } n \text{ large enough)} \end{aligned}$$

Thus,

$$\sum_{n=i}^j |\langle f, u_n \rangle u_n(s)| \leq 2 \sum_{n=i}^j |\langle LLf, u_n \rangle| \left\langle G_2^l(s, \cdot), u_n(\cdot) \right\rangle$$

and by Cauchy-Schwarz inequality,

$$\sum_{n=i}^j |\langle f, u_n \rangle u_n(s)| \leq \left(\sum_{n=i}^j \langle LLf, u_n \rangle^2 \right)^{1/2} \cdot \left(\sum_{n=i}^j \left\langle G_2^l(s, \cdot), u_n(\cdot) \right\rangle^2 \right)^{1/2}$$

By Bessel's inequality,

$$\begin{aligned} \sum_{n=1}^{\infty} \langle LLf, u_n \rangle^2 &\leq \|LLf\|_{L^2}^2 < +\infty \\ \sum_{n=1}^{\infty} \langle G_2^l(s, \cdot), u_n(\cdot) \rangle^2 &\leq \|G_2^l(s, \cdot)\|_{L^2}^2 < +\infty \end{aligned}$$

Since $\|G_2^l(s, \cdot)\|_{L^2}^2$ is continuous by Proposition 2.19, we have $\sup_{s \in K} \|G_2^l(s, \cdot)\|_{L^2}^2 \leq M$ for some $0 < M < +\infty$ when s is restricted to a compact subset K of $[0, +\infty)$. Thus,

$$\sum_{n=i}^j |\langle f, u_n \rangle u_n(s)| \leq M^{1/2} \left(\sum_{n=i}^j \langle LLf, u_n \rangle^2 \right)^{1/2} \quad \forall s \in K$$

Since $\sum_{n=1}^{\infty} \langle LLf, u_n \rangle^2$ is a convergent sum, this concludes the proof. $\square$

Theorem 3.11 (Expansion theorem for L^2 functions)

Let L be a SL operator on $[0, +\infty)$ with boundary conditions (2.5) and (3.1).

Then there exists a countable sequence of eigenvalues $(\lambda_n)_{n=1}^{\infty}$ such that

$$|\lambda_{n+1}| \geq |\lambda_n| \quad \forall n \in \mathbb{N} \quad \text{and} \quad \lim_{n \rightarrow +\infty} |\lambda_n| = +\infty$$

as well as a sequence of associated eigenfunctions

$$(u_n)_{n=1}^{\infty} := \left(\frac{\phi(s, \lambda_n)}{\|\phi(s, \lambda_n)\|_{L^2}} \right)_{n=1}^{\infty} \subset C_{L^2}^2([0, +\infty))$$

such that $U = \{u_n\}_{n=1}^{\infty}$ forms an orthonormal set in $(L^2([0, +\infty)), \langle \cdot, \cdot \rangle_{L^2})$ and such that any $f \in L^2([0, +\infty))$ can be represented as the following Fourier series in $(L^2([0, +\infty)), \|\cdot\|_{L^2})$:

$$f = \lim_{n \rightarrow +\infty} \sum_{k=1}^n \langle f, u_k \rangle u_k \quad (3.4)$$

Proof. See the second part of Theorem 1.15. $\square$

Remark 3.12

Consider the sequence of eigenvalues $(\lambda_n)_{n=1}^{\infty}$ and eigenvectors $(u_n)_{n=1}^{\infty}$ from Theorem 3.10. We set

$$\rho(\lambda) = \begin{cases} - \sum_{n: \lambda < \lambda_n \leq 0} \frac{1}{\|\phi(s, \lambda_n)\|_{L^2}^2} & \text{if } \lambda \leq 0 \\ \sum_{n: \lambda > \lambda_n > 0} \frac{1}{\|\phi(s, \lambda_n)\|_{L^2}^2} & \text{if } \lambda > 0 \end{cases}$$

which is a non-decreasing and right-continuous function. This function defines a Lebesgue-Stieltjes measure on $\mathbb{R}$. Observe that for consecutive eigenvalues $\lambda_n < \lambda_m$, ρ is constant on

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(λ_n, λ_m) . Therefore, for any $g : \mathbb{R} \rightarrow \mathbb{C}$ which is integrable on $\mathbb{R}$ with respect to the measure $d\rho$ (i.e. $g \in L^1(\mathbb{R}, \mathbb{C}, \rho)$), we have

$$\int_{-\infty}^{\infty} g(\lambda) d\rho(\lambda) = \sum_{n=1}^{\infty} \frac{g(\lambda_n)}{\|\phi(s, \lambda_n)\|_{L^2}^2}$$

Conversely, if the above sum converges absolutely, then $g : \mathbb{R} \rightarrow \mathbb{C}$ is integrable on $\mathbb{R}$ with respect to the measure ρ .

Let $f \in L^2([0, +\infty))$. We define the *generalized Fourier transform* of f as :

$$F(\lambda) = \int_0^{+\infty} f(x) \phi(x, \lambda) dx = \langle f(\cdot), \phi(\cdot, \lambda) \rangle_{L^2}, \quad \lambda \in \mathbb{R}$$

Then the following Parseval's identity holds for f :

$$\begin{aligned} \int_0^{+\infty} f(s)^2 ds &= \sum_{n=1}^{\infty} \langle f, u_n \rangle^2 \\ &= \sum_{n=1}^{\infty} \frac{F(\lambda_n)^2}{\|\phi(s, \lambda_n)\|_{L^2}^2} \\ &= \int_{-\infty}^{+\infty} F(\lambda)^2 d\rho(\lambda) \end{aligned} \quad (3.5)$$

which is equivalent to the following expansion for f :

$$\begin{aligned} f(s) &= \lim_{n \rightarrow +\infty} \sum_{k=1}^n \langle f, u_k \rangle u_k(s) \\ &= \lim_{n \rightarrow +\infty} \sum_{k=1}^n \frac{F(\lambda_k) \phi(s, \lambda_k)}{\|\phi(s, \lambda_k)\|_{L^2}^2} \\ &= \lim_{n \rightarrow +\infty} \int_{-n}^n F(\lambda) \phi(s, \lambda) d\rho(\lambda) \end{aligned} \quad (3.6)$$

where the partial sum and partial integral converge to f in $L^2([0, +\infty))$. Moreover, if f satisfies some additional smoothness (i.e. Theorem 3.3 hypotheses), then the partial sum and partial integral converges absolutely, uniformly to f on every compact subset of $[0, +\infty)$. We will see that in the general case, (3.5) and (3.6) remain true.

Similarly, since $f \mapsto \int_0^{+\infty} G^l(s, t) f(t) dt$ is L^2 -bounded, we also have that for $f \in L^2([0, +\infty))$:

$$\begin{aligned} \int_0^{+\infty} G^l(s, t) f(t) dt &= \sum_{n=1}^{\infty} \frac{\langle f, u_n \rangle u_n(s)}{\lambda_n - i} \\ &= \sum_{n=1}^{\infty} \frac{F(\lambda_n) \phi(s, \lambda_n)}{(\lambda_n - i) \|\phi(s, \lambda_n)\|_{L^2}^2} \\ &= \int_{-\infty}^{+\infty} F(\lambda) \frac{\phi(s, \lambda)}{\lambda - i} d\rho(\lambda) \end{aligned} \quad (3.7)$$

where the partial sum and partial integral converge absolutely (and not only in L^2 -sense)

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because $F(\lambda), \frac{\phi(s, \lambda)}{\lambda - i} \in L^2(\mathbb{R}, \mathbb{C}, \rho)$ by Parseval's identity (3.5) applied to f and $G^l(s, t)$ (as a function of t). Again, this is equivalent to Parseval's identity :

$$\begin{aligned} \int_0^{+\infty} \left| \int_0^{+\infty} G^l(s, t) f(t) dt \right|^2 ds &= \sum_{n=1}^{\infty} \frac{\langle f, u_n \rangle^2}{|\lambda_n - i|^2} \\ &= \sum_{n=1}^{\infty} \frac{F(\lambda_n)^2}{(1 + \lambda_n)^2 \|\phi(s, \lambda_n)\|_{L^2}^2} \\ &= \int_{-\infty}^{+\infty} \frac{F(\lambda)^2}{(1 + \lambda)^2} d\rho(\lambda) \end{aligned} \tag{3.8}$$

In the general case, (3.7) and (3.8) remain true.

4 The general case : Transform Theory

Assume that L is a SL operator on $[0, +\infty)$. Our goal is to extend the Parseval's identity and the integral expansion theorem as in Remark 3.12 to the limit-point case. The measure ρ will be obtained as the limit of a sequence of measures for the compact case. The books [Sar98] and [Lev55] are used as our main references for this chapter.

For the rest of this Chapter 4, L is a SL operator on $[0, +\infty)$ with boundary condition (2.5) and boundary condition (2.10) (when L is restricted to $[0, a]$), where $h, j \in [0, \pi)$ are fixed.

4.1 Construction of the spectral measure

For $a > 0$, we restrict L to the compact interval $[0, a]$. Then u is an eigenvalue of the operator L with boundary conditions (2.5), (2.10) if and only if it is equal to $u = c \cdot \phi(s, \lambda)$ for some $\lambda \in \mathbb{R}$, $c \in \mathbb{C}$ and if it satisfies (2.10).

Let $(\lambda_{a,n})_{n=1}^{\infty}$ be the sequence of eigenvalues given by Theorem 1.18 for L on $[0, a]$ with boundary conditions (2.5), (2.10) and let $(\phi(s, \lambda_{a,n}))_{n=1}^{\infty}$ be the associated (non-normalized) eigenfunctions.

We set

$$\rho_a(\lambda) = \begin{cases} - \sum_{n: \lambda < \lambda_{a,n} \leq 0} \frac{1}{\|\phi(s, \lambda_{a,n})\|_{L^2([0,a])}^2} & \text{if } \lambda \leq 0 \\ \sum_{n: \lambda > \lambda_{a,n} > 0} \frac{1}{\|\phi(s, \lambda_{a,n})\|_{L^2([0,a])}^2} & \text{if } \lambda > 0 \end{cases}$$

which is a non-decreasing and right-continuous function. This function defines a Lebesgue-Stieltjes measure on $\mathbb{R}$. Observe that for consecutive eigenvalues $\lambda_{a,n} < \lambda_{a,m}$, ρ_a is constant on $[\lambda_{a,n}, \lambda_{a,m})$. Therefore, for any $g : \mathbb{R} \rightarrow \mathbb{C}$ which is integrable on $\mathbb{R}$ with respect to the measure ρ_a (i.e. $g \in L^1(\mathbb{R}, \mathbb{C}, \rho_a)$), we have

$$\int_{-\infty}^{\infty} g(\lambda) d\rho_a(\lambda) = \sum_{n=1}^{\infty} \frac{g(\lambda_{a,n})}{\|\phi(s, \lambda_{a,n})\|_{L^2([0,a])}^2}$$

Conversely, if the above sum converges absolutely, then $g : \mathbb{R} \rightarrow \mathbb{C}$ is integrable on $\mathbb{R}$ with

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respect to the measure ρ_a .

Notation

We write $\alpha_{a,n} = \|\phi(s, \lambda_{a,n})\|_{L^2([0,a])}$.

For a measure $d\mu$ on $\mathbb{R}$ and a compact interval $[a, b]$, we write

$$\int_a^b f d\mu := \int_{[a,b]} f d\mu = \int_{-\infty}^{+\infty} f \cdot \chi_{[a,b]} d\mu$$

if $f \cdot \chi_{[a,b]} \in L^1(\mathbb{R}, \mathbb{C}, \mu)$.

Similarly to Remark 3.12, for all $f \in L^2([0, a])$, Parseval's identity holds :

$$\int_0^a f(s)^2 ds = \sum_{n=1}^{\infty} \frac{1}{\alpha_{a,n}^2} \langle f, \phi(s, \lambda_{a,n}) \rangle_{L^2([0,a])}^2 = \int_{-\infty}^{+\infty} F(\lambda)^2 d\rho_a(\lambda) \quad (4.1)$$

where

$$F(\lambda) = \int_0^a f(x) \phi(x, \lambda) dx = \langle f(\cdot), \phi(\cdot, \lambda) \rangle_{L^2([0,a])}, \quad \lambda \in \mathbb{R}$$

is the generalized Fourier transform of f .

Moreover,

$$f(s) = \lim_{n \rightarrow +\infty} \sum_{k=1}^n \frac{1}{\alpha_{a,k}^2} \langle f, \phi(s, \lambda_{a,k}) \rangle_{L^2([0,a])} \phi(s, -\lambda_{a,k}) = \lim_{n \rightarrow +\infty} \int_{-n}^n F(\lambda) \phi(s, \lambda) d\rho_a(\lambda) \quad (4.2)$$

as a limit in $L^2([0, a])$. Besides, if $f \in C^2([0, a])$ with $R_1 f = R_2 f = 0$, then the convergence is also absolute and uniform.

We now prove two results which will allow us to have explicit estimates on the growth of $\rho_a(\lambda)$.

Lemma 4.1

Let $\lambda \in \mathbb{R} \setminus \{0\}$ and let $\alpha := \sqrt{|\lambda|}$. Then there exists four functions $u_{\pm}, v_{\pm} \in C^2([0, +\infty), \mathbb{C})$ (which depend only on p) such that $u_{\pm}(0) = 1, u'_{\pm}(0) = 0, v_{\pm}(0) = 0, v'_{\pm}(0) = 1$ and

$$\begin{aligned} \phi(s, \lambda) = & -\sin(h) u(\alpha t) + \frac{\cos(h)}{\alpha} v(\alpha t) \\ & + \frac{1}{\alpha} \int_0^t (u(\alpha \tau) v(\alpha \tau) - v(\alpha \tau) u(\alpha \tau)) p(s(\tau)) q(s(\tau)) \phi(s(\tau), \lambda) d\tau, \quad t \geq 0 \end{aligned} \quad (4.3)$$

where $u = u_+, v = v_+$ if $\lambda > 0$ and $u = u_-, v = v_-$ if $\lambda < 0$. Moreover, the relationship between $s \geq 0$ and $t \geq 0$ is given by

$$t(s) = \int_0^s \frac{1}{p(\tau)} d\tau$$

Proof. We consider

$$t(s) = \int_0^s \frac{1}{p(\tau)} d\tau$$

Then $t(s) : [0, +\infty) \rightarrow [0, +\infty)$ is bijective, increasing, of class $C^2([0, +\infty))$ and fixes 0. Let $s(t)$ denote its inverse. We do a change of variables and observe that

$$\Phi(t, \lambda) := \phi(s(t), \lambda), \quad t \geq 0$$

solves

$$\frac{d^2 u}{dt^2}(t) - p(s(t))q(s(t))u(t) + p(s(t))\lambda u(t) = 0$$

on $[0, +\infty)$ with boundary conditions

$$\Phi(0, \lambda) = -\sin(h), \quad \frac{d\Phi}{dt}(0, \lambda) = \cos(h)$$

Let $\{u(t), v(t)\}$ be a fundamental system for

$$\frac{d^2 u}{dt^2}(t) + \varepsilon p(s(t))u(t) = 0, \quad t \geq 0$$

satisfying

$$u(0) = 1, u'(0) = 0 \quad v(0) = 0, v'(0) = 1$$

where we choose $\varepsilon = 1$ if $\lambda > 0$ and $\varepsilon = -1$ if $\lambda < 0$.

Then $\{\hat{u}(t) := u(\alpha t), \hat{v}(t) := \alpha^{-1} v(\alpha t)\}$ is a fundamental system for

$$\frac{d^2 u}{dt^2}(t) + \lambda p(s(t))u(t) = 0, \quad t \geq 0$$

satisfying

$$\hat{u}(0) = 1, \hat{u}'(0) = 0 \quad \hat{v}(0) = 0, \hat{v}'(0) = 1$$

Similarly to Proposition 2.26, we can write

$$\Phi(t, \lambda) = c \cdot \hat{u}(t) + d \cdot \hat{v}(t) + \int_0^t (\hat{u}(\tau) \hat{v}(t) - \hat{v}(\tau) \hat{u}(t)) p(s(\tau)) q(s(\tau)) \Phi(\tau, \lambda) d\tau$$

where $c = -\sin(h)$ and $d = \cos(h)$ using the initial conditions on $\Phi(t, \lambda)$, $\hat{u}$ and $\hat{v}$. Thus,

$$\begin{aligned} \Phi(t, \lambda) &= -\sin(h) u(\alpha t) + \frac{\cos(h)}{\alpha} v(\alpha t) \\ &\quad + \frac{1}{\alpha} \int_0^t (u(\alpha \tau) v(\alpha t) - v(\alpha \tau) u(\alpha t)) p(s(\tau)) q(s(\tau)) \Phi(\tau, \lambda) d\tau \end{aligned}$$

□

Remark 4.2

If $p \equiv 1$, then $u_+(t) = \cos(t)$, $u_-(t) = \frac{1}{2}(\exp(t) + \exp(-t))$ and $v_+(t) = \sin(t)$, $v_-(t) = \frac{1}{2}(\exp(t) - \exp(-t))$.

Proposition 4.3

There exists two constants $N_0 \geq 1$ and $1 > C > 0$ (which depend only on p, q and h) such that for any $N \geq N_0$, for $\delta = C/N^{1/2}$, we have

$$0 \leq s \leq \delta, \lambda \in [-N, N] \Rightarrow -\phi(s, \lambda) \geq \frac{1}{2} \sin(h) > 0 \quad \text{if } h \in (0, \pi) \quad (4.4)$$

$$0 \leq s \leq \delta, \lambda \in [-N, N] \Rightarrow \frac{d\phi}{ds}(s, \lambda) \geq \frac{1}{2} \quad \text{if } h = 0 \quad (4.5)$$

Proof. Let $s(t)$ denotes the inverse of $t(s) = \int_0^s p(\tau)^{-1} d\tau$. Consider the family of functions

$$\mathcal{F} = \{u_+(t), u'_+(t), u_-(t), u'_-(t), v_+(t), v'_+(t), v_-(t), v'_-(t), p(s(t)), q(s(t))\}$$

where $u_{\pm}, v_{\pm}$ are given by Lemma 4.1. Define

$$M_1 := \prod_{f \in \mathcal{F}} \left(1 + \max_{t \in [0, 1]} |f(t)| \right), \quad M_2 := 2M_1 \exp(2M_1^4), \quad M_3 := 2M_1^4 M_2, \quad M_4 := \left(\min_{t \in [0, 1]} p(t) \right)^{-1} \cdot M_3$$

and let $N_0 \geq 1$ be defined as

$$\begin{aligned} N_0 &:= \max\{1, 2M_3 \sin(h)^{-1}\} & \text{if } h \in (0, \pi) \\ N_0 &:= 1 & \text{if } h = 0 \end{aligned}$$

Let $N \geq N_0$ be fixed. We distinguish three subsets of λ : $S_1 = [-N, -N_0]$, $S_2 = [-N_0, N_0]$ and $S_3 = (N_0, N]$. For each subset, we will get a constant $1 > C_i > 0$ such (4.4) or (4.5) hold when δ is replaced by $\delta_i = C_i/N^{1/2}$ and $[-N, N]$ is replaced by S_i . Then, it suffices to take $C = \min\{C_1, C_2, C_3\}$.

The case $\lambda \in S_2$ is a direct consequence of the uniform continuity of $\phi(s, \lambda)$ and $\frac{d\phi}{ds}(s, \lambda)$ on $[0, 1] \times [-N_0, N_0]$.

We now treat the case $\lambda \in S_1$. The case of $\lambda \in S_3$ is treated similarly.

By Lemma 4.1, for $s \geq 0$ and $\lambda < 0$, we can write

$$\begin{aligned} \phi(s, \lambda) &= -\sin(h)u(\alpha t) + \frac{\cos(h)}{\alpha}v(\alpha t) \\ &\quad + \frac{1}{\alpha} \int_0^t (u(\alpha \tau)v(\alpha \tau) - v(\alpha \tau)u(\alpha \tau)) p(s(\tau))q(s(\tau))\phi(s(\tau), \lambda) d\tau \end{aligned} \quad (4.6)$$

where $u = u_-$, $v = v_-$, $\alpha := \sqrt{|\lambda|}$ and $t = t(s) = \int_0^s p(\tau)^{-1} d\tau$. Let $1 > \gamma > 0$ be such that

$$0 \leq t \leq \gamma \Rightarrow u(t) \geq \frac{3}{2} \quad \text{if } h \in (0, \pi) \quad (4.7)$$

$$0 \leq t \leq \gamma \Rightarrow v'(t) \geq \frac{3}{2} \quad \text{if } h = 0 \quad (4.8)$$

We restrict s to $[0, 1]$ where the following estimate on $t(s)$ is useful :

$$0 \leq s \leq 1 \Rightarrow 0 \leq t(s) = \int_0^s p(\tau)^{-1} d\tau \leq s \cdot \left(\min_{s \in [0,1]} p(s) \right)^{-1} \quad (4.9)$$

Let

$$1 > C := \min\{1, \min_{s \in [0,1]} p(s)\} \cdot \gamma > 0$$

Using (4.9), we find that if $\lambda \leq -1$:

$$0 \leq s \leq \frac{C}{\alpha} \Rightarrow 0 \leq t(s) \leq \frac{\gamma}{\alpha} \quad (4.10)$$

Thus, for $0 \leq s \leq C/N^{1/2}$ and $\lambda \in [-N, -1)$, we have that

1. $0 \leq s \leq C/N^{1/2} \leq C/\alpha \leq 1$, hence $0 \leq \alpha t(s) \leq \gamma \leq 1$ as well by equation (4.10).
2. For $\tau \in [0, t(s)] : 0 \leq s(\tau) \leq s(t(s)) = s \leq 1$ by monotonicity of $t \mapsto s(t)$.
3. For $\tau \in [0, t(s)] : 0 \leq \alpha \tau \leq \alpha t(s) \leq \gamma \leq 1$ using (1.)

In particular, we have the following upper bound on $\phi(s, \lambda)$ when $0 \leq s \leq C/N^{1/2}$ and $\lambda \in [-N, -1)$:

$$|\phi(s, \lambda)| \leq 2M_1 + 2M_1^4 \int_0^t |\phi(s(\tau), \lambda)| d\tau = 2M_1 + 2M_1^4 \int_0^s p(\tau)^{-1} |\phi(\tau, \lambda)| d\tau$$

Grönwall's inequality then implies that

$$|\phi(s, \lambda)| \leq 2M_1 \exp\left(2M_1^4 \int_0^s \frac{1}{p(\tau)} d\tau\right) = 2M_1 \exp(2M_1^4 t(s)) \leq 2M_1 \exp(2M_1^4) = M_2 \quad (4.11)$$

for $0 \leq s \leq C/N^{1/2}$ and $\lambda \in [-N, -1)$.

Assume that $h \neq 0$. Observe that for $0 \leq s \leq C/N^{1/2}$ and $\lambda \in [-N, -N_0)$,

$$\left| \frac{\cos(h)}{\alpha} v(\alpha t) \right| \leq \frac{M_1}{N_0} \leq \frac{1}{2} \sin(h)$$

by definition of N_0 and

$$\left| \frac{1}{\alpha} \int_0^t (u(\alpha t) v(\alpha \tau) - v(\alpha t) u(\alpha \tau)) p(s(\tau)) q(s(\tau)) \phi(s(\tau), \lambda) d\tau \right| \leq \frac{M_3}{N_0} \leq \frac{1}{2} \sin(h)$$

by equation (4.11) and definition of N_0 . We finally get from equation (4.6), the above two estimates and equation (4.7) that for $0 \leq s \leq C/N^{1/2}$ and $\lambda \in [-N, -N_0)$,

$$-\phi(s, \lambda) - \frac{1}{2} \sin(h) \geq \left(u(\alpha t) - \frac{1}{2} \right) \sin(h) - \sin(h) \geq \frac{1}{2} \sin(h) > 0$$

If $h = 0$, we have

$$\frac{d\phi}{ds}(s, \lambda) = v'(\alpha t) + \int_0^t (u'(\alpha t)v(\alpha\tau) - v'(\alpha t)u(\alpha\tau)) p(s(\tau))q(s(\tau))\phi(s(\tau), \lambda) d\tau$$

For $0 \leq s \leq \min\{C, (4M_4)^{-1}\}/N^{1/2}$ and $\lambda \in [-N, -1)$, equation (4.9) implies

$$\left| \int_0^t (u'(\alpha t)v(\alpha\tau) - v'(\alpha t)u(\alpha\tau)) p(s(\tau))q(s(\tau))\phi(s(\tau), \lambda) d\tau \right| \leq 2tM_3 \leq 2sM_4 \leq \frac{1}{2}$$

so that

$$\frac{d\phi}{ds}(s, \lambda) - \frac{1}{2} \geq \left(v'(\alpha t) - \frac{1}{2} \right) - \frac{1}{2} \geq \frac{1}{2} > 0$$

□

Proposition 4.4

For each $N > 0$, the family of functions $\{\rho_a\}_{a \geq 1}$ has a uniformly bounded total variation on $[-N, N]$, i.e.

$$\sup_{a \geq 1} V_{-N}^N(\rho_a) < +\infty$$

More precisely, there exists constants $N_0 \geq 1$ and $C > 0$ (which depend only on p, q and h) such that for all $N \geq N_0$:

$$\sup_{a \geq 1} V_{-N}^N(\rho_a) \leq \frac{N^{1/2}}{C} \quad \text{if } h \in (0, \pi) \quad (4.12)$$

$$\sup_{a \geq 1} V_{-N}^N(\rho_a) \leq \frac{N^{3/2}}{C} \quad \text{if } h = 0 \quad (4.13)$$

Proof. Since ρ_a is monotone,

$$V_{-N}^N(\rho_a) = \rho_a(N) - \rho_a(-N)$$

If $0 < M \leq N$: $V_{-N}^N(\rho_a) \leq V_{-M}^M(\rho_a)$, so assume without loss of generality that $N \geq N_0$, where $N_0 \geq 1$ together with $1 > C > 0$ are given by Proposition 4.3. Assume that $h \neq 0$ first (i.e $h \in (0, \pi)$). Let $a \geq 1$ be fixed. For $1 > \delta = C/N^{1/2} > 0$, we have for all $s \in [0, a]$, for all $\lambda \in [-N, N]$:

$$0 \leq s \leq \delta \Rightarrow -\phi(s, \lambda) \geq \frac{1}{2} \sin(h) > 0$$

Let's define $f \in L^2([0, a])$, $f(s) = 1/\delta$ if $0 \leq s \leq \delta$, $f(s) = 0$ otherwise. By Parseval's identity (4.1):

$$\begin{aligned} \frac{N^{1/2}}{C} &= \frac{1}{\delta} = \int_0^a f(s)^2 ds = \int_{-\infty}^{+\infty} F(\lambda)^2 d\rho_a(\lambda) = \int_{-\infty}^{+\infty} \left(\int_0^\delta \frac{1}{\delta} \phi(x, \lambda) dx \right)^2 d\rho_a(\lambda) \\ &\geq \int_{-N}^N \left(\int_0^\delta \frac{1}{\delta} \phi(x, \lambda) dx \right)^2 d\rho_a(\lambda) \geq \int_{-N}^N \frac{1}{4} \sin(h)^2 d\rho_a(\lambda) \\ &= \frac{1}{4} \sin(h)^2 (\rho_a(N) - \rho_a(-N)) \end{aligned}$$

Similarly, if $h = 0$ and $a \geq 1$ is fixed, for $1 > \delta = C/N^{1/2} > 0$, we have for all $s \in [0, a]$, for all $\lambda \in [-N, N]$:

$$0 \leq s \leq \delta \Rightarrow \frac{d\phi}{ds}(s, \lambda) \geq \frac{1}{2}$$

which implies for all $s \in [0, a]$, for all $\lambda \in [-N, N]$:

$$0 \leq s \leq \delta \Rightarrow \phi(s, \lambda) \geq \frac{s}{2}$$

by the mean value theorem. Let's define $f \in L^2([0, a])$, $f(s) = 1/\delta^2$ if $0 \leq s \leq \delta$, $f(s) = 0$ otherwise. By Parseval's identity (4.1) :

$$\begin{aligned} \frac{N^{3/2}}{C^3} &= \frac{1}{\delta^3} = \int_0^a f(s)^2 ds = \int_{-\infty}^{+\infty} F(\lambda)^2 d\rho_a(\lambda) = \int_{-\infty}^{+\infty} \left(\int_0^\delta \frac{1}{\delta^2} \phi(x, \lambda) dx \right)^2 d\rho_a(\lambda) \\ &\geq \int_{-N}^N \left(\int_0^\delta \frac{1}{\delta^2} \phi(x, \lambda) dx \right)^2 d\rho_a(\lambda) \geq \int_{-N}^N \left(\int_0^\delta \frac{x}{2\delta^2} dx \right)^2 d\rho_a(\lambda) \\ &= \frac{1}{16} (\rho_a(N) - \rho_a(-N)) \end{aligned}$$

□

Proposition 4.5

For each $N > 0$, the family of functions $\{\rho_a\}_{a \geq 1}$ is uniformly bounded on $[-N, N]$.

Proof. Observe that for every $a \geq 1$, $\rho_a(0) = 0$. Thus, for any $\lambda \in [-N, N]$:

$$|\rho_a(\lambda)| = |\rho_a(\lambda) - \rho_a(0)| \leq V_{-N}^N(\rho_a)$$

Since the family $\{\rho_a\}_{a \geq 1}$ has a uniformly bounded total variation, this finishes the proof. □

Theorem 4.6 (Helly's Selection Theorem)

Let $I \subset \mathbb{R}$ be any interval and let $(\mu_n)_{n=1}^\infty : I \rightarrow \mathbb{R}$ be a sequence of functions such that for any compact interval $K \subset I$:

1. The sequence $(\mu_n)_{n=1}^\infty$ is uniformly bounded on K :

$$\sup_{n \in \mathbb{N}} \sup_{\lambda \in K} |\mu_n(\lambda)| < +\infty$$

2. The sequence $(\mu_n)_{n=1}^\infty$ has a uniformly bounded total variation on K :

$$\sup_{n \in \mathbb{N}} V_K(\mu_n) < +\infty$$

Then there exists a subsequence $(\mu_{n_k})_{k=1}^\infty$ of (μ_n) and a function $\mu : I \rightarrow \mathbb{R}$ such that (μ_{n_k}) converges pointwisely to μ on I and such that μ is of bounded variation on every compact interval $K \subset I$.

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Moreover, if the functions $(\mu_n)_{n=1}^\infty$ are resp. monotone / uniformly bounded / right-continuous on I , then μ is resp. monotone / bounded / right-continuous on I .

Proof. See [Nat16], p.222. □

Proposition 4.7 (Integration Theorem)

Let $(\mu_n)_{n=1}^\infty, \mu : \mathbb{R} \rightarrow \mathbb{R}$ be non-decreasing and right-continuous functions such that for all continuity points λ of μ : $\lim_{n \rightarrow +\infty} \mu_n(\lambda) = \mu(\lambda)$. Let $D \subset \mathbb{R}$ be the countable set made of all the points of discontinuity of μ and each μ_n . Let $a < b \in D^c$. Then the sequence of Stieltjes measures $(\mu_n)_{n=1}^\infty$ converges weakly on $[a, b]$ to the measure μ in the sense that

$$\lim_{n \rightarrow +\infty} \int_a^b f(\lambda) d\mu_n(\lambda) = \int_a^b f(\lambda) d\mu(\lambda)$$

for every continuous functions $f \in C^0([a, b])$.

Proof. Recall that a step function $g : \mathbb{R} \rightarrow \mathbb{R}$ is a function of the form

$$g(x) = \sum_{i=1}^n a_i \chi_{A_i}(x)$$

where $a_i \in \mathbb{R}$ and the A_i 's are disjoint intervals. Since $[a, b] \setminus D \subset [a, b]$ is dense and f is uniformly continuous on $[a, b]$, for all $m \in \mathbb{N}$, there exists

$$a = x_0^{(m)} < x_1^{(m)} < \dots < x_{k_m}^{(m)} = b$$

points of continuity of μ and each μ_n such that for all $i \in \{0, 1, \dots, k_m - 1\}$:

$$x \in (x_i^{(m)}, x_{i+1}^{(m)}] \Rightarrow |f(x) - f(x_{i+1}^{(m)})| \leq 1/m$$

It follows that the sequence

$$f_m(x) = f(a) \chi_{\{a\}}(x) + \sum_{i=0}^{k_m-1} f(x_{i+1}^{(m)}) \chi_{(x_i^{(m)}, x_{i+1}^{(m)}]}(x)$$

approximate uniformly f on $[a, b]$. Moreover, since the $x_i^{(m)}$'s are points of continuity of μ and each μ_n , it is easy to check that for any $m \in \mathbb{N}$:

$$\lim_{n \rightarrow +\infty} \int_a^b f_m(\lambda) d\mu_n(\lambda) = \int_a^b f_m(\lambda) d\mu(\lambda)$$

Let $\varepsilon > 0$, let $m \in \mathbb{N}$ be chosen large enough so that

$$\sup_{\lambda \in [a, b]} |f(\lambda) - f_m(\lambda)| \leq \frac{\varepsilon}{3 \cdot (\mu([a, b]) + 1)}$$

For this $m \in \mathbb{N}$, let $N \in \mathbb{N}$ be large enough so that

$$\begin{aligned} n > N &\Rightarrow \left| \int_a^b f_m(\lambda) d\mu_n(\lambda) - \int_a^b f_m(\lambda) d\mu(\lambda) \right| \leq \frac{\varepsilon}{3} \\ n > N &\Rightarrow \left| \frac{\mu_n([a, b])}{\mu([a, b]) + 1} \right| \leq 1 \end{aligned}$$

Then for any $n > N$, the triangle inequality implies

$$\begin{aligned} \left| \int_a^b f(\lambda) d\mu_n(\lambda) - \int_a^b f(\lambda) d\mu(\lambda) \right| &\leq \left| \int_a^b f(\lambda) d\mu_n(\lambda) - \int_a^b f_m(\lambda) d\mu_n(\lambda) \right| \\ &\quad + \left| \int_a^b f_m(\lambda) d\mu_n(\lambda) - \int_a^b f_m(\lambda) d\mu(\lambda) \right| \\ &\quad + \left| \int_a^b f_m(\lambda) d\mu(\lambda) - \int_a^b f(\lambda) d\mu(\lambda) \right| \\ &\leq \sup_{\lambda \in [a, b]} |f(\lambda) - f_m(\lambda)| \cdot (\mu_n([a, b]) + \mu([a, b])) + \frac{\varepsilon}{3} \\ &\leq \varepsilon \end{aligned}$$

This concludes the proof. $\square$

Corollary 4.8

Let $(\mu_n)_{n=1}^\infty, \mu : \mathbb{R} \rightarrow \mathbb{R}$ be non-decreasing and right-continuous functions such that for all continuity points λ of μ : $\lim_{n \rightarrow +\infty} \mu_n(\lambda) = \mu(\lambda)$. Then the sequence of Stieltjes measures $(\mu_n)_{n=1}^\infty$ converges vaguely on $\mathbb{R}$ to the measure μ in the sense that

$$\lim_{n \rightarrow +\infty} \int_{-\infty}^{+\infty} f(\lambda) d\mu_n(\lambda) = \int_{-\infty}^{+\infty} f(\lambda) d\mu(\lambda)$$

for every continuous and compactly supported functions $f \in C^0(\mathbb{R})$.

Proof. Let $K = \overline{\text{supp}(f)}$ be the compact support of f . Since the set of discontinuities of μ and each μ_n is countable, we can find $a < b$ points of continuity of μ and each μ_n such that $K \subset [a, b]$. Then, we apply Proposition 4.7. $\square$

Proposition 4.9 (Existence of a spectral measure)

There exists an increasing sequence $(a_n)_{n=1}^\infty \subset \mathbb{N}$ and a non-decreasing, right-continuous function $\rho : \mathbb{R} \rightarrow \mathbb{R}$ such that $(\rho_{a_n})_{n=1}^\infty$ converges pointwisely to ρ on $\mathbb{R}$.

Proof. Apply Theorem 4.6 to $I = \mathbb{R}$ and the sequence $(\rho_a)_{a=1}^\infty$ to get a subsequence converging pointwisely to a non-decreasing and right-continuous function ρ . $\square$

Remark 4.10

The estimates (4.12) and (4.13) from Proposition 4.4 also hold for ρ because of pointwise convergence.

Notation

For the rest of this Chapter 4, we fix a spectral function ρ , a holomorphic function $m(\lambda)$ on $\mathbb{C} \setminus \mathbb{R}$ and an increasing sequence $(a_n)_{n=1}^\infty \subset \mathbb{N}$ such that $(\rho_{a_n})_{n=1}^\infty$ converges pointwisely to ρ on $\mathbb{R}$ and $(l_{a_n}(\lambda))_{n=1}^\infty$ converges uniformly to $m(\lambda)$ on every compact subset of $\mathbb{C} \setminus \mathbb{R}$.

Moreover, for $a > 0$ and $\lambda \in \mathbb{C} \setminus \mathbb{R}$, we will write

$$G_a(s, t, \lambda) := G^{l_a(\lambda)}(s, t, \lambda) := \begin{cases} \eta(s, \lambda) \beta^l(t, \lambda) & 0 \leq s \leq t \leq a \\ \beta^l(s, \lambda) \eta(t, \lambda) & a \geq s > t \geq 0 \\ 0 & s, t > a \end{cases}$$

$$G(s, t, \lambda) := G^{m(\lambda)}(s, t, \lambda) := \begin{cases} \eta(s, \lambda) \beta^m(t, \lambda) & 0 \leq s \leq t \\ \beta^m(s, \lambda) \eta(t, \lambda) & s > t \geq 0 \end{cases}$$

Observe that for a fixed $z \in \mathbb{C} \setminus \mathbb{R}$, the eigenfunction $\phi(s, \lambda_{a,n})$ solves $Lu + zu = (z - \lambda_{a,n})u$ on $[0, a]$ with $R_1 u = R_2 u = 0$. This implies that

$$\phi(s, \lambda_{a,n}) = (\lambda_{a,n} - z) \int_0^a G_a(s, t, z) \phi(t, \lambda_{a,n}) dt, \quad s \in [0, a] \quad (4.14)$$

Thus, similarly to Remark 3.12, if $f \in L^2([0, a])$ we have for $s \in [0, a]$:

$$\begin{aligned} \int_0^a G_a(s, t, z) f(t) dt &= \sum_{n=1}^\infty \frac{\langle f, \phi(s, \lambda_{a,n}) \rangle_{L^2([0, a])}}{\alpha_{a,n}^2 (\lambda_{a,n} - z)} \phi(s, \lambda_{a,n}) \\ &= \int_{-\infty}^{+\infty} F(\lambda) \frac{\phi(s, \lambda)}{\lambda - z} d\rho_a(\lambda) \end{aligned} \quad (4.15)$$

where the integral converges absolutely. Equivalently, for $s \in [0, a]$:

$$\int_0^a \left| \int_0^a G_a(s, t, z) f(t) dt \right|^2 ds = \sum_{n=1}^\infty \frac{\langle f, \phi(s, \lambda_{a,n}) \rangle_{L^2([0, a])}^2}{\alpha_{a,n}^2 \cdot |\lambda_{a,n} - z|^2} = \int_{-\infty}^{+\infty} \frac{F(\lambda)^2}{|\lambda - z|^2} d\rho_a(\lambda) \quad (4.16)$$

4.2 The integral expansion theorem

Now that we have proved the existence of a limit spectral measure, we define the Fourier transform for $L^2([0, +\infty))$ functions, which allows us generalize Parseval's identity and the integral expansion theorem. The latter comes down to proving that the inverse Fourier transform of the Fourier transform of f is exactly f .

Proposition 4.11 (Parseval's identity for compactly supported functions)

Let $f \in C^\infty([0, \infty))$ be a smooth, compactly supported function such that

$$K := \overline{\text{supp}(f)} = \overline{\{s \in [0, +\infty) \mid f(s) \neq 0\}} \subset (0, +\infty) \quad (4.17)$$

Let F be the generalized Fourier transform of f , i.e.

$$F(\lambda) = \int_0^{+\infty} f(x)\phi(x, \lambda)dx = \int_K f(x)\phi(x, \lambda)dx, \quad \lambda \in \mathbb{R}$$

Then $F \in L^2(\mathbb{R}, \rho)$ and Parseval's identity holds :

$$\int_0^{+\infty} f(s)^2 ds = \int_{-\infty}^{+\infty} F(\lambda)^2 d\rho(\lambda) \quad (4.18)$$

Proof. The Parseval's identity (4.1) on the compact case ensures that for any $a > 0$,

$$\int_0^a f(s)^2 ds = \int_{-\infty}^{+\infty} F_a(\lambda)^2 d\rho_a(\lambda) \quad (4.19)$$

where

$$F_a(\lambda) = \int_0^a f(x)\phi(x, \lambda)dx$$

Let $a > \sup K$. Since both $f(x)$ and $\phi(x, \lambda)$ satisfy (2.5) : $(f\phi(\cdot, \lambda))_0 = 0$, and since $f \equiv 0$ in a neighborhood of a : $(f\phi(\cdot, \lambda))_a = 0$. Thus,

$$F_a(\lambda) = \frac{1}{\lambda} \int_0^a f(x)L(\phi(x, \lambda))dx = \frac{1}{\lambda} \int_0^a L(f)(x)\phi(x, \lambda)dx$$

where we used Green's identity.

Observe that for all $a > \sup K$: $F_a(\lambda) \equiv F(\lambda)$. Then for any $a > \sup K$ and any points of continuity $M < N$ of ρ and each ρ_{a_n} , equality (4.19) implies that

$$\begin{aligned} \left| \int_0^a f(s)^2 ds - \int_M^N F(\lambda)^2 d\rho_a(\lambda) \right| &= \left| \int_{\mathbb{R} \setminus [M, N]} F(\lambda)^2 d\rho_a(\lambda) \right| \\ &\leq \frac{1}{\min\{N^2, M^2\}} \int_{\mathbb{R} \setminus [M, N]} \left(\int_0^a L(f)(x)\phi(x, \lambda)dx \right)^2 d\rho_a(\lambda) \\ &\leq \frac{1}{\min\{N^2, M^2\}} \int_{-\infty}^{+\infty} \left(\int_0^a L(f)(x)\phi(x, \lambda)dx \right)^2 d\rho_a(\lambda) \end{aligned}$$

and Parseval's identity (4.1) applied to $L(f)$ implies that

$$\left| \int_0^a f(s)^2 ds - \int_M^N F(\lambda)^2 d\rho_a(\lambda) \right| \leq \frac{1}{\min\{N^2, M^2\}} \int_0^a L(f)(s)^2 ds$$

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Letting $a \rightarrow +\infty$ along the sequence $(a_n)_{n=1}^{\infty}$ yields

$$\left| \int_0^{+\infty} f(s)^2 ds - \int_M^N F(\lambda)^2 d\rho(\lambda) \right| \leq \frac{1}{\min\{N^2, M^2\}} \int_0^{+\infty} L(f)(s)^2 ds$$

and letting $M \rightarrow -\infty, N \rightarrow +\infty$ along continuity points of ρ and each ρ_{a_n} finishes the proof. $\square$

Proposition 4.12 (Convergence of the truncated Fourier transform)

Let $f \in L^2([0, \infty))$. Let F_n be the truncated Fourier transform of f , i.e.

$$F_n(\lambda) = \int_0^n f(x)\phi(x, \lambda) dx$$

Then $(F_n)_{n=1}^{\infty}$ converges in $L^2(\mathbb{R}, \rho)$ to a limit $F \in L^2(\mathbb{R}, \rho)$, called the generalized Fourier transform of f . Moreover, Parseval's identity holds :

$$\int_0^{+\infty} f(s)^2 ds = \int_{-\infty}^{+\infty} F(\lambda)^2 d\rho(\lambda) \quad (4.20)$$

Proof. Let $(g_m)_{m=1}^{\infty} \subset C^\infty([0, \infty))$ be a sequence of smooth, compactly supported functions such that

$$\overline{\text{supp}(g_m)} \subset (0, +\infty) \quad \forall m \in \mathbb{N}, \quad \lim_{m \rightarrow +\infty} \int_0^{+\infty} |g_m - f|^2 ds = 0 \quad (4.21)$$

For each $m \in \mathbb{N}$, let $G_m(\lambda) \in L^2(\mathbb{R}, \rho)$ be the generalized Fourier transform of g_m . Observe that for any $n, m \in \mathbb{N}$, $g_n - g_m$ also has a compact support $\overline{\text{supp}(g_n - g_m)} \subset (0, +\infty)$ and its generalized Fourier transform is $G_n - G_m$.

Now, Parseval's identity (4.18) applied to g_n implies that

$$\int_0^{+\infty} g_m(s)^2 ds = \int_{-\infty}^{+\infty} G_m(\lambda)^2 d\rho(\lambda) \quad (4.22)$$

Since g_m converges to f in $(L^2([0, +\infty)), \|\cdot\|_{L^2})$, the left-hand side of (4.22) converges to $\int_0^{+\infty} f^2 ds$ as $m \rightarrow +\infty$.

Moreover, $(G_m)_{m=1}^{\infty} \subset L^2(\mathbb{R}, \rho)$ is a Cauchy sequence and thus converges to some G in $L^2(\mathbb{R}, \rho)$. Indeed, Parseval's identity (4.18) applied to $g_n - g_m$ yields

$$\int_{-\infty}^{+\infty} (G_n(\lambda) - G_m(\lambda))^2 d\rho(\lambda) = \int_0^{+\infty} (g_n(s) - g_m(s))^2 ds \rightarrow 0 \text{ as } n, m \rightarrow +\infty$$

Therefore, the right-hand side of (4.22) converges to $\int_{-\infty}^{+\infty} G^2 d\rho$ as $m \rightarrow +\infty$.

Thus, for any $f \in L^2([0, \infty))$, we have constructed $G \in L^2(\mathbb{R}, \rho)$ such that

$$\int_0^{+\infty} f(s)^2 ds = \int_{-\infty}^{+\infty} G(\lambda)^2 d\rho(\lambda)$$

It is easy to check that the constructed G is independent of the chosen sequence $(g_n)_{n=1}^\infty$ and is linear in f . We will denote it by G_f .

We now prove that for $f \in L^2([0, \infty))$, G_f is in fact the limit of $(F_n)_{n=1}^\infty$ in $L^2(\mathbb{R}, \rho)$.

Indeed, let $f \in L^2([0, \infty))$ with associated $G_f \in L^2(\mathbb{R}, \rho)$. Observe that for $f_n = f \cdot \chi_{[0, n]}$, $n \in \mathbb{N}$, G_{f_n} is exactly F_n and by linearity, G_{f-f_n} is exactly $G_f - F_n$. Thus, we obtain that

$$\int_0^{+\infty} (f(s) - f_n(s))^2 ds = \int_{-\infty}^{+\infty} (G_f(\lambda) - F_n(\lambda))^2 d\rho(\lambda)$$

Letting $n \rightarrow +\infty$ finishes the proof. $\square$

Notation

We write

$$F(\lambda) = \lim_{n \rightarrow +\infty} \int_0^n f(x) \phi(x, \lambda) dx = \int_0^{+\infty} f(x) \phi(x, \lambda) dx$$

where the convergence is in $L^2(\mathbb{R}, \rho)$.

Remark 4.13

If $f \in L^2([0, +\infty), \mathbb{C})$, then by linearity, the generalized Fourier transform

$$F(\lambda) = \lim_{n \rightarrow +\infty} \int_0^n f(x) \phi(x, \lambda) dx$$

is a well-defined $L^2(\mathbb{R}, \mathbb{C}, \rho)$ -function and Parseval's identity

$$\int_0^{+\infty} |f(s)|^2 ds = \int_{-\infty}^{+\infty} |F(\lambda)|^2 d\rho(\lambda) \quad (4.23)$$

holds.

Corollary 4.14 (Generalized Parseval's identity)

Let $f, g \in L^2([0, \infty), \mathbb{C})$ with respective generalized Fourier transform F, G . Then the generalized Parseval's identity holds :

$$\int_0^{+\infty} f(s) \overline{g(s)} ds = \int_{-\infty}^{+\infty} F(\lambda) \overline{G(\lambda)} d\rho(\lambda) \quad (4.24)$$

Proof. The generalized Fourier transform of $f \pm g$ is $F \pm G$. Apply Parseval's identity to both $f + g$ and $f - g$ and then subtract one equation from the other. $\square$

Proposition 4.15 (Convergence of the truncated inverse Fourier transform)

Let $F \in L^2(\mathbb{R}, \rho)$. Let f_n be the truncated inverse Fourier transform of F , i.e.

$$f_n(s) = \int_{-n}^n F(\mu) \phi(s, \mu) d\rho(\mu)$$

Then $(f_n)_{n=1}^\infty \subset L^2([0, +\infty))$ and this sequence converges in $L^2([0, +\infty))$ to a limit $f \in L^2([0, +\infty))$, called the generalized inverse Fourier transform of F .

Moreover, the following inequality holds :

$$\int_0^{+\infty} f(s)^2 ds \leq \int_{-\infty}^{+\infty} F(\lambda)^2 d\rho(\lambda) \quad (4.25)$$

Proof. We show that $(f_n)_{n=1}^\infty \subset L^2([0, +\infty))$ and then proves that it is a Cauchy sequence. Indeed, for any compactly supported $g \in L^2([0, +\infty))$ with Fourier transform G , Fubini-Tonelli Theorem implies for all $m \geq n \in \mathbb{N}$:

$$\begin{aligned} \int_0^{+\infty} (f_m(s) - f_n(s)) g(s) ds &= \int_0^{+\infty} \left(\int_{|n| < |\mu| \leq |m|} F(\mu) \phi(s, \mu) d\rho(\mu) \right) g(s) ds \\ &= \int_{|n| < |\mu| \leq |m|} F(\mu) \left(\int_0^{+\infty} g(s) \phi(s, \mu) ds \right) d\rho(\mu) \\ &= \int_{|n| < |\mu| \leq |m|} F(\mu) G(\mu) d\rho(\mu) \end{aligned}$$

Moreover, Cauchy-Schwarz inequality implies

$$\begin{aligned} \left| \int_0^{+\infty} (f_m(s) - f_n(s)) g(s) ds \right| &\leq \left(\int_{|n| < |\mu| \leq |m|} F(\mu)^2 d\rho(\mu) \right)^{1/2} \left(\int_{|n| < |\mu| \leq |m|} G(\mu)^2 d\rho(\mu) \right)^{1/2} \\ &\leq \left(\int_{|n| < |\mu| \leq |m|} F(\mu)^2 d\rho(\mu) \right)^{1/2} \left(\int_{-\infty}^{+\infty} G(\mu)^2 d\rho(\mu) \right)^{1/2} \end{aligned}$$

and Parseval's identity applied to g yields

$$\left| \int_0^{+\infty} (f_m(s) - f_n(s)) g(s) ds \right| \leq \left(\int_{|n| < |\mu| \leq |m|} F(\mu)^2 d\rho(\mu) \right)^{1/2} \left(\int_0^{+\infty} g(s)^2 ds \right)^{1/2}$$

Let $b > 0$ be arbitrary. Taking $g(s) = f_m(s) - f_n(s)$ if $0 \leq s \leq b$ and $g(s) = 0$ otherwise, we get

$$\int_0^b (f_m(s) - f_n(s))^2 ds \leq \left(\int_{|n| < |\mu| \leq |m|} F(\mu)^2 d\rho(\mu) \right)^{1/2} \left(\int_0^b (f_m(s) - f_n(s))^2 ds \right)^{1/2}$$

or equivalently

$$\int_0^b (f_m(s) - f_n(s))^2 ds \leq \int_{|n| < |\mu| \leq |m|} F(\mu)^2 d\rho(\mu)$$

Letting $b \rightarrow +\infty$,

$$\int_0^{+\infty} (f_m(s) - f_n(s))^2 ds \leq \int_{|n| < |\mu| \leq |m|} F(\mu)^2 d\rho(\mu) \quad (4.26)$$

Note that the above reasoning can be replicated to prove that

$$\int_0^{+\infty} (f_n(s))^2 ds \leq \int_{-n}^n F(\mu)^2 d\rho(\mu)$$

showing in particular that $(f_n)_{n=1}^\infty \subset L^2([0, +\infty))$.

Letting $n, m \rightarrow +\infty$ in (4.26) shows that $(f_n)_{n=1}^\infty \subset L^2([0, +\infty))$ is a Cauchy sequence. $\square$

Notation

We write

$$f(s) = \lim_{n \rightarrow +\infty} \int_{-n}^n F(\mu) \phi(s, \mu) d\rho(\mu)$$

where the convergence is in $L^2([0, +\infty))$.

Remark 4.16

If $F \in L^2(\mathbb{R}, \mathbb{C}, \rho)$, then by linearity, the generalized inverse Fourier transform

$$f(s) = \lim_{n \rightarrow +\infty} \int_{-n}^n F(\mu) \phi(s, \mu) d\rho(\mu)$$

is a well-defined $L^2([0, +\infty), \mathbb{C})$ -function.

Theorem 4.17 (Expansion theorem)

Let $f \in L^2([0, +\infty), \mathbb{C})$ with generalized Fourier transform F and let f_n be the truncated inverse Fourier transform of F , i.e.

$$f_n(s) = \int_{-n}^n F(\mu) \phi(s, \mu) d\rho(\mu)$$

Then $(f_n)_{n=1}^\infty$ converges to f in $L^2([0, +\infty), \mathbb{C})$.

In other words, the generalized inverse Fourier transform of F is exactly f .

Proof. Since $(f_n)_{n=1}^\infty \in L^2([0, +\infty), \mathbb{C})$, Fubini-Tonelli Theorem implies that for all compactly supported $g \in L^2([0, +\infty), \mathbb{C})$ with Fourier transform G , for all $n \in \mathbb{N}$:

$$\int_0^{+\infty} f_n(s) g(s) ds = \int_{-n}^n F(\mu) G(\mu) d\rho(\mu)$$

Moreover, the generalized Parseval's identity applied to f and g implies that

$$\int_0^{+\infty} f(s) g(s) ds = \int_{-\infty}^{+\infty} F(\mu) G(\mu) d\rho(\mu)$$

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Thus,

$$\int_0^{+\infty} (f(s) - f_n(s))g(s)ds = \int_{|\mu|>n} F(\mu)G(\mu)d\rho(\mu)$$

so that by Cauchy-Schwarz inequality and Parseval's identity applied to g :

$$\left| \int_0^{+\infty} (f(s) - f_n(s))g(s)ds \right| \leq \left(\int_{|\mu|>n} F(\mu)^2 d\rho(\mu) \right)^{1/2} \left(\int_0^{+\infty} g(s)^2 ds \right)^{1/2}$$

Let $b > 0$ be arbitrary. Taking $g(s) = f(s) - f_n(s)$ if $0 \leq s \leq b$ and $g(s) = 0$ otherwise, we finally obtain

$$\int_0^b (f(s) - f_n(s))^2 ds \leq \int_{|\mu|>n} F(\mu)^2 d\rho(\mu)$$

so that we get the desired result if we let $b \rightarrow +\infty$ and then $n \rightarrow +\infty$. $\square$

Remark 4.18

We will prove in Section 4.5 that the Fourier transform map $f \mapsto F$ is surjective and thus defines a one-to-one linear isometry between $L^2([0, +\infty), \mathbb{F})$ and $L^2(\mathbb{R}, \mathbb{F}, \rho)$.

With some additional smoothness properties, the expansion theorem holds almost everywhere or everywhere on $[0, +\infty)$.

Corollary 4.19

Let $f \in L^2([0, +\infty), \mathbb{C})$ with generalized Fourier transform F and assume that

$$\int_{-\infty}^{\infty} F(\lambda)\phi(s, \lambda)d\rho(\lambda)$$

converges absolutely (i.e. $F(\lambda)\phi(s, \lambda) \in L^1(\mathbb{R}, \mathbb{C}, d\rho)$ for any fixed $s \geq 0$). Then

$$f(s) = \int_{-\infty}^{\infty} F(\lambda)\phi(s, \lambda)d\rho(\lambda) \tag{4.27}$$

almost everywhere on $[0, +\infty)$.

Proof. Theorem 4.17 implies that

$$f(s) = \lim_{n \rightarrow +\infty} \int_{-n}^n F(\mu)\phi(s, \mu)d\rho(\mu)$$

in $L^2([0, +\infty), \mathbb{C})$ -sense, so there exists an increasing subsequence $(n_k)_{k=1}^{\infty} \subset \mathbb{N}$ such that

$$f(s) = \lim_{k \rightarrow +\infty} \int_{-n_k}^{n_k} F(\mu)\phi(s, \mu)d\rho(\mu) \text{ for almost all } s \in [0, +\infty)$$

as a pointwise limit of functions. Since $F(\lambda)\phi(s, \lambda) \in L^1(\mathbb{R}, \mathbb{C}, d\rho)$, the pointwise limit in the right-hand side is exactly

$$\int_{-\infty}^{\infty} F(\mu)\phi(s, \mu) d\rho(\mu)$$

□

Corollary 4.20

Let $f \in L^2([0, +\infty), \mathbb{C})$ be continuous with generalized Fourier transform F . Assume that

$$\int_{-\infty}^{\infty} F(\lambda)\phi(s, \lambda) d\rho(\lambda)$$

converges absolutely and uniformly on every compact subset of $[0, +\infty)$. Then

$$f(s) = \int_{-\infty}^{\infty} F(\lambda)\phi(s, \lambda) d\rho(\lambda) \quad \forall s \geq 0 \quad (4.28)$$

Proof. By Corollary 4.19, it suffices to show that

$$\hat{f}(s) = \int_{-\infty}^{\infty} F(\lambda)\phi(s, \lambda) d\rho(\lambda)$$

is continuous on $[0, +\infty)$. Indeed, let $s_0 \in [0, +\infty)$ be fixed and let $\varepsilon > 0$. By absolute and uniform convergence on $[0, s_0 + 1]$, there exists $N \in \mathbb{N}$ such that for all $s \in [0, s_0 + 1]$

$$\int_{|\lambda| > N} |F(\lambda)\phi(s, \lambda)| d\rho(\lambda) \leq \frac{\varepsilon}{4} \quad (4.29)$$

By uniform continuity of $\phi(s, \lambda)$ on $[0, s_0 + 1] \times [-N, N]$, there exists $\delta > 0$ small enough such that for all $s \in [0, s_0 + 1]$, for all $\lambda \in [-N, N]$:

$$|s - s_0| \leq \delta \Rightarrow |\phi(s, \lambda) - \phi(s_0, \lambda)| \leq \frac{\varepsilon}{4N^{1/2} \cdot (\|F\|_{L^2} + 1)} \quad (4.30)$$

Thus, for all $s \in [0, +\infty)$ such that $|s - s_0| \leq \min\{\delta, 1\}$, we have

$$\begin{aligned} |\hat{f}(s) - \hat{f}(s_0)| &= \left| \int_{-\infty}^{\infty} F(\lambda)(\phi(s, \lambda) - \phi(s_0, \lambda)) d\rho(\lambda) \right| \\ &\leq \left| \int_{-N}^N \dots d\rho(\lambda) \right| + \left| \int_{|\lambda| > N} \dots d\rho(\lambda) \right| = (1) + (2) \end{aligned}$$

The first term is bounded using Cauchy-Schwarz inequality and then our estimate (4.30) :

$$\begin{aligned} \left| \int_{-N}^N \dots d\rho(\lambda) \right| &\leq \left(\int_{-N}^N |F(\lambda)|^2 d\rho(\lambda) \right)^{1/2} \cdot \left(\int_{-N}^N (\phi(s, \lambda) - \phi(s_0, \lambda))^2 d\rho(\lambda) \right)^{1/2} \\ &\leq \varepsilon/2 \end{aligned}$$

The second term is bounded by our first estimate (4.29). This proves that $\hat{f}$ is continuous. □

Chapter 4. The general case : Transform Theory

Our final goal in this section will be to prove an integral expansion theorem for

$$\int_0^{+\infty} G(s, t, z) f(t) dt$$

when $z \in \mathbb{C} \setminus \mathbb{R}$. First, we prove that for $z \in \mathbb{C} \setminus \mathbb{R}$ fixed, the map

$$T_z : L^2([0, +\infty), \mathbb{C}) \rightarrow L^2([0, +\infty), \mathbb{C})$$

$$f(t) \mapsto (T_z f)(s) := \int_0^{+\infty} G(s, t, z) f(t) dt$$

is well-defined and continuous. This was already proven in the case $z = i$ in Corollary 2.16.

Proposition 4.21 ($T_z f$ is well-defined)

For any fixed $z \in \mathbb{C} \setminus \mathbb{R}$ and $f \in L^2([0, +\infty), \mathbb{C})$:

$$s \mapsto \int_0^{+\infty} G(s, t, z) f(t) dt$$

is well-defined, i.e. the integral converges absolutely.

Proof. We have

$$\int_0^{+\infty} G(s, t, z) f(t) dt = \beta^m(s, z) \int_0^s \eta(t, z) f(t) dt + \eta(s, z) \int_s^{+\infty} \beta^m(t, z) f(t) dt$$

which is well-defined since $\beta^m, f \in L^2([0, +\infty), \mathbb{C})$. □

Lemma 4.22

Let $z \in \mathbb{C} \setminus \mathbb{R}$ and $f \in L^2([0, +\infty), \mathbb{C})$ be fixed. For any sequence $(f_n)_{n=1}^\infty \subset L^2([0, +\infty), \mathbb{C})$ converging to f in $L^2([0, +\infty), \mathbb{C})$:

$$\lim_{n \rightarrow +\infty} \int_0^{a_n} |G_{a_n}(s, t, z) f_n(t)| dt = \int_0^{+\infty} |G(s, t, z) f(t)| dt$$

uniformly on every compact subset of $[0, +\infty)$.

Proof. Let $K \subset [0, +\infty)$ be a compact subset. By definition of G and G_{a_n} , we have

$$\int_0^{+\infty} |G(s, t, z) f(t)| dt = |\beta^m(s, z)| \int_0^s |\eta(t, z) f(t)| dt + |\eta(s, z)| \int_s^{+\infty} |\beta^m(t, z) f(t)| dt, \quad s \geq 0$$

and for n large enough,

$$\int_0^{a_n} |G_{a_n}(s, t, z) f_n(t)| dt = |\beta^{l_{an}}(s, z)| \int_0^s |\eta(t, z) f_n(t)| dt + |\eta(s, z)| \int_s^{a_n} |\beta^{l_{an}}(t, z) f_n(t)| dt, \quad s \in K$$

Moreover, we have the following hypotheses :

1. $\beta^{l_{a_n}}(s, z) \rightarrow \beta^m(s, z)$ uniformly on every compact subset of $[0, +\infty)$.
2. $\beta^{l_{a_n}}(s, z) \cdot \chi_{[0, a_n]}(s) \rightarrow \beta^m(s, z)$ in $L^2([0, +\infty), \mathbb{C})$ by Corollary 2.11.
3. $f_n \rightarrow f$ in $L^2([0, +\infty), \mathbb{C})$ by hypothesis.

which is sufficient to prove the statement. $\square$

Proposition 4.23 (T_z is continuous)

Let $z \in \mathbb{C} \setminus \mathbb{R}$ be fixed. Then for any $f \in L^2([0, +\infty), \mathbb{C})$:

$$\int_0^{+\infty} |(T_z f)(s)|^2 ds = \int_0^{+\infty} \left| \int_0^{+\infty} G(s, t, z) f(t) dt \right|^2 ds \leq \frac{1}{z_2^2} \int_0^{+\infty} |f(t)|^2 dt \quad (4.31)$$

Proof. For $a > 0$, it follows from Parseval's identity (4.1) and (4.16) applied to $f|_{[0, a]}$ that

$$\begin{aligned} \int_0^a \left| \int_0^a G_a(s, t, z) f(t) dt \right|^2 ds &= \sum_n \frac{\langle f, \phi(s, \lambda_{a, n}) \rangle_{L^2([0, a])}^2}{\alpha_{a, n}^2 \cdot |\lambda_{a, n} - z|^2} \\ &\leq \sum_n \frac{\langle f, \phi(s, \lambda_{a, n}) \rangle_{L^2([0, a])}^2}{\alpha_{a, n}^2 \cdot z_2^2} \\ &= \frac{1}{z_2^2} \int_0^a |f(t)|^2 dt \end{aligned}$$

where we used the fact that the $\lambda_{a, n}$'s are real. In particular, for $a > b > 0$:

$$\int_0^b \left| \int_0^a G_a(s, t, z) f(t) dt \right|^2 ds \leq \int_0^a \left| \int_0^a G_a(s, t, z) f(t) dt \right|^2 ds \leq \frac{1}{z_2^2} \int_0^a |f(t)|^2 dt$$

Letting $a \rightarrow +\infty$ along the sequence $(a_n)_{n=1}^\infty$, Lemma 4.22 implies

$$\int_0^b \left| \int_0^{+\infty} G(s, t, z) f(t) dt \right|^2 ds \leq \frac{1}{z_2^2} \int_0^{+\infty} |f(t)|^2 dt$$

We conclude by letting $b \rightarrow +\infty$. $\square$

Corollary 4.24

Let L be in the limit-point case and let $z \in \mathbb{C} \setminus \mathbb{R}$ be fixed. For $g \in C_{L^2}^0([0, +\infty), \mathbb{C})$,

$$u(s) = \int_0^{+\infty} G(s, t, z) g(t) dt$$

is of class $C_{L^2}^2([0, +\infty), \mathbb{C})$ and is the unique solution to

$$Lu + zu = (\eta \beta^{m(z)})_0 g = -g$$

on $[0, +\infty)$ that is square-integrable and satisfies boundary condition (2.5).

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Proof. The solution u is in $L^2([0, +\infty), \mathbb{C})$ because of Proposition 4.23. Then, we proceed as in the proof of Theorem 2.17 to show that $u \in C^2([0, +\infty), \mathbb{C})$, $Lu + iu = -g$ and $R_1 u = 0$. The unicity follows from the fact that we are in the limit-point case. $\square$

To prove the expansion theorem for $T_z f$, we will need the two following Lemmas :

Lemma 4.25

For any fixed $z \in \mathbb{C} \setminus \mathbb{R}$:

$$\begin{aligned} \lim_{n \rightarrow +\infty} \int_0^{a_n} |G_{a_n}(s, t, z)|^2 dt &= \int_0^{+\infty} |G(s, t, z)|^2 dt \\ \lim_{n \rightarrow +\infty} \int_0^{a_n} \left| \frac{dG_{a_n}}{ds}(s, t, z) \right|^2 dt &= \int_0^{+\infty} \left| \frac{dG}{ds}(s, t, z) \right|^2 dt \end{aligned}$$

uniformly on every compact subset of $[0, +\infty)$.

Proof. The proof is analogous to the proof of Lemma 4.22. $\square$

Lemma 4.26

Let $z \in \mathbb{C} \setminus \mathbb{R}$ and $K \subset [0, +\infty)$ be compact. There exists $C = C(z, K) > 0$ such that for all $n \in \mathbb{N}$ and $s \in K$:

$$\int_{-\infty}^{+\infty} \frac{\phi(s, \lambda)^2}{|\lambda - z|^2} d\rho_{a_n}(\lambda) \leq C \quad \int_{-\infty}^{+\infty} \frac{\frac{d\phi}{ds}(s, \lambda)^2}{|\lambda - z|^2} d\rho_{a_n}(\lambda) \leq C \quad (4.32)$$

Moreover, for all $s \in K$:

$$\int_{-\infty}^{+\infty} \frac{\phi(s, \lambda)^2}{|\lambda - z|^2} d\rho(\lambda) \leq C, \quad \int_{-\infty}^{+\infty} \frac{\frac{d\phi}{ds}(s, \lambda)^2}{|\lambda - z|^2} d\rho(\lambda) \leq C \quad (4.33)$$

Proof. Fix $k \in \{0, 1\}$.

For any $j, n \in \mathbb{N}$, using (4.14) (and differentiating the equality if $k = 1$) yields

$$\frac{d^k \phi}{ds^k}(s, \lambda_{a_n, j}) = (\lambda_{a_n, j} - z) \int_0^a \frac{d^k G_{a_n}}{ds^k}(s, t, z) \phi(t, \lambda_{a_n, j}) dt$$

and Parseval's Identity (4.1) applied to $f(t) = \frac{d^k G_{a_n}}{ds^k}(s, t, z)$ yields

$$\int_0^{a_n} \left| \frac{d^k G_{a_n}}{ds^k}(s, t, z) \right|^2 dt = \sum_j \frac{\frac{d^k \phi}{ds^k}(s, \lambda_{a_n, j})^2}{\alpha_{a_n, n}^2 \cdot |\lambda_{a_n, j} - z|^2} = \int_{-\infty}^{+\infty} \frac{\frac{d^k \phi}{ds^k}(s, \lambda)^2}{|\lambda - z|^2} d\rho_{a_n}(\lambda)$$

This finishes the proof of (4.32) since the left-hand side converges uniformly by Lemma 4.25.

As for (4.33), observe that for all continuity points $M < N$ of ρ and each ρ_{a_n} , for all $n \in \mathbb{N}$:

$$\int_M^N \frac{\frac{d^k \phi}{ds^k}(s, \lambda)^2}{|\lambda - z|^2} d\rho_{a_n}(\lambda) \leq C$$

Letting $n \rightarrow +\infty$, we have that for all continuity points $M < N$ of ρ and each ρ_{a_n} :

$$\int_M^N \frac{\frac{d^k \phi}{ds^k}(s, \lambda)^2}{|\lambda - z|^2} d\rho(\lambda) \leq C$$

We conclude the proof of (4.33) by letting $M \rightarrow -\infty, N \rightarrow +\infty$. $\square$

Corollary 4.27

Let $z \in \mathbb{C} \setminus \mathbb{R}$. Then there exists $C > 0$ such that

$$\int_{-\infty}^{+\infty} \frac{1}{|\lambda - z|^2} d\rho_{a_n}(\lambda) \leq C \quad \forall n \in \mathbb{N}, \quad \int_{-\infty}^{+\infty} \frac{1}{|\lambda - z|^2} d\rho(\lambda) \leq C \quad (4.34)$$

Proof. This follows from Lemma 4.26 applied to z and $K = \{0\}$. $\square$

Proposition 4.28

Let $z \in \mathbb{C} \setminus \mathbb{R}$. Then

$$\lim_{n \rightarrow +\infty} \int_{-\infty}^{+\infty} \frac{1}{|\lambda - z|^2} d\rho_{a_n}(\lambda) = \int_{-\infty}^{+\infty} \frac{1}{|\lambda - z|^2} d\rho(\lambda) \quad (4.35)$$

Proof. Let $\varepsilon > 0$. First, we show that for $K > 0$ large enough,

$$\int_{-\infty}^{-K} \frac{1}{|\lambda - z|^2} d\rho_{a_n}(\lambda) \leq \frac{\varepsilon}{8}, \quad \int_K^{+\infty} \frac{1}{|\lambda - z|^2} d\rho_{a_n}(\lambda) \leq \frac{\varepsilon}{8} \quad (4.36)$$

independently of $n \in \mathbb{N}$. Let $N \geq K$. Integrating by parts yields

$$\int_K^N \frac{1}{|\lambda - z|^2} d\rho_{a_n}(\lambda) = \int_K^N \rho_{a_n}(\lambda) d\left(-\frac{1}{|\lambda - z|^2}\right) + \frac{\rho_{a_n}(N)}{|N - z|^2} - \lim_{\substack{\lambda \rightarrow K \\ \lambda < K}} \frac{\rho_{a_n}(\lambda)}{|K - z|^2}$$

By Proposition 4.4, we can find $N_0 \geq 1$ and $1 > C > 0$ such that for all $\lambda \geq N_0$:

$$\frac{\rho_{a_n}(\lambda)}{|\lambda - z|^2} \leq \frac{1}{C\lambda^{1/2}}$$

uniformly in $n \in \mathbb{N}$. In particular, for $N \geq K \geq N_0$:

$$\frac{\rho_{a_n}(N)}{|N - z|^2} \leq \frac{1}{CN^{1/2}}, \quad \lim_{\substack{\lambda \rightarrow K \\ \lambda < K}} \frac{\rho_{a_n}(\lambda)}{|K - z|^2} \leq \frac{1}{CK^{1/2}}$$

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uniformly in $n \in \mathbb{N}$. It remains to estimate

$$\int_K^N \rho_{a_n}(\lambda) d\left(-\frac{1}{|\lambda - z|^2}\right)$$

Since $\lambda \mapsto -(|\lambda - z|)^{-2}$ is differentiable, we have

$$\int_K^N \rho_{a_n}(\lambda) d\left(-\frac{1}{|\lambda - z|^2}\right) = \int_K^N 2\rho_{a_n}(\lambda) \frac{\lambda - z_1}{|\lambda - z|^4} d\lambda \leq \tilde{C} \int_K^{+\infty} \frac{1}{\lambda^{3/2}} d\lambda$$

where $\tilde{C}$ does not depend on $n \in \mathbb{N}$. The integrals on the right-hand side are standard Lebesgue integrals. Letting $N \rightarrow +\infty$ and choosing $K \geq N_0$ large enough proves the claim.

Now, let $\varepsilon > 0$ and let $K > 0$ be as in (4.36). Fix two points of continuity $M < -K < 0 < K < N$ of ρ and each ρ_{a_n} such that

$$\int_{[M, N]^c} \frac{1}{|\lambda - z|^2} d\rho(\lambda) \leq \frac{\varepsilon}{4} \quad (4.37)$$

This is possible thanks to Corollary 4.27. For this pair (M, N) , there exists $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$:

$$\left| \int_M^N \frac{1}{|\lambda - z|^2} d\rho(\lambda) - \int_M^N \frac{1}{|\lambda - z|^2} d\rho_{a_n}(\lambda) \right| \leq \frac{\varepsilon}{2} \quad (4.38)$$

by Proposition 4.7. Finally, combining (4.36), (4.37) and (4.38), we obtain

$$\left| \int_{-\infty}^{+\infty} \frac{1}{|\lambda - z|^2} d\rho(\lambda) - \int_{-\infty}^{+\infty} \frac{1}{|\lambda - z|^2} d\rho_{a_n}(\lambda) \right| \leq \varepsilon \quad \forall n \geq n_0$$

which concludes the proof. $\square$

Theorem 4.29 (Expansion theorem for $T_z f$)

Let $z \in \mathbb{C} \setminus \mathbb{R}$ be fixed. Let $f \in L^2([0, +\infty), \mathbb{C})$. Then

$$(T_z f)(s) := \int_0^{+\infty} G(s, t, z) f(t) dt = \int_{-\infty}^{+\infty} F(\lambda) \frac{\phi(s, \lambda)}{\lambda - z} d\rho(\lambda), \quad s \geq 0 \quad (4.39)$$

where both integrals converge absolutely.

Proof. The first integral converges absolutely because of Proposition 4.21 and the second converges absolutely since $F \in L^2(\mathbb{R}, \mathbb{F}, \rho)$ and $\phi(s, \lambda)/(\lambda - z) \in L^2(\mathbb{R}, \mathbb{C}, \rho)$ by Lemma 4.26.

It suffices to prove the theorem for a smooth $f \in C^\infty([0, +\infty), \mathbb{C})$ with compact support $K := \overline{\text{supp}(f)} \subset (0, +\infty)$ and then use a standard density argument since T_z is continuous and $\phi(s, \lambda)/(\lambda - z)$ is square-integrable.

Let $s \geq 0$ be fixed and let $a > \sup K$. By (4.15), we have

$$\int_0^a G_a(s, t, z) f(t) dt = \int_{-\infty}^{+\infty} F(\lambda) \frac{\phi(s, \lambda)}{\lambda - z} d\rho_a(\lambda) \quad (4.40)$$

where

$$F(\lambda) \equiv F_a(\lambda) = \int_0^a f(x)\phi(x, \lambda)dx = \frac{1}{\lambda} \int_0^a L(f)(x)\phi(x, \lambda)dx$$

by Green's identity (see for example the proof of Proposition 4.18).

The left-hand side of (4.40) converges to $(T_z f)(s)$ as $a \rightarrow +\infty$ along the subsequence $(a_n)_{n=1}^\infty$ by Lemma 4.22. Now, we study the right-hand side. Let $M < N$ be points of continuity of ρ and each ρ_{a_n} . Write

$$\int_{-\infty}^{+\infty} F(\lambda) \frac{\phi(s, \lambda)}{\lambda - z} d\rho_a(\lambda) = \int_M^N \dots d\rho_a(\lambda) + \int_{[M, N]^c} \dots d\rho_a(\lambda) = (1) + (2)$$

Letting $a \rightarrow +\infty$ along the subsequence $(a_n)_{n=1}^\infty$, the first term converges to

$$\lim_{n \rightarrow +\infty} \int_M^N F(\lambda) \frac{\phi(s, \lambda)}{\lambda - z} d\rho_{a_n}(\lambda) = \int_M^N F(\lambda) \frac{\phi(s, \lambda)}{\lambda - z} d\rho(\lambda)$$

and then letting $M \rightarrow -\infty$, $N \rightarrow +\infty$, we get

$$\lim_{\substack{M \rightarrow -\infty \\ N \rightarrow +\infty}} \lim_{n \rightarrow +\infty} \int_M^N F(\lambda) \frac{\phi(s, \lambda)}{\lambda - z} d\rho_{a_n}(\lambda) = \int_{-\infty}^{+\infty} F(\lambda) \frac{\phi(s, \lambda)}{\lambda - z} d\rho(\lambda)$$

As for the second term, notice that by Cauchy-Schwarz inequality

$$\begin{aligned} \int_{[M, N]^c} \left| F(\lambda) \frac{\phi(s, \lambda)}{\lambda - z} \right| d\rho_a(\lambda) &\leq \left(\int_{[M, N]^c} |F(\lambda)|^2 d\rho_a(\lambda) \right)^{1/2} \cdot \left(\int_{[M, N]^c} \left| \frac{\phi(s, \lambda)}{\lambda - z} \right|^2 d\rho_a(\lambda) \right)^{1/2} \\ &\leq C^{1/2} \left(\int_{[M, N]^c} |F(\lambda)|^2 d\rho_a(\lambda) \right)^{1/2} \\ &\leq \frac{C^{1/2}}{\min\{|M|, |N|\}} \left(\int_{[M, N]^c} \left| \int_0^a L(f)(x)\phi(x, \lambda)dx \right|^2 d\rho_a(\lambda) \right)^{1/2} \\ &\leq \frac{C^{1/2}}{\min\{|M|, |N|\}} \left(\int_{-\infty}^{+\infty} \left| \int_0^a L(f)(x)\phi(x, \lambda)dx \right|^2 d\rho_a(\lambda) \right)^{1/2} \end{aligned}$$

where $C = C(z, \{s\})$ is the constant from Lemma 4.26. Thus, Parseval's identity (4.1) applied to $L(f)$ implies

$$\int_{[M, N]^c} \left| F(\lambda) \frac{\phi(s, \lambda)}{\lambda - z} \right| d\rho_a(\lambda) \leq \frac{C^{1/2}}{\min\{|M|, |N|\}} \left(\int_0^a |L(f)(s)|^2 ds \right)^{1/2} \leq \frac{C^{1/2}}{\min\{|M|, |N|\}} \left(\int_0^{+\infty} |L(f)(s)|^2 ds \right)^{1/2}$$

i.e.

$$\int_{[M, N]^c} \left| F(\lambda) \frac{\phi(s, \lambda)}{\lambda - z} \right| d\rho_a(\lambda)$$

converges uniformly in a to 0 as $M \rightarrow -\infty$, $N \rightarrow +\infty$. Thus, we have

$$\lim_{a \rightarrow +\infty} \lim_{\substack{M \rightarrow -\infty \\ N \rightarrow +\infty}} \int_{[M, N]^c} \left| F(\lambda) \frac{\phi(s, \lambda)}{\lambda - z} \right| d\rho_a(\lambda) = \lim_{\substack{M \rightarrow -\infty \\ N \rightarrow +\infty}} \lim_{a \rightarrow +\infty} \int_{[M, N]^c} \left| F(\lambda) \frac{\phi(s, \lambda)}{\lambda - z} \right| d\rho_a(\lambda) = 0$$

This concludes the proof. $\square$

Remark 4.30

A similar proof shows that for $z \in \mathbb{C} \setminus \mathbb{R}$ fixed and $f \in L^2([0, +\infty), \mathbb{C})$:

$$\int_0^{+\infty} \beta^{m(z)}(t, z) f(t) dt = \int_{-\infty}^{+\infty} F(\lambda) \frac{1}{\lambda - z} d\rho(\lambda), \quad s \geq 0$$

where both integrals converge absolutely.

4.3 The spectral measure and the m -function

In this Section 4.3, we give formulas to express $\rho(\lambda)$ in terms of $m(z)$ and vice-versa.

First recall that since $m(z)$ is the limit-point or belongs to the limit-circle, we have

$$\int_0^{+\infty} |\beta^{m(z)}(s, z)|^2 ds = \frac{\Im m(z)}{z_2} \quad (4.41)$$

by (2.19) and (2.20). Moreover, we have the following lemma :

Lemma 4.31

Let $z \in \mathbb{C} \setminus \mathbb{R}$ be fixed. Then

$$\int_0^{+\infty} |\beta^{m(z)}(s, z)|^2 ds = \int_{-\infty}^{+\infty} \frac{d\rho(\lambda)}{(z_1 - \lambda)^2 + z_2^2} \quad (4.42)$$

Proof. Let $a > 0$. Since $\phi(s, \lambda_{a,n})$ and $\beta^{l_a(z)}(s, z)$ both satisfy (2.10), Green's identity implies that

$$\int_0^a \phi(t, \lambda_{a,n}) \beta^{l_a(z)}(t, z) dt = \frac{-(\phi \beta^{l_a(z)})_0}{\lambda_{a,n} - z} = \frac{1}{\lambda_{a,n} - z}$$

and Parseval's identity (4.1) applied to $f(t) = \beta^{l_a(z)}(t, z)$ implies

$$\int_0^a |\beta^{l_a(z)}(t, z)|^2 dt = \sum_j \frac{1}{\alpha_{a,j}^2 \cdot |\lambda_{a,j} - z|^2} = \int_{-\infty}^{+\infty} \frac{1}{|\lambda - z|^2} d\rho_a(\lambda)$$

Letting $a \rightarrow +\infty$ along the subsequence $(a_n)_{n=1}^\infty$, we conclude using Corollary 2.11 and Proposition 4.35. $\square$

This leads to the following theorem :

Theorem 4.32 (Inversion formula)

Let $\mu < \lambda$ be points of continuity of ρ . Then we have the following inversion formula :

$$\lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} \int_{\mu}^{\lambda} \Im m(x + i\varepsilon) dx = \pi (\rho(\lambda) - \rho(\mu)) \quad (4.43)$$

Proof. For $\varepsilon > 0$, we have

$$\Im m(x + i\varepsilon) = \int_{-\infty}^{+\infty} \frac{\varepsilon}{(x - \sigma)^2 + \varepsilon^2} d\rho(\sigma), \quad x \in \mathbb{R}$$

Integrating the equality on $[\mu, \lambda]$ with respect to x , we get

$$\int_{\mu}^{\lambda} \Im m(x + i\varepsilon) dx = \int_{\mu}^{\lambda} \left(\int_{-\infty}^{+\infty} \frac{\varepsilon}{(x - \sigma)^2 + \varepsilon^2} d\rho(\sigma) \right) dx$$

By Tonelli's theorem,

$$\begin{aligned} \int_{\mu}^{\lambda} \Im m(x + i\varepsilon) dx &= \int_{-\infty}^{+\infty} \left(\int_{\mu}^{\lambda} \frac{\varepsilon}{(x - \sigma)^2 + \varepsilon^2} dx \right) d\rho(\sigma) \\ &= \int_{-\infty}^{+\infty} \left[\arctan\left(\frac{\lambda - \sigma}{\varepsilon}\right) - \arctan\left(\frac{\mu - \sigma}{\varepsilon}\right) \right] d\rho(\sigma) \end{aligned}$$

And by the Monotone Convergence Theorem,

$$\begin{aligned} \lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} \int_{\mu}^{\lambda} \Im m(x + i\varepsilon) dx &= \int_{-\infty}^{+\infty} \lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} \left[\arctan\left(\frac{\lambda - \sigma}{\varepsilon}\right) - \arctan\left(\frac{\mu - \sigma}{\varepsilon}\right) \right] d\rho(\sigma) \\ &= \int_{-\infty}^{+\infty} \left(\pi \cdot \chi_{(\mu, \lambda)}(\sigma) + \frac{\pi}{2} \cdot \chi_{\{\mu\}}(\sigma) + \frac{\pi}{2} \cdot \chi_{\{\lambda\}}(\sigma) \right) d\rho(\sigma) \\ &= \pi \cdot \rho((\mu, \lambda)) + \frac{\pi}{2} \cdot \rho(\{\mu\}) + \frac{\pi}{2} \cdot \rho(\{\lambda\}) \\ &= \pi (\rho(\lambda) - \rho(\mu)) \end{aligned}$$

since μ and λ are points of continuity of ρ . □

Remark 4.33

In fact, the proof shows that for any $\mu < \lambda \in \mathbb{R}$:

$$\lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} \int_{\mu}^{\lambda} \Im m(x + i\varepsilon) dx = \frac{\pi}{2} \left(\rho(\lambda) + \lim_{\substack{x \rightarrow \lambda \\ x < \lambda}} \rho(x) \right) - \frac{\pi}{2} \left(\rho(\mu) + \lim_{\substack{x \rightarrow \mu \\ x < \mu}} \rho(x) \right) \quad (4.44)$$

Theorem 4.34 (Herglotz representation of $m(z)$)

For any $z, z_0 \in \mathbb{C} \setminus \mathbb{R}$:

$$m(z) - m(z_0) = \int_{-\infty}^{+\infty} \left(\frac{1}{\lambda - z} - \frac{1}{\lambda - z_0} \right) d\rho(\lambda) \quad (4.45)$$

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Proof. Let $z_0 \in \mathbb{C} \setminus \mathbb{R}$ be fixed. Observe that

$$\lambda \mapsto \frac{1}{\lambda - z} - \frac{1}{\lambda - z_0} = \frac{z - z_0}{(\lambda - z)(\lambda - z_0)} \in L^2(\mathbb{R}, \mathbb{C}, \rho)$$

so the integral

$$\int_{-\infty}^{+\infty} \left(\frac{1}{\lambda - z} - \frac{1}{\lambda - z_0} \right) d\rho(\lambda)$$

is well-defined. Moreover,

$$z \mapsto f(z) := \int_{-\infty}^{+\infty} \left(\frac{1}{\lambda - z} - \frac{1}{\lambda - z_0} \right) d\rho(\lambda)$$

is holomorphic on $\mathbb{C} \setminus \mathbb{R}$ and we can interchange derivative and integral. Indeed, let $y_0 \in \mathbb{C} \setminus \mathbb{R}$ be fixed and let $\delta > 0$ be such that $K := \overline{B(y_0, 2\delta)} \subset \mathbb{C} \setminus \mathbb{R}$. Then for any $y \in B(y_0, \delta)$, we have

$$|y - y_0|^{-1} \cdot \left| \frac{1}{\lambda - y} - \frac{1}{\lambda - y_0} \right| = \frac{1}{|\lambda - y| \cdot |\lambda - y_0|} \leq \frac{2}{\lambda^2 + \delta^2}$$

for all $|\lambda|$ large enough, so by the Dominated Convergence Theorem

$$\begin{aligned} \lim_{y \rightarrow y_0} \frac{f(y) - f(y_0)}{y - y_0} &= \lim_{y \rightarrow y_0} \int_{-\infty}^{+\infty} (y - y_0)^{-1} \left(\frac{1}{\lambda - y} - \frac{1}{\lambda - y_0} \right) d\rho(\lambda) \\ &= \int_{-\infty}^{+\infty} \lim_{y \rightarrow y_0} (y - y_0)^{-1} \left(\frac{1}{\lambda - y} - \frac{1}{\lambda - y_0} \right) d\rho(\lambda) \\ &= \int_{-\infty}^{+\infty} \frac{1}{(\lambda - y_0)^2} d\rho(\lambda) \end{aligned}$$

Thus, $m(z) - m(z_0)$ and $f(z)$ are two holomorphic functions on $\mathbb{C} \setminus \mathbb{R}$ that have the same imaginary part. It follows from Cauchy-Riemann equations that there exists two constants $c_+, c_- \in \mathbb{R}$ such that

$$m(z) - m(z_0) = \int_{-\infty}^{+\infty} \left(\frac{1}{\lambda - z} - \frac{1}{\lambda - z_0} \right) d\rho(\lambda) + c_+ \text{ if } \Im z > 0 \quad (4.46)$$

$$m(z) - m(z_0) = \int_{-\infty}^{+\infty} \left(\frac{1}{\lambda - z} - \frac{1}{\lambda - z_0} \right) d\rho(\lambda) + c_- \text{ if } \Im z < 0 \quad (4.47)$$

Without loss of generality, $\Im z_0 > 0$. By choosing $z = z_0$ in (4.46), we find $c_+ = 0$ and by choosing $z = \overline{z_0}$ in (4.47), we deduce that $c_- = 0$ since both the integral on the right-hand side of (4.47) and $m(\overline{z_0}) - m(z_0)$ are pure imaginary numbers (recall that $m(\overline{z_0}) = \overline{m(z_0)}$). $\square$

4.4 The spectrum of L

In this Section 4.4, we define the spectrum of L , give a sufficient criterion for L to have a discrete spectrum and show that in the limit-circle case, this definition is consistent with the definition from Chapter 3 (Definition 3.1) and that the expansion Theorem 4.17 is exactly the expansion Theorem 3.11.

Definition 4.1 (Spectrum, point spectrum, continuous spectrum)

The *spectrum* of L is defined as the set

$$\rho(L) := \{\lambda \in \mathbb{R} \mid \rho \text{ is not constant in a neighborhood of } \lambda\}$$

We consider two subsets of the spectrum, namely the *point spectrum* and the *continuous spectrum*:

$$\rho_{\text{point}}(L) := \{\lambda \in \mathbb{R} \mid \lambda \text{ is a point of discontinuity of } \rho\}$$

$$\rho_{\text{cont}}(L) := \overline{\{\lambda \in \mathbb{R} \mid \rho \text{ is continuous but not constant in a neighborhood of } \lambda\}}$$

The remainder of the spectrum is called the *residual spectrum* and points in $\rho_{\text{cont}}(L) \setminus \rho_{\text{point}}(L)$ form the *essential spectrum*.

Proposition 4.35

Let $\lambda^* \in \mathbb{R}$ be fixed. Then

$$\lambda^* \in \rho_{\text{point}}(L) \Rightarrow \phi(s, \lambda^*) \in L^2([0, +\infty))$$

Proof. Assume that $\lambda^* \in \rho_{\text{point}}(L)$. Let $f \in L^2([0, +\infty))$ be compactly supported. Then its Fourier transform

$$F(\lambda) = \int_0^{+\infty} f(x) \phi(x, \lambda) dx$$

is continuous on $\mathbb{R}$ and we have

$$(\rho(\lambda^*) - \lim_{\substack{\lambda \rightarrow \lambda^* \\ \lambda < \lambda^*}} \rho(\lambda)) F(\lambda^*)^2 = \int_{\lambda^*}^{+\infty} F(\lambda)^2 d\rho(\lambda) \leq \int_{-\infty}^{+\infty} F(\lambda)^2 d\rho(\lambda) = \int_0^{+\infty} f(s)^2 ds$$

Setting $f(s) = \phi(s, \lambda^*) \cdot \chi_{[0, n]}(s)$, we get

$$(\rho(\lambda^*) - \lim_{\substack{\lambda \rightarrow \lambda^* \\ \lambda < \lambda^*}} \rho(\lambda)) \left(\int_0^n \phi(s, \lambda^*)^2 ds \right)^2 d\rho(\lambda) \leq \int_0^n \phi(s, \lambda^*)^2 ds$$

so we deduce that

$$\int_0^{+\infty} \phi(s, \lambda^*)^2 ds \leq \frac{1}{\rho(\lambda^*) - \lim_{\substack{\lambda \rightarrow \lambda^* \\ \lambda < \lambda^*}} \rho(\lambda)} < +\infty$$

by letting $n \rightarrow +\infty$. □

Remark 4.36

In the limit-point case, the converse is true. See [Tit62], Chapter 3, Section 10. Moreover, we can express the L^2 -norm of $\phi(s, \lambda^*)$ in terms of the jump discontinuity. See Corollary 4.56.

Definition 4.2

Points in $\rho_{\text{point}}(L)$ are called the *eigenvalues* of L and the associated $\phi(s, \lambda)$, $\lambda \in \rho_{\text{point}}(L)$, are called the *eigenfunctions* of L .

Proposition 4.37 (Sufficient criterion for a discrete spectrum)

Let L be a SL operator on $[0, +\infty)$ such that $\lim_{s \rightarrow +\infty} q(s) = +\infty$. Then L has a discrete spectrum (i.e. $\rho(L) = \rho_{\text{point}}(L)$).

Proof. This is a consequence of Sturm Oscillation Theorem. See [Sar98], Lemma 3.1.1 for more details. $\square$

For the rest of this Section 4.4, we assume that L is in the limit-circle case. Our goal is to show that the spectrum of L is only made of a point spectrum, which is exactly the set of eigenvalues of L as defined in Chapter 3. Moreover, we prove that $\rho(\lambda)$ coincides, up to the addition of a constant, with the function $\hat{\rho}(\lambda)$ from Remark 3.12.

Proposition 4.38

The functions

$$z \mapsto 1 - (z - i) \int_0^{+\infty} \phi(t, z) \beta^{m(i)}(t, i) dt, \quad z \mapsto m(i) + (z - i) \int_0^{+\infty} \theta(t, z) \beta^{m(i)}(t, i) dt$$

are holomorphic on $\mathbb{C}$.

Proof. First, observe that the proof of Theorem 2.28 shows that for any compact subset $K \subset \mathbb{C}$, $\|\phi(\cdot, z)\|_{L^2}$ and $\|\theta(\cdot, z)\|_{L^2}$ are uniformly bounded on K .

By the Cauchy-Schwarz inequality and Corollary 2.11, on every compact subset $K \subset \mathbb{C}$, the functions in the statement are respectively the uniform limit of the sequences of holomorphic functions

$$z \mapsto 1 - (z - i) \int_0^{a_n} \phi(t, z) \beta^{l_{a_n}(i)}(t, i) dt, \quad z \mapsto m(i) + (z - i) \int_0^{a_n} \theta(t, z) \beta^{l_{a_n}(i)}(t, i) dt$$

Morera's theorem then ensures that the limits are holomorphic. $\square$

Proposition 4.39

For any $z \in \mathbb{C}$:

$$1 - (z - i) \int_0^{+\infty} \phi(t, z) \beta^{m(i)}(t, i) dt = \lim_{s \rightarrow +\infty} -(\phi_z \beta^{m(i)})_s := -(\phi_z \beta^{m(i)})_\infty \quad (4.48)$$

Moreover, if $\{\lambda_n\}_{n=1}^\infty$ denotes the set of eigenvalues of L in the sense of Chapter 3, then for any $\lambda \in \mathbb{R}$:

$$1 - (\lambda - i) \int_0^{+\infty} \phi(t, \lambda) \beta^{m(i)}(t, i) dt = 0 \iff \lambda \in \{\lambda_n\}_{n=1}^\infty \quad (4.49)$$

In particular,

$$z \mapsto 1 - (z - i) \int_0^{+\infty} \phi(t, z) \beta^{m(i)}(t, i) dt$$

is holomorphic, non-constant and its number of zeros is at most countable.

Proof. Equation (4.48) follows from Green's identity and Equivalence (4.49) follows from the definition of an eigenvalue in Chapter 3. $\square$

Proposition 4.40

For any $z \in \mathbb{C} \setminus \mathbb{R}$:

$$\left(1 - (z - i) \int_0^{+\infty} \phi(t, z) \beta^{m(i)}(t, i) dt \right) m(z) = m(i) + (z - i) \int_0^{+\infty} \theta(t, z) \beta^{m(i)}(t, i) dt$$

Proof. Since $\beta^{l_{a_n}(z)}(s, z)$ and $\beta^{l_{a_n}(i)}(s, i)$ both satisfy (2.10), Green's identity implies that

$$(z - i) \int_0^{a_n} \beta^{l_{a_n}(z)}(t, z) \beta^{l_{a_n}(i)}(t, i) dt = -(\beta^{l_{a_n}(z)} \beta^{l_{a_n}(i)})_0 = l_{a_n}(z) - l_{a_n}(i)$$

Letting $n \rightarrow +\infty$ and using Corollary 2.11, we find

$$(z - i) \int_0^{+\infty} \beta^{m(z)}(t, z) \beta^{m(i)}(t, i) dt = m(z) - m(i) \quad (4.50)$$

Writing $\beta^{m(z)}(s, z) = \theta(s, z) + m(z)\phi(s, z)$ and since $\theta, \phi \in L^2([0, +\infty), \mathbb{C})$ in the limit-circle case, we can write

$$\int_0^{+\infty} \beta^{m(z)}(t, z) \beta^{m(i)}(t, i) dt = \int_0^{+\infty} \theta(t, z) \beta^{m(i)}(t, i) dt + m(z) \int_0^{+\infty} \phi(t, z) \beta^{m(i)}(t, i) dt$$

$\square$

Corollary 4.41

Let L be in the limit-circle case. Let $\{\lambda_n\}_{n=1}^{\infty}$ be the set of eigenvalues of L in the sense of Chapter 3. Then $m(z)$ has an analytic continuation to $\mathbb{C} \setminus \{\lambda_n\}_{n=1}^{\infty}$ such that $m(z)$ is real on $\mathbb{R} \setminus \{\lambda_n\}_{n=1}^{\infty}$ and each λ_n is either a pole or a removable singularity of $m(z)$.

Proof. By Proposition 4.39 and 4.40, the meromorphic function

$$z \mapsto \frac{m(i) + (z - i) \int_0^{+\infty} \theta(t, z) \beta^{m(i)}(t, i) dt}{1 - (z - i) \int_0^{+\infty} \phi(t, z) \beta^{m(i)}(t, i) dt}$$

extends $m(z)$ holomorphically on $\mathbb{R} \setminus \{\lambda_n\}_{n=1}^{\infty}$.

Therefore, $m(z)$ is an holomorphic function on $\mathbb{C} \setminus \{\lambda_n\}_{n=1}^{\infty}$ such that $m(\bar{z}) = \overline{m(z)}$ and $\Im m(z) > 0$ if $\Im z > 0$. By continuity of $\Im m(z)$, $m(z)$ must be real on $\mathbb{R} \setminus \{\lambda_n\}_{n=1}^{\infty}$. Moreover, if λ_n is

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not a removable singularity, it follows from the Argument Principle (see [Sim04], Proposition 13.8.1) that λ_n is a pole. $\square$

Corollary 4.42

For any interval $[\mu, \lambda] \subset \mathbb{R} \setminus \{\lambda_n\}_{n=1}^\infty$, $\rho(\lambda)$ is constant on $[\mu, \lambda]$.

In particular, each discontinuity point of $\rho(\lambda)$ belongs to $\{\lambda_n\}_{n=1}^\infty$.

Proof. This follows from Theorem 4.32 and the fact that $m(z)$ has a well-defined continuous extension on a neighborhood of $[\mu, \lambda]$ and is real-valued on $[\mu, \lambda]$. $\square$

Corollary 4.43

The set of discontinuities of $\rho(\lambda)$ is exactly $\{\lambda_n\}_{n=1}^\infty$.

Moreover, for any $n \in \mathbb{N}$:

$$\rho(\lambda_n) - \lim_{\substack{\lambda \rightarrow \lambda_n \\ \lambda < \lambda_n}} \rho(\lambda) = \frac{1}{\|\phi(s, \lambda_n)\|_{L^2}^2} \quad (4.51)$$

Proof. Let $\{\mu_n\}_{n=1}^M \subset \{\lambda_n\}_{n=1}^\infty$, $M \in \mathbb{N} \cup \{+\infty\}$, be the set of discontinuities of $\rho(\lambda)$. Then any $f \in L^2([0, +\infty), \mathbb{C})$ can be represented as the following limit in $L^2([0, +\infty), \mathbb{C})$:

$$\begin{aligned} f &= \lim_{n \rightarrow +\infty} \int_{-n}^n F(\lambda) \phi(s, \lambda) d\rho(\lambda) = \lim_{k \rightarrow M} \sum_{n=1}^k F(\mu_n) \phi(s, \mu_n) (\rho(\mu_n) - \lim_{\substack{\mu \rightarrow \mu_n \\ \mu < \mu_n}} \rho(\mu)) \\ &= \lim_{k \rightarrow M} \sum_{n=1}^k \alpha_n \phi(s, \mu_n), \quad \alpha_n := \langle f, \phi(s, \mu_n) \rangle_{L^2} \cdot (\rho(\mu_n) - \lim_{\substack{\mu \rightarrow \mu_n \\ \mu < \mu_n}} \rho(\mu)) \end{aligned} \quad (4.52)$$

where $F \in L^2(\mathbb{R}, \mathbb{C}, d\rho)$ denotes the Fourier transform of f . Moreover, the orthogonality of the eigenfunctions $\{\phi(s, \mu_n)\}_{n=1}^M$ implies that

$$\alpha_n = \frac{\langle f, \phi(s, \mu_n) \rangle_{L^2}}{\|\phi(s, \mu_n)\|_{L^2}^2} \quad \forall n = 1, 2, \dots, M$$

Assume that there is some $n \in \mathbb{N}$ such that $\lambda_n \notin \{\mu_m\}_{m=1}^M$. Then taking $f = \phi(s, \lambda_n)$ in (4.52), we deduce $f \equiv 0$ since the eigenfunction $\phi(s, \lambda_n)$ is orthogonal to the other eigenfunctions $\{\phi(s, \mu_m)\}_{m=1}^M$, which is a contradiction. $\square$

Corollary 4.44

In the limit-circle case, the function $\rho(\lambda)$ coincides, up to the addition of a constant, with the function $\hat{\rho}(\lambda)$ from Remark 3.12.

Corollary 4.45 (Orthogonality of eigenfunctions)

Let L be in the limit-point or limit-circle case. Then for any eigenvalues $\lambda \neq \mu \in \rho_{\text{point}}(L)$, the eigenfunctions $\phi(s, \lambda)$ and $\phi(s, \mu)$ are $L^2([0, +\infty))$ -orthogonal.

Proof. In the limit-circle case, this follows from Corollary 4.43 and Proposition 3.8.

In the limit-point case, since $\phi(s, \lambda)$ and $\phi(s, \mu)$ are respectively the unique square-integrable solutions to $Lu + iu = (i - \lambda)u$ and $Lu + iu = (i - \mu)u$ satisfying (2.5), we have

$$\begin{aligned}\phi(s, \lambda) &= (\lambda - i) \int_0^{+\infty} G(s, t, i) \phi(t, \lambda) dt \\ \phi(s, \mu) &= (\mu - i) \int_0^{+\infty} G(s, t, i) \phi(t, \mu) dt\end{aligned}$$

Then, it suffices to repeat the proof of Proposition 3.8. □

Corollary 4.46

If L has a discrete spectrum, the expansion Theorem 4.17 takes a discrete form as in the compact case or in the limit-circle case. More precisely, the set of normalized eigenfunctions of L forms an orthonormal basis of $L^2([0, +\infty), \mathbb{F})$.

4.5 Surjectivity of the Fourier transform

In this Section 4.5, we prove that the Fourier transform is surjective and hence defines a bijective linear isometry between $L^2([0, +\infty), \mathbb{F})$ and $L^2(\mathbb{R}, \mathbb{F}, \rho)$. This will be a consequence of a reverse generalized Parseval's identity. In order to prove this identity, we will need multiple technical lemmas that allows us to prove the identity for a dense subset of functions. Our main reference for this chapter is [Tit62].

Proposition 4.47

For any fixed $v_1 < v_2 \in \mathbb{R}$,

$$\int_{v_1}^{v_2} \Im m(x + i\varepsilon) dx$$

is uniformly bounded with respect to $\varepsilon \in (0, 1]$.

Proof. This follows from Equation (4.44). □

Corollary 4.48

For any $v_1 < v_2 \in \mathbb{R}$, there exists $C > 0$ such that for all $0 < \varepsilon \leq 1$, we have

$$\int_{v_1}^{v_2} |m(x + i\varepsilon)| dx \leq \frac{C}{\varepsilon^{1/2}}$$

Proof. Recall that for any $z \in \mathbb{C} \setminus \mathbb{R}$:

$$\int_0^{+\infty} |\beta^{m(z)}(s, z)|^2 ds = \frac{\Im m(z)}{z_2}$$

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Using this, Equation (4.50) and Cauchy-Schwarz inequality implies that for any $z \in \mathbb{C} \setminus \mathbb{R}$:

$$|m(z)| \leq |m(i)| + |z - i| \cdot \frac{|\Im m(z)|^{1/2}}{|z_2|^{1/2}} \cdot |\Im m(i)|^{1/2}$$

Thus, for $0 < \varepsilon \leq 1$, we obtain

$$\begin{aligned} \int_{v_1}^{v_2} |m(x + i\varepsilon)| dx &\leq C_1 + C_2 \int_{v_1}^{v_2} \frac{\Im m(x + i\varepsilon)^{1/2}}{\varepsilon^{1/2}} dx \\ &\leq C_1 + C_2 (v_2 - v_1)^{1/2} \left(\int_{v_1}^{v_2} \frac{\Im m(x + i\varepsilon)}{\varepsilon} dx \right)^{1/2} \\ &\leq \frac{C_3}{\varepsilon^{1/2}} \left(\int_{v_1}^{v_2} \Im m(x + i\varepsilon) dx \right)^{1/2} \end{aligned}$$

for some positive constant C_3 depending only on v_1 , v_2 and $m(i)$.

We conclude by Proposition 4.47. □

Proposition 4.49

Let

$$k(\lambda) := \lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} \int_0^\lambda \Im m(x + i\varepsilon) dx, \quad \lambda \in \mathbb{R}$$

which is well-defined, non-decreasing and right-continuous by Equation (4.44). Let $a < b$ be points of continuity of $\rho(\lambda)$. Then for any continuous function $f \in C^0([a, b])$, we have

$$\int_a^b f(\lambda) dk(\lambda) = \pi \int_a^b f(\lambda) d\rho(\lambda)$$

Proof. Observe that by Equation (4.44), $k(\lambda)$ and $\rho(\lambda)$ have the same points of continuity. Moreover, for any points of continuity $a < b$ of $\rho(\lambda)$, we also have

$$k([a, b]) = \pi \cdot \rho([a, b])$$

Approximate f by a sequence of step functions as in the proof of the Theorem 4.7 and conclude with the above observation. □

Lemma 4.50

For any $u \in \mathbb{R}$ and $s \geq 0$:

$$\lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} \int_0^u \Im(\beta^{m(x+i\varepsilon)}(s, x + i\varepsilon)) dx = \int_0^u \phi(s, \lambda) dk(\lambda) \quad (4.53)$$

In particular, if $u_1 < u_2$ are points of continuity of $\rho(\lambda)$, we have

$$\lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} \int_{u_1}^{u_2} \Im(\beta^{m(x+i\varepsilon)}(s, x + i\varepsilon)) dx = \pi \int_{u_1}^{u_2} \phi(s, \lambda) d\rho(\lambda) \quad (4.54)$$

Proof. Let $u \in \mathbb{R}$ and $s \geq 0$ be fixed. For any $z \in \mathbb{C} \setminus \mathbb{R}$,

$$\Im(\beta^{m(z)}(s, z)) = \Im(\theta(s, z)) + \Re(m(z))\Im(\phi(s, z)) + \Im(m(z))\Re(\phi(s, z))$$

so we can separate

$$\int_0^u \Im(\beta^{m(x+i\varepsilon)}(s, x+i\varepsilon)) dx$$

into three parts. Since $\theta(s, z)$ and $\phi(s, z)$ are holomorphic when s is fixed and are real when z is real, the mean value theorem implies that for any $x \in [0, u]$ and $\varepsilon \in [0, 1]$:

$$|\Im(\theta(s, x+i\varepsilon)) - 0| \leq C\varepsilon, \quad |\Im(\phi(s, x+i\varepsilon)) - 0| \leq C\varepsilon \quad (4.55)$$

where

$$C := \max\left\{\max_{z \in [0, u] \times [0, 1]} |\theta_z(s, z)|, \max_{z \in [0, u] \times [0, 1]} |\phi_z(s, z)|\right\}$$

Thus,

$$\lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} \int_0^u \Im(\theta(s, x+i\varepsilon)) dx = 0$$

and since

$$\left| \int_0^u \Re(m(x+i\varepsilon))\Im(\phi(s, x+i\varepsilon)) dx \right| \leq \left(\int_0^u |m(x+i\varepsilon)|^2 dx \right)^{1/2} \cdot \left(\int_0^u |\Im(\phi(s, x+i\varepsilon))|^2 dx \right)^{1/2}$$

we also have

$$\lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} \int_0^u \Re(m(x+i\varepsilon))\Im(\phi(s, x+i\varepsilon)) dx = 0$$

by Corollary 4.48 and estimate (4.55).

Finally, we show that

$$\lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} \int_0^u \Im(m(x+i\varepsilon))\Re(\phi(s, x+i\varepsilon)) dx = \int_0^u \phi(s, \lambda) dk(\lambda)$$

Indeed, integration by parts yields

$$\begin{aligned} \int_0^u \Im(m(x+i\varepsilon))\Re(\phi(s, x+i\varepsilon)) dx &= \left[\left(\int_0^x \Im m(t+i\varepsilon) dt \right) \Re(\phi(s, x+i\varepsilon)) \right]_0^{x=u} \\ &\quad - \int_0^u \left(\int_0^x \Im m(t+i\varepsilon) dt \right) \Re(\phi_z(s, x+i\varepsilon)) dx \end{aligned}$$

By Corollary 4.47 and the Dominated Convergence Theorem, we can interchange limit and integral in the right-hand side when letting $\varepsilon \rightarrow 0$, $\varepsilon > 0$. Thus,

$$\begin{aligned} \lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} \int_0^u \Im(m(x+i\varepsilon))\Re(\phi(s, x+i\varepsilon)) dx &= [k(x)\phi(s, x)]_0^{x=u} - \int_0^u k(x)\phi_z(s, x) dx \\ &= \int_0^u \phi(s, \lambda) dk(\lambda) \end{aligned}$$

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using integration by parts once again. □

Lemma 4.51

For any compact subset $K \subset \mathbb{R}$,

$$\int_0^{+\infty} \left| \int_0^u \Im(\beta^{m(x+i\varepsilon)}(t, x+i\varepsilon)) dx \right|^2 dt$$

is uniformly bounded with respect to $u \in K$ and $\varepsilon \in (0, 1]$.

Proof. Let $z \in \mathbb{C} \setminus \mathbb{R}$ and $a > 0$. Since $\phi(s, \lambda_{a,n})$ and $\beta^{l_a(z)}(s, z)$ both satisfy (2.10), Green's identity implies that

$$\int_0^a \phi(t, \lambda_{a,n}) \beta^{l_a(z)}(t, z) dt = \frac{-(\phi \beta^{l_a(z)})_0}{\lambda_{a,n} - z} = \frac{1}{\lambda_{a,n} - z}$$

Letting $z = x + i\varepsilon$, $\varepsilon > 0$, and taking the imaginary part, we get

$$\int_0^a \phi(t, \lambda_{a,n}) \Im(\beta^{l_a(x+i\varepsilon)}(t, x+i\varepsilon)) dt = \frac{\varepsilon}{(\lambda_{a,n} - x)^2 + \varepsilon^2} \quad (4.56)$$

Fix $K \subset \mathbb{R}$ a compact subset and let $u \in K$. Integrating (4.56) on $[0, u]$ with respect to x and using Fubini-Tonelli Theorem yields

$$\int_0^a \phi(t, \lambda_{a,n}) \left(\int_0^u \Im(\beta^{l_a(x+i\varepsilon)}(t, x+i\varepsilon)) dx \right) dt = \int_0^u \frac{\varepsilon}{(\lambda_{a,n} - x)^2 + \varepsilon^2} dx$$

Parseval's identity (4.1) then implies

$$\begin{aligned} \int_0^a \left| \int_0^u \Im(\beta^{l_a(x+i\varepsilon)}(t, x+i\varepsilon)) dx \right|^2 dt &= \sum_n \frac{1}{\alpha_{a,n}^2} \int_0^u \frac{\varepsilon}{(\lambda_{a,n} - x)^2 + \varepsilon^2} dx \\ &= \int_{-\infty}^{+\infty} \left(\int_0^u \frac{\varepsilon}{(\lambda - x)^2 + \varepsilon^2} dx \right) d\rho_a(\lambda) \\ &= \int_{-\infty}^{+\infty} \left[\arctan\left(\frac{u - \lambda}{\varepsilon}\right) - \arctan\left(\frac{0 - \lambda}{\varepsilon}\right) \right] d\rho_a(\lambda) \end{aligned}$$

In particular, for $0 < b < a$, we have

$$\int_0^b \left| \int_0^u \Im(\beta^{l_a(x+i\varepsilon)}(t, x+i\varepsilon)) dx \right|^2 dt \leq \int_{-\infty}^{+\infty} \left[\arctan\left(\frac{u - \lambda}{\varepsilon}\right) - \arctan\left(\frac{0 - \lambda}{\varepsilon}\right) \right] d\rho_a(\lambda)$$

Let $0 < \varepsilon \leq 1$ and $b > 0$ be fixed. Since $\Im(\beta^{l_{a_n}(x+i\varepsilon)}(t, x+i\varepsilon)) \cdot \chi_{[0, a_n]}(t)$ converges uniformly to $\Im(\beta^{m(x+i\varepsilon)}(t, x+i\varepsilon))$ with respect to $t \in [0, b]$, $x \in [0, u]$, we have

$$\lim_{n \rightarrow +\infty} \int_0^b \left| \int_0^u \Im(\beta^{l_{a_n}(x+i\varepsilon)}(t, x+i\varepsilon)) dx \right|^2 dt = \int_0^b \left| \int_0^u \Im(\beta^{m(x+i\varepsilon)}(t, x+i\varepsilon)) dx \right|^2 dt$$

Moreover, there exists $C = C(K) > 0$ and $N \in \mathbb{N}$ such that for $|\lambda| > N$ and $0 < \varepsilon \leq 1$, we have

$$\left| \arctan\left(\frac{u-\lambda}{\varepsilon}\right) - \arctan\left(\frac{0-\lambda}{\varepsilon}\right) \right| \leq \frac{C}{|\lambda|^2 + 1}$$

Therefore, a similar argument to the proof of Proposition 4.28 shows that

$$\begin{aligned} & \lim_{n \rightarrow +\infty} \int_{-\infty}^{+\infty} \left[\arctan\left(\frac{u-\lambda}{\varepsilon}\right) - \arctan\left(\frac{0-\lambda}{\varepsilon}\right) \right] d\rho_{a_n}(\lambda) \\ &= \lim_{n \rightarrow +\infty} \int_{-\infty}^{+\infty} \left[\arctan\left(\frac{u-\lambda}{\varepsilon}\right) - \arctan\left(\frac{0-\lambda}{\varepsilon}\right) \right] d\rho(\lambda) \end{aligned}$$

Thus, we have proved that for any $0 < \varepsilon \leq 1$ and $b > 0$:

$$\int_0^b \left| \int_0^u \Im(\beta^{m(x+i\varepsilon)}(t, x+i\varepsilon)) dx \right|^2 dt \leq \int_{-\infty}^{+\infty} \left[\arctan\left(\frac{u-\lambda}{\varepsilon}\right) - \arctan\left(\frac{0-\lambda}{\varepsilon}\right) \right] d\rho(\lambda)$$

Letting $b \rightarrow +\infty$ and then noticing that the right-hand side increases as $\varepsilon \rightarrow 0, \varepsilon > 0$, we have

$$\int_0^{+\infty} \left| \int_0^u \Im(\beta^{m(x+i\varepsilon)}(t, x+i\varepsilon)) dx \right|^2 dt \leq \lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} \int_{-\infty}^{+\infty} \left[\arctan\left(\frac{u-\lambda}{\varepsilon}\right) - \arctan\left(\frac{0-\lambda}{\varepsilon}\right) \right] d\rho(\lambda)$$

for any $\varepsilon \in (0, 1]$. This concludes the proof since the right-hand side is uniformly bounded with respect to $u \in K$ (see the proof of Theorem 4.32). $\square$

Corollary 4.52

For any $u \in \mathbb{R}$, for any $g \in L^2([0, +\infty))$, we have

$$\lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} \int_0^{+\infty} g(t) \left(\int_0^u \Im(\beta^{m(x+i\varepsilon)}(t, x+i\varepsilon)) dx \right) dt = \int_0^{+\infty} g(t) \left(\int_0^u \phi(t, \lambda) dk(\lambda) \right) dt \quad (4.57)$$

Proof. Let $u \in \mathbb{R}$ be fixed. The following family of functions indexed by $\varepsilon \in (0, 1]$

$$f_\varepsilon(s) := \int_0^u \Im(\beta^{m(x+i\varepsilon)}(s, x+i\varepsilon)) dx, \quad s \geq 0$$

converges pointwisely as $\varepsilon \rightarrow 0, \varepsilon > 0$, to

$$\lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} \int_0^u \Im(\beta^{m(x+i\varepsilon)}(s, x+i\varepsilon)) dx = \int_0^u \phi(s, \lambda) dk(\lambda) =: f(s), \quad s \geq 0$$

by Lemma 4.50.

Moreover, the family $(f_\varepsilon(s))_{0 < \varepsilon \leq 1}$ is uniformly bounded with respect to the $L^2([0, +\infty))$ -norm by Lemma 4.51. In particular, Fatou's Lemma implies that the pointwise limit $f(s)$ is square-integrable.

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We now prove that for any $g \in L^2([0, +\infty))$:

$$\lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} \int_0^{+\infty} |g(t)f_\varepsilon(t) - g(t)f(t)| dt = 0$$

In fact, it suffices to prove the equality for continuous and compactly supported $g \in C^0([0, +\infty))$ and then use a standard density argument. So assume that g is continuous and compactly supported, say $\text{supp}(g) \subset [-K, K]$ for some $K > 0$.

Define

$$C := \max\{ \sup_{0 < \varepsilon \leq 1} \|f_\varepsilon\|_{L^2([0, +\infty))}, \|f\|_{L^2([0, +\infty))} \}$$

Let $\delta > 0$ be arbitrary. By Egorov's Theorem, there exists a Lebesgue measurable subset $A \subset [-K, K]$ for which

$$\lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} f_\varepsilon(s) = f(s) \text{ uniformly on } [-K, K] \setminus A, \quad \int_A g(t)^2 dt \leq \frac{\delta}{4C}$$

Let $\gamma > 0$ be small enough so that

$$0 < \varepsilon \leq \gamma \implies \sup_{s \in [-K, K] \setminus A} |f_\varepsilon(s) - f(s)| \leq \frac{\delta}{4K \cdot (\|g\|_\infty + 1)}$$

Thus, for $0 < \varepsilon \leq \gamma$, we have

$$\begin{aligned} \int_0^{+\infty} |g(t)f_\varepsilon(t) - g(t)f(t)| dt &= \int_{[-K, K] \setminus A} |g(t)f_\varepsilon(t) - g(t)f(t)| dt + \int_A |g(t)f_\varepsilon(t) - g(t)f(t)| dt \\ &\leq \|g\|_\infty \int_{[-K, K] \setminus A} |f_\varepsilon(t) - f(t)| dt \\ &\quad + \left(\int_A g(t)^2 dt \right)^{1/2} \left(\int_A (f_\varepsilon(t) - f(t))^2 dt \right)^{1/2} \\ &\leq \frac{\delta}{2} + \left(\int_A g(t)^2 dt \right)^{1/2} (\|f_\varepsilon\|_{L^2([0, +\infty))} + \|f\|_{L^2([0, +\infty))}) \\ &\leq \delta \end{aligned}$$

where we used Cauchy-Schwarz inequality and the triangle inequality for $L^2([0, +\infty))$ -norms. Since $\delta > 0$ was arbitrary, this concludes the proof. $\square$

Lemma 4.53

For $u > 0$, let

$$\omega(s, u) := \int_0^u \phi(s, \lambda) d\rho(\lambda) \in L^2([0, +\infty))$$

be the inverse Fourier transform of $\chi_{[0, u]}(\lambda)$. For any points of continuity $u_1 \leq u_2$ and $v_1 \leq v_2 \in \mathbb{R}$ of $\rho(\lambda)$, we have

$$\int_0^{+\infty} (\omega(s, u_2) - \omega(s, u_1))(\omega(s, v_2) - \omega(s, v_1)) ds = \rho(w_2) - \rho(w_1) \quad (4.58)$$

where $w_1 = \max\{u_1, v_1\}$, $w_2 = \min\{u_2, v_2\}$.

Proof. For $a > 0$, define

$$\omega_a(s, u) := \int_0^u \phi(s, \lambda) d\rho_a(\lambda) \in L^2([0, +\infty))$$

By Fubini-Tonelli Theorem, we have that

$$\begin{aligned} \int_0^a (\omega_a(t, u_2) - \omega_a(t, u_1)) \beta^{l_a(z)}(t, z) dt &= \int_{u_1}^{u_2} \left(\int_0^a \phi(s, \lambda) \beta^{l_a(z)}(t, z) dt \right) d\rho_a(\lambda) \\ &= \sum_{n: u_1 \leq \lambda_{a,n} \leq u_2} \frac{1}{\alpha_{a,n}^2} \int_0^a \phi(s, \lambda_{a,n}) \beta^{l_a(z)}(t, z) dt \end{aligned}$$

for any $a > 0$ and $z \in \mathbb{C} \setminus \mathbb{R}$. Since $\phi(s, \lambda_{a,n})$ and $\beta^{l_a(z)}(s, z)$ both satisfy (2.10), Green's identity implies that the right-hand side is exactly

$$\begin{aligned} \int_0^a (\omega_a(t, u_2) - \omega_a(t, u_1)) \beta^{l_a(z)}(t, z) dt &= \sum_{n: u_1 \leq \lambda_{a,n} \leq u_2} \frac{-(\phi \beta^{l_a(z)})_0}{\alpha_{a,n}^2 \cdot (\lambda_{a,n} - z)} \\ &= \sum_{n: u_1 \leq \lambda_{a,n} \leq u_2} \frac{1}{\alpha_{a,n}^2 \cdot (\lambda_{a,n} - z)} \\ &= \int_{u_1}^{u_2} \frac{d\rho_a(\lambda)}{\lambda - z} \end{aligned}$$

Letting $a \rightarrow +\infty$ along the subsequence $(a_n)_{n=1}^\infty$, Corollary 2.11 implies that

$$\int_0^{+\infty} (\omega(t, u_2) - \omega(t, u_1)) \beta^{m(z)}(t, z) dt = \int_{u_1}^{u_2} \frac{d\rho(\lambda)}{\lambda - z} \quad (4.59)$$

for any $z \in \mathbb{C} \setminus \mathbb{R}$. Taking the imaginary part, this becomes

$$\int_0^{+\infty} (\omega(t, u_2) - \omega(t, u_1)) \Im(\beta^{m(z)}(t, z)) dt = \int_{u_1}^{u_2} \frac{z_2}{(\lambda - z_1)^2 + z_2^2} d\rho(\lambda) \quad (4.60)$$

We write $z = x + i\varepsilon$, $\varepsilon > 0$, and integrate (4.60) on $[v_1, v_2]$ with respect to x . The same argument

as in the proof of Theorem 4.32 shows that

$$\lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} \int_{v_1}^{v_2} \int_{u_1}^{u_2} \frac{\varepsilon}{(\lambda - x)^2 + \varepsilon^2} d\rho(\lambda) dx = \pi(\rho(w_2) - \rho(w_1))$$

Moreover, by Fubini-Tonelli Theorem :

$$\begin{aligned} & \int_{v_1}^{v_2} \int_0^{+\infty} (\omega(t, u_2) - \omega(t, u_1)) \Im(\beta^{m(x+i\varepsilon)}(t, x+i\varepsilon)) dt dx \\ &= \int_0^{+\infty} (\omega(t, u_2) - \omega(t, u_1)) \left(\int_{v_1}^{v_2} \Im(\beta^{m(x+i\varepsilon)}(t, x+i\varepsilon)) dx \right) dt \end{aligned}$$

Finally, Corollary 4.52 implies that

$$\begin{aligned} & \lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} \int_0^{+\infty} (\omega(t, u_2) - \omega(t, u_1)) \left(\int_{v_1}^{v_2} \Im(\beta^{m(x+i\varepsilon)}(t, x+i\varepsilon)) dx \right) dt \\ &= \lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} \int_0^{+\infty} (\omega(t, u_2) - \omega(t, u_1)) (\omega(t, v_2) - \omega(t, v_1)) dt \end{aligned}$$

□

Theorem 4.54 (Reverse generalized Parseval's identity)

Let $F, G \in L^2(\mathbb{R}, \mathbb{C}, d\rho)$ with respective generalized inverse Fourier transform f, g . Then

$$\int_{-\infty}^{+\infty} F(\lambda) \overline{G(\lambda)} d\rho(\lambda) = \int_0^{+\infty} f(s) \overline{g(s)} ds$$

Proof. Lemma 4.52 implies that the reverse generalized Parseval's identity holds when F and G are of the form $\chi_{[u,v]}(t)$ for points of continuity $u \leq v$ of $\rho(\lambda)$. Such functions are dense in $L^2(\mathbb{R}, \mathbb{C}, \rho)$. Let $(F_n)_{n=1}^\infty, (G_n)_{n=1}^\infty \subset L^2(\mathbb{R}, \mathbb{C}, d\rho)$ be sequences of such functions converging respectively to F and G in $L^2(\mathbb{R}, \mathbb{C}, d\rho)$. By inequality (4.25), the associated sequences of inverse Fourier transform $(f_n)_{n=1}^\infty, (g_n)_{n=1}^\infty \subset L^2([0, +\infty), \mathbb{C})$ converges respectively to f and g in $L^2([0, +\infty), \mathbb{C})$. Thus, for every $n \in \mathbb{N}$, we have

$$\int_{-\infty}^{+\infty} F_n(\lambda) \overline{G_n(\lambda)} d\rho(\lambda) = \int_0^{+\infty} f_n(s) \overline{g_n(s)} ds$$

and it is easy to check that the theorem is proved by letting $n \rightarrow +\infty$. □

Corollary 4.55 (Surjectivity of the Fourier transform)

Let $F \in L^2(\mathbb{R}, \mathbb{C}, d\rho)$ with generalized inverse Fourier transform $f \in L^2([0, +\infty), \mathbb{C})$. Then the generalized Fourier transform of f is exactly F .

Proof. Let $\hat{F}$ be the Fourier transform of f . By hypothesis and by Theorem 4.17, $F - \hat{F}$ has inverse Fourier transform $0 \in L^2([0, +\infty), \mathbb{C})$. By the reverse Parseval's identity (Theorem 4.54),

we have that

$$\int_{-\infty}^{+\infty} |F(\lambda) - \hat{F}(\lambda)|^2 d\rho(\lambda) = \int_0^{+\infty} 0^2 ds = 0$$

which concludes the proof. $\square$

Corollary 4.56 (Jump discontinuities of $\rho(\lambda)$)

Let μ^* be an eigenvalue of L , i.e. a point of discontinuity of $\rho(\lambda)$. Then the Fourier transform of $\phi(s, \mu^*)$ is exactly

$$\lim_{n \rightarrow +\infty} \int_0^n \phi(t, \mu^*) \phi(t, \lambda) dt = (\rho(\mu^*) - \lim_{\substack{x \rightarrow \mu^* \\ x < \mu^*}} \rho(x))^{-1} \chi_{\{\mu^*\}}(\lambda) \quad (4.61)$$

In particular,

$$\int_0^{+\infty} \phi(t, \mu^*)^2 dt = (\rho(\mu^*) - \lim_{\substack{x \rightarrow \mu^* \\ x < \mu^*}} \rho(x))^{-1} \quad (4.62)$$

Proof. The inverse Fourier transform of $\chi_{\{\mu^*\}}(\lambda)$ is

$$\int_{\{\mu^*\}} \phi(s, \lambda) d\rho(\lambda) = (\rho(\mu^*) - \lim_{\substack{x \rightarrow \mu^* \\ x < \mu^*}} \rho(x)) \phi(s, \mu^*)$$

Taking the Fourier transform, Corollary 4.55 proves (4.61).

As for (4.62), take $(n_k)_{k=1}^{\infty} \subset \mathbb{N}$ an increasing sequence such that

$$\lim_{k \rightarrow +\infty} \int_0^{n_k} \phi(t, \mu^*) \phi(t, \lambda) dt = (\rho(\mu^*) - \lim_{\substack{x \rightarrow \mu^* \\ x < \mu^*}} \rho(x)) \phi(s, \mu^*)$$

pointwisely ρ -almost everywhere on $\mathbb{R}$. Since $\{\mu^*\}$ has non-zero measure with respect to $d\rho$, we can take $\lambda = \mu^*$ and conclude. $\square$

Corollary 4.57

For any fixed $s \geq 0$ and $z \in \mathbb{C} \setminus \mathbb{R}$, the Fourier transform of $G(s, t, z)$ (as a function of t) is exactly

$$\lim_{n \rightarrow +\infty} \int_0^n G(s, t, z) \phi(t, \lambda) dt = \frac{\phi(s, \lambda)}{\lambda - z}$$

Proof. Let $g(s, \lambda, z)$ be the Fourier transform of $G(s, t, z)$. By Theorem 4.29, the generalized Parseval's identity (Corollary 4.24) and the surjectivity of the Fourier transform (Corollary 4.55), we have

$$\int_{-\infty}^{+\infty} F(\lambda) g(s, \lambda, z) d\rho(\lambda) = \int_{-\infty}^{+\infty} F(\lambda) \frac{\phi(s, \lambda)}{\lambda - z} d\rho(\lambda)$$

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for any $F \in L^2(\mathbb{R}, \mathbb{C}, \rho)$. The fundamental lemma of calculus of variations (see [Dac09], Theorem 1.24) allows us to conclude. $\square$

Corollary 4.58

For any fixed $s \geq 0$ and $z \in \mathbb{C} \setminus \mathbb{R}$, the Fourier transform of $\beta^{m(z)}(s, z)$ is exactly

$$\lim_{n \rightarrow +\infty} \int_0^n \beta^{m(z)}(z, z) \phi(t, \lambda) dt = \frac{1}{\lambda - z}$$

Proof. See Remark 4.30 and the proof of Corollary 4.57. $\square$

5 Examples

5.1 Fourier equation

We consider the classical Fourier equation

$$Lu(s) + \lambda u(s) := \frac{d^2 u}{ds^2}(s) + \lambda u(s) = 0, \quad s \geq 0$$

with fundamental system of solutions

$$\{u^{(1)}(s, \lambda) := \cos(\sqrt{\lambda}s), u^{(2)}(s, \lambda) := \lambda^{-1/2} \sin(\sqrt{\lambda}s)\}$$

where $\sqrt{\lambda} = \lambda^{1/2}$ denotes the principal square root of λ . It has already been established in Proposition 2.29 that this problem is in the limit-point case.

5.1.1 Finding the limit-point

Let $h \in [0, \pi)$ be fixed. We form

$$\begin{aligned} \eta(s, \lambda) &:= -\sin(h) \cdot \cos(\sqrt{\lambda}s) + \cos(h) \cdot \lambda^{-1/2} \sin(\sqrt{\lambda}s) \\ \theta(s, \lambda) &:= \cos(h) \cdot \cos(\sqrt{\lambda}s) + \sin(h) \cdot \lambda^{-1/2} \sin(\sqrt{\lambda}s) \end{aligned}$$

Let $z \in \mathbb{C} \setminus \mathbb{R}$ with $\Im z > 0$ be fixed. We are looking for $m(z) \in \mathbb{C}$, $\Im m(z) > 0$, such that $\theta(s, z) + m(z)\eta(s, z) \in L^2([0, +\infty), \mathbb{C})$. Observe that

$$\theta(s, z) + m(z)\eta(s, z) = a \cdot \exp(i\sqrt{z}s) + b \cdot \exp(-i\sqrt{z}s), \quad a = a(z), b = b(z) \in \mathbb{C}$$

Since $\Im \sqrt{z} > 0$ for $\Im z > 0$, we have that the above solution is in $L^2([0, +\infty), \mathbb{C})$ if and only if $b = 0$. Moreover, the initial conditions on θ and η implies $a = \cos(h) - m(z) \sin(h)$. Thus, $m(z)$

satisfies the system of equations

$$\begin{cases} 2^{-1}(\cos(h) - i \cdot \sin(h)z^{-1/2} - m(z)\sin(h) - i \cdot m(z)\cos(h)z^{-1/2}) &= \cos(h) - m(z)\sin(h) \\ 2^{-1}(\cos(h) + i \cdot \sin(h)z^{-1/2} - m(z)\sin(h) + i \cdot m(z)\cos(h)z^{-1/2}) &= 0 \end{cases}$$

This leads to

$$m(z) = \frac{\cos(h) + i \cdot \sin(h)z^{-1/2}}{\sin(h) - i \cdot \cos(h)z^{-1/2}} = -\frac{\sin(h) - i \cdot \cos(h)\sqrt{z}}{\cos(h) + i \cdot \sin(h)\sqrt{z}}$$

and this formula also holds when $\Im z < 0$.

5.1.2 Finding the spectral function

Assume first that $\cot(h) \leq 0$ (i.e. $h \in [\pi/2, \pi)$) or $h = 0$. Then $\Im m(z)$ is well-defined and continuous on

$$\{z \in \mathbb{C} \mid \Im z \geq 0, \Re z < 0\} \cup \{z \in \mathbb{C} \mid \Im z \geq 0, \Re z > 0\}$$

and for real values of z , we have

$$\Im m(z) = \begin{cases} \frac{\sqrt{z}}{\cos(h)^2 + z \sin(h)^2} & \text{if } z > 0 \\ 0 & \text{if } z < 0 \end{cases}$$

For $\mu \leq \lambda \in \mathbb{R}$, consider the non-decreasing and right-continuous function $k(u)$ defined (up to an additive constant) by

$$k(\lambda) - k(\mu) := \lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} \int_{\mu}^{\lambda} \Im m(x + i\varepsilon) dx$$

Explicitly, for $\mu \leq \lambda \in \mathbb{R}$, we have

$$k(\lambda) - k(\mu) = \int_{\max\{\mu, 0\}}^{\min\{\lambda, 0\}} \frac{\sqrt{x}}{\cos(h)^2 + x \sin(h)^2} dx$$

Setting $k(0) = 0$ to fix notations, we find that $k(u) : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and is equal to

$$k(u) = \begin{cases} 2 \cos(h)^{-2} \tan(h)^{-3} (\tan(h)\sqrt{u} - \arctan(\tan(h)\sqrt{u})) & \text{if } h \in (\pi/2, \pi), u > 0 \\ \frac{2}{3} u^{3/2} & \text{if } h = 0, u > 0 \\ 2\sqrt{u} & \text{if } h = \pi/2, u > 0 \\ 0 & \text{if } u \leq 0 \end{cases}$$

Since $k(u)$ and $\rho(u)$ have the same points of continuity and since $k(\lambda) - k(\mu) = \rho(\lambda) - \rho(\mu)$ at points of continuity $\mu \leq \lambda$, we can set $\rho(u) \equiv k(u)$ as our spectral function on $\mathbb{R}$. Note that in this case, in order to get the expansion theorem, it is not needed to compute $\rho(u)$ explicitly. Indeed, $\rho(u)$ is absolutely continuous, so we only need its derivative almost everywhere (i.e.

$\Im m(z)$ for real z) in order to consider the Stieltjes integral with measure ρ .

Assume now that $\cot(h) > 0$, $h \neq 0$ (i.e. $h \in (0, \pi/2)$). Then $\Im m(z)$ is well-defined and continuous on

$$\{z \in \mathbb{C} \mid \Im z \geq 0, \Re z < -\cot(h)^2\} \cup \{z \in \mathbb{C} \mid \Im z \geq 0, \Re z > -\cot(h)^2\}$$

and for real values of z , we have

$$\Im m(z) = \begin{cases} \frac{\sqrt{z}}{\cos(h)^2 + z \sin(h)^2} & \text{if } z > 0 \\ 0 & \text{if } z \leq 0, z \neq -\cot(h)^2 \end{cases}$$

For $\mu \leq \lambda \in \mathbb{R}$, consider the non-decreasing and right-continuous function $k(u)$ defined (up to an additive constant) by

$$k(\lambda) - k(\mu) := \lim_{\substack{\varepsilon \rightarrow 0 \\ \varepsilon > 0}} \int_{\mu}^{\lambda} \Im m(x + i\varepsilon) dx$$

Then $k(u)$, and thus $\rho(u)$, is constant on $(-\infty, -\cot(h)^2)$, constant on $[-\cot(h)^2, 0]$ and continuous on $[-\cot(h)^2, +\infty)$.

To fix notations, we set $\rho(u) = 0$ on $(-\infty, -\cot(h)^2)$. As mentioned in Remark 4.36, we know that $\rho(u)$ necessarily has a jump discontinuity at $-\cot(h)^2$ because

$$\phi(s, -\cot(h)^2) = \eta(s, -\cot(h)^2) = -\sin(h) \exp(-\cot(h)s)$$

is square-integrable with squared L^2 -norm equaling

$$\sin(h)^2 \int_0^{+\infty} \exp(-2\cot(h)s) ds = \frac{\sin(h)^3}{2\cos(h)} = \rho(-\cot(h)^2)^{-1}$$

by Corollary 4.56. For $u > 0$, we proceed as in the previous case and obtain finally

$$\rho(u) = \begin{cases} 2\cos(h)^{-2} \tan(h)^{-3} (\tan(h)\sqrt{u} - \arctan(\tan(h)\sqrt{u})) + \rho(0) & \text{if } u > 0 \\ \frac{2\cos(h)}{\sin(h)^3} & \text{if } -\cot(h)^2 \leq u \leq 0 \\ 0 & \text{if } u < -\cot(h)^2 \end{cases}$$

5.1.3 Expansion theorem

Assume first that $h \in [\pi/2, \pi)$ or $h = 0$. For $f \in L^2([0, +\infty), \mathbb{C})$ and $\lambda \geq 0$, we can write

$$\begin{aligned} F(\lambda) &= \lim_{n \rightarrow +\infty} \int_0^n f(t) \phi(t, \lambda) dt \\ &= \lim_{n \rightarrow +\infty} \int_0^n f(t) \left(-\sin(h) \cdot \cos(\sqrt{\lambda}t) + \cos(h) \cdot \lambda^{-1/2} \sin(\sqrt{\lambda}t) \right) dt \end{aligned}$$

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as a limit of functions in $L^2(\mathbb{R}, \mathbb{C}, \rho)$. Observe that the value of $F(\lambda)$ on $(-\infty, 0)$ does not matter and can be set to be $\equiv 0$ since this set has ρ -measure zero. Since $\rho(u)$ is absolutely continuous on $\mathbb{R}$ with derivative $\Im m(u)$ almost everywhere, the inversion formula yields

$$\begin{aligned} f(s) &= \lim_{n \rightarrow +\infty} \int_0^n F(\mu) \phi(s, \mu) d\rho(\mu) \\ &= \lim_{n \rightarrow +\infty} \int_0^n F(\mu) (-\sin(h) \cdot \cos(\sqrt{\mu}s) + \cos(h) \cdot \mu^{-1/2} \sin(\sqrt{\mu}s)) d\rho(\mu) \\ &= \lim_{n \rightarrow +\infty} \int_0^n F(\mu) (-\sin(h) \cdot \cos(\sqrt{\mu}s) + \cos(h) \cdot \mu^{-1/2} \sin(\sqrt{\mu}s)) \Im m(\mu) d\mu \\ &= \lim_{n \rightarrow +\infty} \int_0^n F(\mu) (-\sin(h) \cdot \cos(\sqrt{\mu}s) + \cos(h) \cdot \mu^{-1/2} \sin(\sqrt{\mu}s)) \frac{\sqrt{\mu}}{\cos(h)^2 + \mu \sin(h)^2} d\mu \end{aligned}$$

as a limit of functions in $L^2([0, +\infty), \mathbb{C})$ and where the last integrals are standard Lebesgue integrals.

Assume now that $h \in (0, \pi/2)$. For $f \in L^2([0, +\infty), \mathbb{C})$, $\lambda > 0$, we can write

$$\begin{aligned} F(\lambda) &= \lim_{n \rightarrow +\infty} \int_0^n f(t) \phi(t, \lambda) dt \\ &= \lim_{n \rightarrow +\infty} \int_0^n f(t) (-\sin(h) \cdot \cos(\sqrt{\lambda}t) + \cos(h) \cdot \lambda^{-1/2} \sin(\sqrt{\lambda}t)) dt \end{aligned}$$

and for $\lambda = -\cot(h)^2$, we necessarily have

$$F(-\cot(h)^2) = -\sin(h) \int_0^{+\infty} f(t) \exp(-\cot(h)t) dt$$

Observe that the value of $F(\lambda)$ on $(-\infty, -\cot(h)^2) \cup (-\cot(h)^2, 0)$ does not matter and can be set to be $\equiv 0$ since this set has ρ -measure zero. Since $\rho(u)$ is absolutely continuous on $[0, +\infty)$ with derivative $\Im m(u)$, the inversion formula yields

$$\begin{aligned} f(s) &= \int_{\{-\cot(h)^2\}} F(\mu) \phi(s, \mu) d\rho(\mu) + \lim_{n \rightarrow +\infty} \int_0^n F(\mu) \phi(s, \mu) d\rho(\mu) \\ &= F(-\cot(h)^2) \phi(s, -\cot(h)^2) \rho(-\cot(h)^2) \\ &\quad + \lim_{n \rightarrow +\infty} \int_0^n F(\mu) (-\sin(h) \cdot \cos(\sqrt{\mu}s) + \cos(h) \cdot \mu^{-1/2} \sin(\sqrt{\mu}s)) \frac{\sqrt{\mu}}{\cos(h)^2 + \mu \sin(h)^2} d\mu \\ &= 2 \cot(h) \exp(-\cot(h)s) \int_0^{+\infty} f(t) \exp(-\cot(h)t) dt \\ &\quad + \lim_{n \rightarrow +\infty} \int_0^n F(\mu) (-\sin(h) \cdot \cos(\sqrt{\mu}s) + \cos(h) \cdot \mu^{-1/2} \sin(\sqrt{\mu}s)) \frac{\sqrt{\mu}}{\cos(h)^2 + \mu \sin(h)^2} d\mu \end{aligned}$$

as a limit of functions in $L^2([0, +\infty), \mathbb{C})$ and where the last integrals are standard Lebesgue integrals.

5.1.4 Inversion formula for the sine and cosine transforms

The choices $h = 0$ and $h = \pi/2$ yields the following expansions for $f \in L^2([0, +\infty), \mathbb{C})$:

$$\begin{aligned} F_{h=0}(\lambda) &= \lim_{n \rightarrow +\infty} \int_0^n f(t) \lambda^{-1/2} \sin(\sqrt{\lambda} t) dt \\ f(s) &= \lim_{n \rightarrow +\infty} \int_0^n F_{h=0}(\mu) \sin(\sqrt{\mu} s) d\mu \end{aligned}$$

and

$$\begin{aligned} F_{h=\pi/2}(\lambda) &= \lim_{n \rightarrow +\infty} - \int_0^n f(t) \cos(\sqrt{\lambda} t) dt \\ f(s) &= \lim_{n \rightarrow +\infty} - \int_0^n F_{h=\pi/2}(\mu) \mu^{-1/2} \cos(\sqrt{\mu} s) d\mu \end{aligned}$$

For simplicity, we assume that $f \in L^1([0, +\infty), \mathbb{C}) \cap L^2([0, +\infty), \mathbb{C})$. Combining both yields the Fourier inversion formula for the sine and cosine transforms. Indeed, we find

$$\begin{aligned} f(s) &= \lim_{n \rightarrow +\infty} \frac{1}{2} \int_0^n \int_0^{+\infty} f(t) \mu^{-1/2} (\sin(\sqrt{\mu} t) \sin(\sqrt{\mu} s) + \cos(\sqrt{\mu} t) \cos(\sqrt{\mu} s)) dt d\mu \\ &= \lim_{n \rightarrow +\infty} \frac{1}{2} \int_0^n \int_0^{+\infty} f(t) \mu^{-1/2} \cos(\sqrt{\mu}(t-s)) dt d\mu \\ &= \lim_{n \rightarrow +\infty} \int_0^{\sqrt{n}} \int_0^{+\infty} f(t) \cos(\mu(t-s)) dt d\mu \end{aligned}$$

as a limit of functions in $L^2([0, +\infty), \mathbb{C})$.

5.2 Laguerre polynomials

5.2.1 Definition and basic properties

Definition 5.1

The *Laguerre polynomial* of degree n , $n \geq 0$, is defined as

$$L_n(x) := e^x \frac{d^n}{dx^n} (x^n e^{-x})$$

More generally, the n -th *generalized Laguerre polynomial*, $n \geq 0$, with parameter $\alpha \in \mathbb{R}$ is defined as

$$L_n^{(\alpha)}(x) := x^{-\alpha} e^x \frac{d^n}{dx^n} (x^{n+\alpha} e^{-x})$$

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Remark 5.1

It follows from Leibniz formula that $L_n^{(\alpha)}(x)$ is a polynomial of degree n . More precisely,

$$L_n^{(\alpha)}(x) = (-1)^n x^n + \sum_{k=0}^{n-1} (-1)^k \binom{n}{k} (n+\alpha)(n-1+\alpha) \cdots (k+1+\alpha) x^k$$

Proposition 5.2

The polynomial $L_n^{(\alpha)}(x)$ satisfies the differential equation

$$xu''(x) + (\alpha + 1 - x)u'(x) + nu(x) = 0, \quad x \geq 0 \quad (5.1)$$

called Laguerre equation.

Proof. Let $y(x) = x^{n+\alpha} e^{-x}$. Then

$$x \frac{dy}{dx}(x) = (\alpha + n - x)y(x)$$

Differentiating this equation $(n+1)$ -times, we get

$$x \frac{d^{n+2}y}{dx^{n+2}}(x) + (n+1) \frac{d^{n+1}y}{dx^{n+1}}(x) = (\alpha + n - x) \frac{d^{n+1}y}{dx^{n+1}}(x) - (n+1) \frac{d^n y}{dx^n}(x)$$

by Leibniz formula. This means that $y^{(n)}(x)$ satisfies the differential equation

$$xy''(x) + (1 + x - \alpha)y'(x) + (n+1)y(x) = 0, \quad x > 0$$

and that $u(x) = x^{-\alpha} e^x y^{(n)}(x) = L_n^{(\alpha)}(x)$ satisfies the differential equation

$$xu''(x) + (\alpha + 1 - x)u'(x) + nu(x) = 0, \quad x > 0$$

By analyticity of $L_n^{(\alpha)}(x)$, this is also satisfied at $x = 0$. □

Remark 5.3

Let $\lambda \in \mathbb{R}$ be fixed and let $u \in C^2((0, +\infty))$. Then $u(x)$ solves

$$xu''(x) + (\alpha + 1 - x)u'(x) + \lambda u(x) = 0, \quad x > 0 \quad (5.2)$$

if and only if it solves

$$(x^{\alpha+1} e^{-x} u'(x))' + \lambda x^{\alpha} e^{-x} u(x) = 0, \quad x > 0 \quad (5.3)$$

5.2.2 The case $\alpha = -1/2$

In this Section 5.2.2, we choose and fix $\alpha = -1/2$.

Proposition 5.4

For $\lambda \in \mathbb{R}$ fixed, the differential equation

$$xu''(x) + \left(\frac{1}{2} - x\right)u'(x) + \lambda u(x) = 0, \quad x > 0 \quad (5.4)$$

has solutions

$$u_1(x) = \sum_{k=0}^{\infty} a_k x^k, \quad a_0 \in \mathbb{R}, \quad a_{k+1} = \frac{k - \lambda}{(k+1)(k+1/2)} a_k \quad k \geq 0 \quad (5.5)$$

$$u_2(x) = x^{1/2} \sum_{k=0}^{\infty} b_k x^k, \quad b_0 \in \mathbb{R}, \quad b_{k+1} = \frac{k - \lambda + 1/2}{(k+1)(k+3/2)} b_k \quad k \geq 0 \quad (5.6)$$

where the infinite sums converge absolutely on $\mathbb{R}$.

Moreover, if $\lambda \in \mathbb{N}$, then $u_1(x)$ is a polynomial and for $a_0 = L_n^{(-1/2)}(0)$, we have $u_1(x) = L_n^{(-1/2)}(x)$.

Proof. The solutions are obtained using Frobenius method (see for example [Tes12], Chapter 4), but this does not really matter. Checking the statement is straightforward. $\square$

Remark 5.5

The relationship between $u_1(x)$ and a_0 , respectively between $u_2(x)$ and b_0 , is linear.

Proposition 5.6

Let $\lambda \in \mathbb{R}$ and consider the differential equation

$$xu''(x) + \left(\frac{1}{2} - x\right)u'(x) + \lambda u(x) = 0, \quad x > 0$$

This differential equation is equivalent to the Sturm-Liouville problem

$$v''(t) + \left(\frac{1}{4} - \frac{t^2}{16}\right)v(t) + \lambda v(t) = 0, \quad t > 0 \quad (5.7)$$

The relationship between solutions of (5.4) and (5.7) is given by

$$x = t^2/4, \quad v(t) = e^{-x/2}u(x), \quad x, t > 0 \quad (5.8)$$

Proof. Equation (5.7) is obtained from (5.3) using a Liouville Transformation (see for example [Eve05], Section 7), but this does not really matter. Again, the statement can be checked easily. $\square$

Remark 5.7

The SL operator defined with $p(t) \equiv 1$ and $q(t) = t^2/16 - 1/4$ is in the limit-point case (c.f. Corollary 2.33) and has a discrete spectrum (c.f. Proposition 4.37).

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Corollary 5.8

For $\lambda \in \mathbb{R}$ fixed, the SL equation

$$v''(t) + \left(\frac{1}{4} - \frac{t^2}{16}\right)v(t) + \lambda v(t) = 0, \quad t \geq 0$$

has solutions

$$v_1(t) = e^{-t^2/8} \sum_{k=0}^{\infty} \frac{a_k}{4^k} t^{2k}, \quad a_0 \in \mathbb{R}, \quad a_{k+1} = \frac{k - \lambda}{(k+1)(k+1/2)} a_k \quad k \geq 0 \quad (5.9)$$

$$v_2(t) = \frac{1}{2} e^{-t^2/8} \sum_{k=0}^{\infty} \frac{b_k}{4^k} t^{2k+1}, \quad b_0 \in \mathbb{R}, \quad b_{k+1} = \frac{k - \lambda + 1/2}{(k+1)(k+3/2)} b_k \quad k \geq 0 \quad (5.10)$$

where the infinite sums converge absolutely on $\mathbb{R}$.

Moreover, if $\lambda \in \mathbb{N}$, then the infinite sum in the expression of $v_1(t)$ is a polynomial and for $a_0 = L_n^{(-1/2)}(0)$, we have $v_1(t) = e^{-t^2/8} L_n^{(-1/2)}(t^2/4)$.

Proof. This follows from Propositions 5.4 and 5.6. □

Proposition 5.9

For $\lambda \in \mathbb{R}$ fixed, the SL equation

$$v''(t) + \left(\frac{1}{4} - \frac{t^2}{16}\right)v(t) + \lambda v(t) = 0, \quad t \geq 0$$

with initial conditions

$$v(0) = 0, \quad v'(0) = 1$$

has a unique solution given by $\phi(t, \lambda) = c \cdot v_1(t)$ for some $c \in \mathbb{R} \setminus \{0\}$. This solution is square-integrable on $[0, +\infty)$ if and only if $\lambda \in \mathbb{N}$.

Proof. We show that

$$v_1(t) = e^{-t^2/8} \sum_{k=0}^{\infty} \frac{a_k}{4^k} t^{2k}, \quad a_0 \in \mathbb{R} \setminus \{0\}, \quad a_{k+1} = \frac{k - \lambda}{(k+1)(k+1/2)} a_k \quad k \geq 0$$

is square-integrable on $[0, +\infty)$ if and only if the sum is finite (which is the case if and only if $\lambda \in \mathbb{N}$). If the sum is finite, this is square-integrable on $[0, +\infty)$ since for any $n \in \mathbb{N}$: $e^{-t^2} t^n \leq e^{-t}$ when $t > 0$ is large enough. If the sum is not finite, then $\lambda \notin \mathbb{N}$ and for all k large enough, the a_k 's have the same sign. Without loss of generality, this sign is positive (otherwise, consider $-v_1(t)$). In particular, there exists $k_0 \in \mathbb{N}$ such that for all $k \geq k_0$, we have

$$a_{k+1} \geq \frac{1}{2(k+1)} a_k > 0$$

which then implies

$$a_k \geq \frac{1}{2^{k-k_0} k(k-1) \cdots (k_0+1)} a_{k_0} \geq \frac{C}{2^k k!} \quad \forall k > k_0$$

for some constant $C = C(k_0) > 0$. Therefore, we can rewrite $v_1(t)$ as

$$\begin{aligned} v_1(t) &= e^{-t^2/8} \sum_{k=0}^{k_0} \left(\frac{a_k}{4^k} t^{2k} - \frac{C}{8^k k!} t^{2k} \right) + e^{-t^2/8} \sum_{k=k_0+1}^{\infty} \frac{a_k}{4^k} t^{2k} + e^{-t^2/8} \sum_{k=0}^{k_0} \frac{C}{8^k k!} t^{2k} \\ &\geq e^{-t^2/8} P(t) + C \end{aligned}$$

where $P(t)$ is a polynomial of degree at most $2k_0$. This proves that $v_1(t)$ is not square-integrable on $[0, +\infty)$. $\square$

Corollary 5.10

The set of normalized eigenfunctions

$$\{v_n(t)\}_{n=0}^{\infty} := \left\{ \frac{e^{-t^2/8} L_n^{(-1/2)}(t^2/4)}{\|e^{-t^2/8} L_n^{(-1/2)}(t^2/4)\|_{L^2([0, +\infty))}} \right\}_{n=0}^{\infty}$$

forms an orthonormal basis for $L^2([0, +\infty), \mathbb{F})$.

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